

Deprotonation of Quinacridone on Ag(100): An XPS/NIXSW study of the phase transition

Masterthesis

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1 Introduction

Organic electronic materials have attracted significant attention in recent years due to their potential applications. The organic electronics include, e.g., organic light-emitting diodes (OLEDs), organic photovoltaics (OPVs), and organic field-effect transistors (OFETs).^[1–6]

Generally, an extended π -conjugated system is considered to be a prerequisite for high electron mobility and thus for good performance in organic electronics.^[7,8] Model molecules with an extended π -conjugated system, such as pentacene, have been widely studied for its application in organic electronic materials.^[9–11]

A structurally analogous molecule is quinacridone (QA), which possesses a comparable molecular structure but contains four distinct functional groups. QA is a planar, prochiral organic molecule ($\text{C}_{20}\text{H}_{12}\text{N}_2\text{O}_2$), which is widely used as pigment (Violet 19) in paints, laquers, plastics, and printing inks.^[12–15]

However, QA is not only an interesting molecule for its application as pigment, but also for its potential application in organic electronics. The functional groups of QA are responsible for the intermolecular hydrogen bonds, which lead to high field effect mobilities and photocurrents.^[16–18] More detailed, QA has been studied in the context of OLEDs,^[19–21] OFETs,^[22,23] and OPVs.^[24–27]

In addition, QA has been shown to form self-ordering molecular layers on metal surfaces, such as the Ag(100), the Ag(111), and the Cu(111) surface.^[28–31] A common structural motif of QA on these surfaces is the formation of 1D molecular chains, which is also observed in the bulk structure of QA.^[32] The formation of these molecular chains is driven by intermolecular hydrogen bonds between the amine and carbonyl groups.^[28–30]

The investigation of QA on the Ag(100) surface revealed two different phases: the α -phase and the β -phase. The metastable α -phase with a point-on-line (pol) structure consists of the previously mentioned homochiral 1D molecular chains, which are parallel to each other. The thermodynamically favored β -phase with a commensurate structure is formed irreversibly by annealing the sample and consists of dimers with alternating handedness. In addition, the β -phase is supposed to have fewer hydrogen bonds than the α -phase, which is energetically unfavorable, but is compensated by a stronger molecule-substrate interaction.^[28,30]

Nevertheless, the phase transition, including the differences between the two phases and the hydrogen bonding, is not fully understood and requires further investigation.

This is due to the fact that the previous studies of QA on the Ag(100) surface were mainly based on scanning tunneling microscopy (STM) and low energy electron diffraction (LEED) measurements, which only provide structural information.^[28,30]

Consequently, this thesis focuses on how the two phases of QA on the Ag(100) surface is affected by the interplay between intermolecular interactions via hydrogen bonds and adsorbate-susbrtate interactions. Especially, the formation of the hydrogen bonds and their change between the two phases, as well as the specific atoms that are involved in the adsorbate-substrate interaction are of interest.

To investigate these questions, x-ray photoelectron spectroscopy (XPS) and normal incidence x-ray standing wavefield (NIXSW) techniques are used. XPS provides insights into the binding energies of the core level electrons, which contain information about the chemical environment of the different atoms in the molecule. Furthermore, NIXSW allows to determine the adsorption height with high precision, which is directly related to the strength of the adsorbate-substrate interaction.

As a result thereof, new insights into the situation of the hydrogen bonds are obtained. Previous studies proposed two hydrogen bonds per molecule in the α -phase and 1.5 hydrogen bonds per molecule in the β -phase.^[28,30] The XPS support the two hydrogen bonds per molecule in the α -phase, but contradict the 1.5 hydrogen bonds per molecule in the β -phase, as the results indicate only one hydrogen bonds per molecule. In combination with the adsorption heights obtained with NIXSW, which show strong changes for the N atoms, a new structure model for QA on the Ag(100) surface is necessary.

2 Literature

2.1 Quinacridone (QA)

The investigated molecule in this thesis is quinacridone (QA). Its full IUPAC name is 5,12-dihydroquinolino[2,3-*b*]acridine-7,14-dione, but it is also known under Pigment Violet 19.^[33] QA is widely used as a red/violet pigment in paints, laquers, plastics, and printing inks.^[12–15]

QA is an planar organic molecule with the chemical formula $C_{20}H_{12}N_2O_2$ and a molar mass of $312.32 \text{ g mol}^{-1}$,^[33] which is known since 1935^[34] and industrially produced since 1958.^[35] It has four different isomers,^[36] but only the linear *trans* isomer is considered in this thesis. The molecular structure of QA is shown in Figure 2.1.

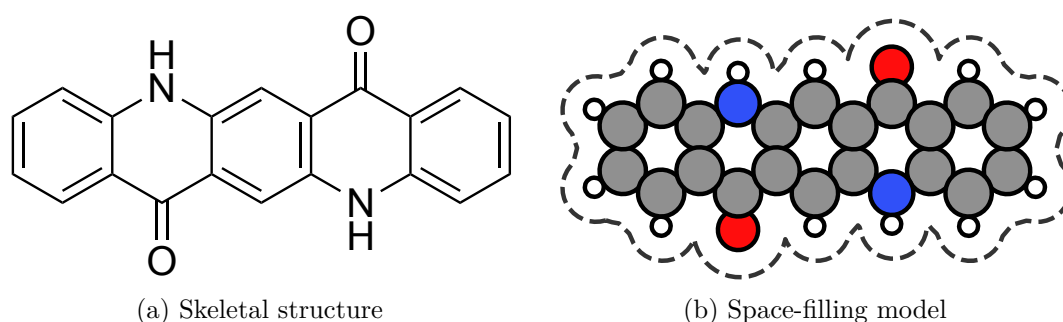


Figure 2.1: Molecular structure of QA with the skeletal structure and space-filling model. The space-filling model is based on the covalent radii of the atoms, with the following color code: C atoms are shown in grey, N atoms in blue, O atoms in red and H atoms in white. The envelope of the van-der-Waals radii of the atoms are shown as grey dashed line.

The molecule is a heterocyclic compound, which consists of five annulated carbon rings with four functional groups, namely two amine and two carbonyl groups. Thus, the molecule is very similar to the well-known molecule pentacene. The functional groups of QA are responsible for the intermolecular hydrogen bonds between the molecules, which are present in its four crystal structures^[32,37,38] or on surfaces.^[28,31,39–42]

2.2 Structures of QA on the Ag(100) surface

QA forms two different phases on the Ag(100) surface. Upon adsorption by room temperature, the molecules arrange in the so-called α -phase with a pol structure. By annealing the sample to 500 K, the molecules rearrange into the commensurate β -phase. These two phases were first observed and characterized by STM and LEED measurements. All information that are presented in the following is based on the work of Humberg et al.^[28,30] if not stated otherwise.

As said above, the α -phase of QA on the Ag(100) surface has forms when QA molecules are deposited onto the Ag(100) surface at room temperature. The α -phase consists of domains of homochiral molecular chains that are parallel to each other. The QA molecules adsorb with its π -system parallel to the surface in a face-on orientation.

The molecular chains are dominantly formed due to intermolecular hydrogen bonds between the H atoms of the amine group of one molecule and the O atoms of the carbonyl group of another molecule. This means that each QA molecule has four hydrogen bonds to its two nearest neighbors, resulting in two hydrogen bonds per molecule, as each hydrogen bond is shared between two molecules.

The evaluation of the spot profile analysis low energy electron diffraction (SPA-LEED) pattern of the α -phase results in the following unit cell matrix:

$$\mathbf{M}_{\alpha} = \begin{pmatrix} 2 & 1.25 \\ -3 & 4.80 \end{pmatrix},$$

which indicates a pol structure, because the values of the first column are integers.^[43] This indicates the importance of the substrate for the formation of the α -phase, as every lattice point of the superstructure matches with a substrate lattice line of the [10] direction.

The unit cell is primitive rectangular ($\alpha = 90^{\circ} \pm 1^{\circ}$) with the intermolecular distance in the molecular chains of $\mathbf{b}_1 = (6.8 \pm 0.1) \text{ \AA}$ and the distance of two parallel molecular chains of $\mathbf{b}_2 = (16.4 \pm 0.1) \text{ \AA}$ and contains one QA molecule. The STM images, the SPA-LEED pattern and the structure model of the α -phase are shown in Figure 2.2.

When the sample is annealed to 500 K, the molecules rearrange irreversibly into the thermodynamically favored β -phase. This means that the α -phase is a metastable phase, which is kinetically stabilized by e.g. the intermolecular hydrogen bonds. The exact mechanism of the phase transition is not known.

Similar to the α -phase, the QA molecules in the β -phase also adsorb with its π -system parallel to the surface in a face-on orientation and in molecular chains that are parallel

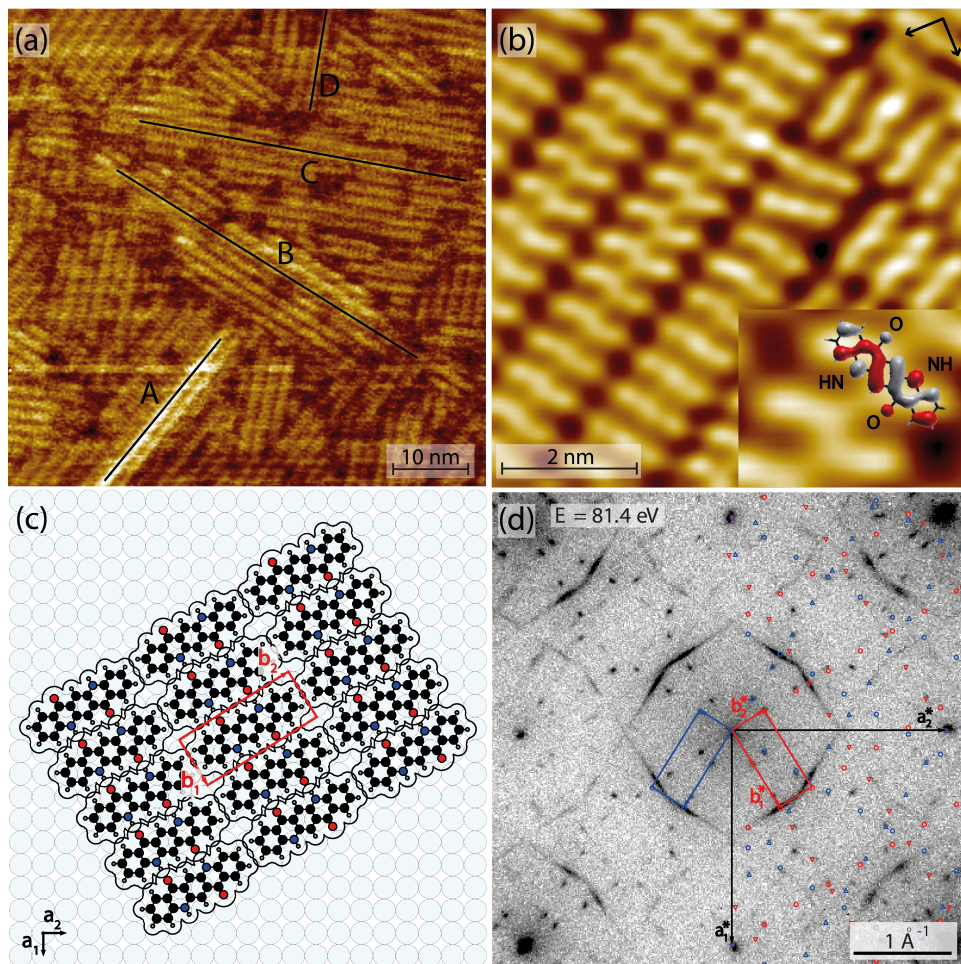


Figure 2.2: Current structure of complete monolayer (ML) of QA on Ag(100) in the α -phase. The image shows an overview STM image (a), a close-up STM image of the molecular arrangement (b), the SPA-LEED pattern (c) and the structure model (d). The image is taken from reference [30].

to each other. In contrast to the α -phase, the molecular chains are not homochiral, but consist of dimers with alternating handedness. In addition, the molecular chains also exhibit periodic indents.

This leads to a different situation of the hydrogen bonds. The two QA molecules of a dimer are connected by two hydrogen bonds, but the dimers are only connected by one hydrogen bond due to the indents. This means that each QA molecule has three hydrogen bonds to its two nearest neighbors, resulting in 1.5 hydrogen bonds per molecule and an energy loss due to the missing hydrogen bonds.

The evaluation of the SPA-LEED pattern of the β -phase results in the following unit cell matrix:

$$\mathbf{M}_{\beta} = \begin{pmatrix} 4 & 3 \\ -5 & 3 \end{pmatrix},$$

which corresponds to a commensurate structure. This indicates strong substrate interactions and is related to generally related to a gain in energy, which compensates the energy loss due to the missing hydrogen bonds.^[43] It is important to note that the unit cell of the β -phase does not take the handedness of the molecules into account, as the unit cell is determined by SPA-LEED, which is not sensitive to the handedness of the molecules, because the enantiomers have very similar structure factors.

The unit cell of the β -phase is oblique ($\alpha = 112.2^\circ \pm 1^\circ$) with the unit cell vectors $\mathbf{b}_1 = (6.9 \pm 0.1) \text{ \AA}$ and $\mathbf{b}_2 = (15.8 \pm 0.1) \text{ \AA}$ and contains two QA molecules. The STM images, the SPA-LEED pattern and the structure model of the β -phase are shown in Figure 2.3.

In addition, it was investigated if the β -phase involves metal-coordinated bonds between the QA molecules, which could be dismissed.^[30,44]

Further questions regarding the two phases are still open, e.g. why the thermodynamically favored β -phase does not form directly upon adsorption at room temperature, and what the exact mechanism of the phase transition is.

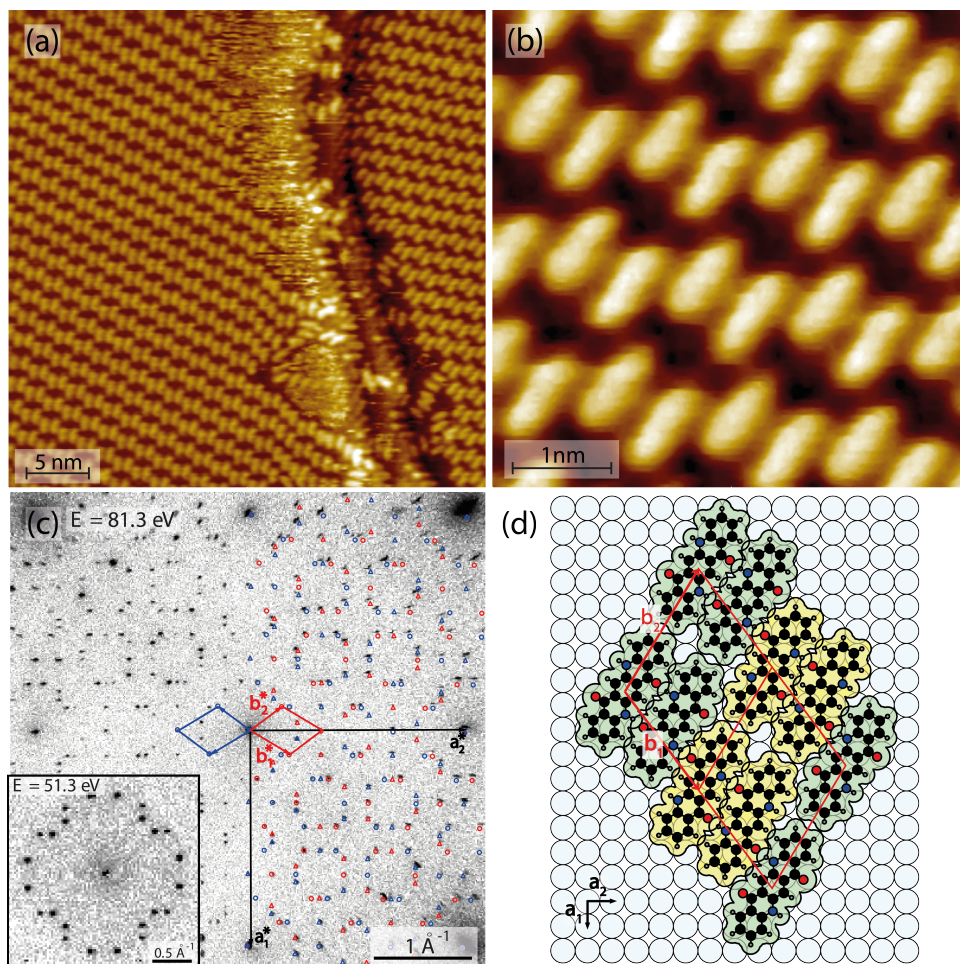


Figure 2.3: Current structure of complete ML of QA on Ag(100) in the β -phase. The image shows an overview STM image (a), a close-up STM image of the molecular arrangement (b), the SPA-LEED image (c) and the structure model (d). The image is taken from reference [30].

3 Theory

The following chapter shortly introduces the theoretical background of the experimental methods used in this work, namely XPS and NIXSW. For more detailed information, please refer to the cited literature.

3.1 X-ray photoelectron spectroscopy (XPS)

X-ray photoelectron spectroscopy (XPS) is based on the photoelectric effect, which is the ejection of electrons from bound states to the continuum by photon excitation.^[45,46] The photon energies of X-rays ($E_\gamma = h\nu$) are similar to the binding energy (BE) E_B of core electrons in atoms. Often, soft X-rays with energies between 100 eV and 2100 eV are used to excite core electrons into the vacuum level. More precisely, E_γ for each core level is chosen to be higher than the respective BE in order to create photoelectrons with a kinetic energy E_{kin} of approximately 100 eV. The specific choice of E_γ leads to a high surface sensitivity of the technique, as the mean free path of the emitted photoelectrons is only a few nanometers.^[47]

The photoemission process is schematically illustrated in Figure 3.1. Incident X-ray photons are absorbed by a core level electron (here: 1s electron), which leads to the ejection of a photoelectron. The kinetic energy of the emitted photoelectron E_{kin} is the difference between the tunable photon energy of the X-rays E_γ and the BE E_B of the core level electron in its initial state. This kinetic energy of the emitted photoelectron is measured by an analyzer and provides information about the chemical environment and the oxidation state of the atom. This phenomenon is referred to as chemical shift.^[48]

The kinetic energy of the emitted photoelectron that is measured by the analyzer E'_{kin} is relative to the Fermi level and given by:

$$E'_{\text{kin}} = E_{\text{kin}} + \phi_{\text{sample}}, \quad (3.1)$$

where E_{kin} is the kinetic energy of the photoelectron and ϕ_{sample} is the work function of the sample. As seen in Figure 3.1, the kinetic energy of the emitted photoelectron can be expressed as:

$$E_{\text{kin}} = E_\gamma - E_B - \phi_{\text{sample}}, \quad (3.2)$$

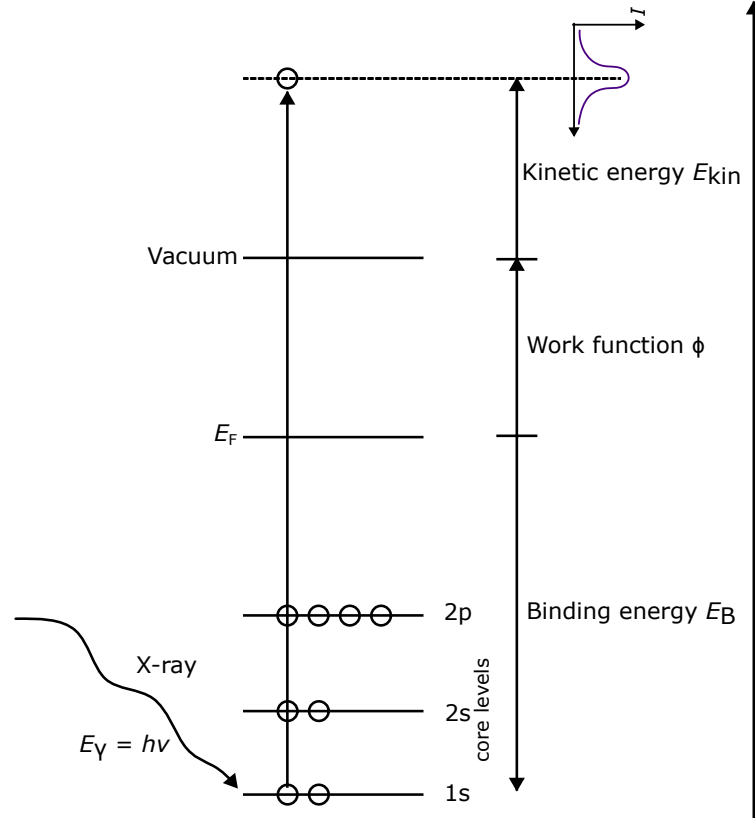


Figure 3.1: Schematic drawing of the photoemission process of an electron in the 1s orbital upon X-ray excitation. The image was adapted using reference [49].

where $E_\gamma = h\nu$ is the photon energy of the incident X-rays and E_B is the BE of the core level electron in its initial state.

E_B of an electron is defined as the energy difference between the final state of the system $E_f(n-1)$ and the initial state of the system $E_i(n)$ with n electrons:

$$E_B = E_f(n-1) - E_i(n). \quad (3.3)$$

As seen in Equation 3.3, BE depends on initial state and final state effects.

The initial state effects occur for the neutral unexcited atome and are determined by the chemical environment of the atom. Final state effects result from the ionized final state due to the photoemission process. Furthermore, as the photoemission process is fast, the system may not be in an adabatic equilibrium.^[50]

In addition, asymmetric line shapes of the photoemission spectra result from coupling of the emitted photoelectron to the conduction electrons.^[51] Moreover, various other effects have a large effect on the line shapes and full width at half maximum (FWHM) of the photoemission spectra, such as phonon broadening, the lifetime of the photohole, the spectral resolution of the exciting X-ray beam and the energy resolution of the analyzer.^[47,51]

3.2 Normal Incidence X-ray Standing Wave (NIXSW)

Normal incidence x-ray standing wavefield (NIXSW) is a technique that is based on XPS and allows to determine the positions of atoms relative to the Bragg planes of a substrate lattice with chemical sensitivity.^[50,52–54]

In contrast to other x-ray standing wavefield (XSW) techniques, NIXSW is measured with an angle of the incoming X-rays that is close to $\theta_B = 90^\circ$ with respect to the Bragg plane. This is necessary for the investigation of molecules on metal substrates. The reason behind this is that the Bragg condition $2d_{hkl} \sin(\theta_B) = n\lambda$ is essentially unaffected by small angular deviations for $\theta_B \approx 90^\circ$, because the gradient with respect to θ_B is very small. This unsensitivity to angular deviations is needed because metal substrates have a lower degree of perfection compared to semiconductor substrates, which is caused by the higher mosaicity of the metal substrate.^[50,52,53]

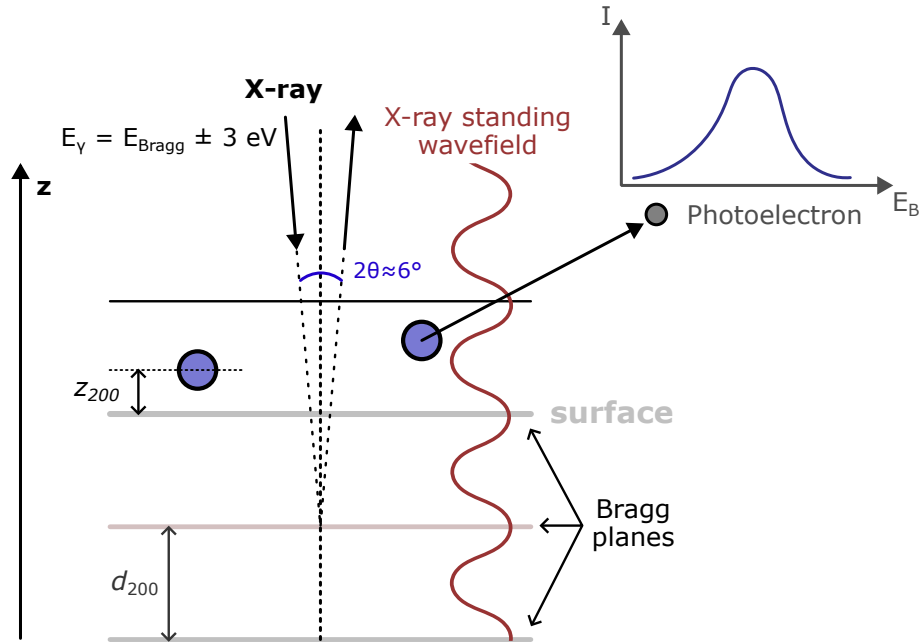
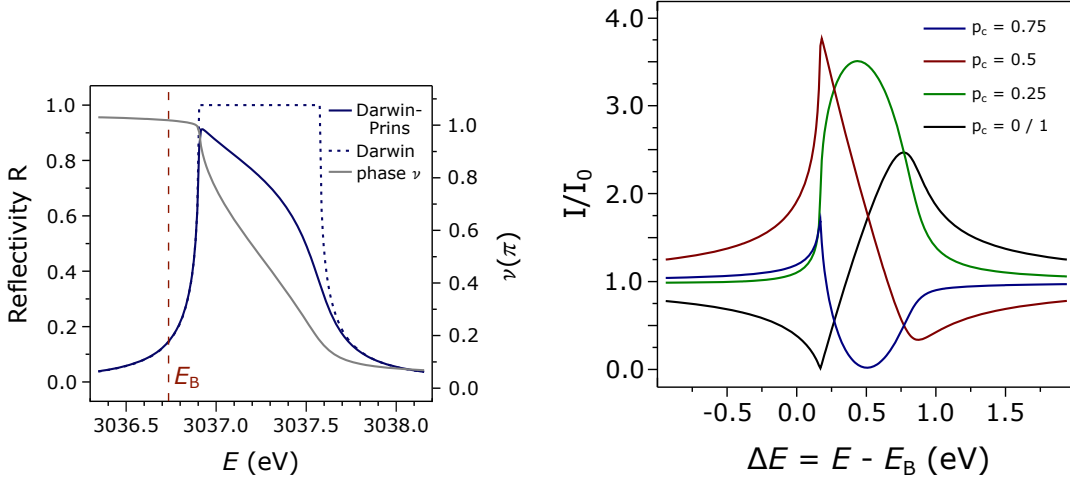


Figure 3.2: Schematic drawing of the principle of NIXSW. The image was adapted using reference [53].

The principle of NIXSW is schematically illustrated in Figure 3.2. An incoming X-ray beam is reflected at the surface of a single crystal substrate and interferes with the incoming beam, which leads to an standing wavefield. The standing wavefield extends into the bulk material and also into the vacuum above the surface.^[53] The periodicity of the standing wavefield is given by the spacing d_{hkl} of the Bragg planes of the substrate lattice. As there is a small difference between the incoming and the reflected X-ray beam, the intensity of the reflected beam as a function of the wavelength λ (or

respective beam energy $E_\gamma = h\frac{c}{\lambda}$) can be measured, which is referred to as reflectivity curve.

The intensity of the reflected X-ray beam depends on the photon energy E_γ . This is illustrated in Figure 3.3a. As the photon energy E_γ is tuned through the Bragg energy E_B , the anti-nodes of the standing wavefield shift with a phase $\nu(E_\gamma)$ from π to 0 with higher photon energies. For low photon energies E_γ , the anti-nodes of the standing wavefield are located between the Bragg planes. As the photon energy E_γ is increased, the anti-nodes of the standing wavefield shift by $d_{\text{hkl}}/2$ and are directly located at the Bragg planes. Accordingly, the absorption of the X-rays by the atoms of the substrate increases with a higher overlap of the anti-nodes of the standing wavefield with the Bragg planes, which leads to a decrease in the intensity of the reflected X-ray beam. This results in the characteristic asymmetric line shape of the reflectivity curve, which is known as the Darwin-Prins curve.^[55,56]



(a) Reflectivity curve showing the intensity of the reflected X-ray beam as a function of photon energy.

(b) Photoelectron yield showing the modulation of photoemission intensity due to the standing wave field.

Figure 3.3: Illustrated as in reference [57].

Depending on the position of the absorbed atoms relative to the Bragg plane, the atoms experience different intensities of the standing wavefield, which can be measured by e.g. XPS. The change of the photoemission intensity due to the standing wavefield is referred to as photoelectron yield and is schematically illustrated in Figure 3.3b. The normalized photoemission yield $\frac{I}{I_0}$ can be expressed as:

$$\frac{I}{I_0} = 1 + R(E_\gamma) + 2\sqrt{R(E_\gamma)}f_c \cos(\nu(E_\gamma) - 2\pi p_c), \quad (3.4)$$

3. Theory

where $R(E_\gamma)$ is the reflectivity, $\nu(E_\gamma)$ is the phase of the standing wavefield, f_c is the coherent fraction and p_c is the coherent position.

The coherent fraction f_c is a measure of the degree of order of the adsorbed atoms and can take values between 0 and 1. A value of $f_c = 1$ indicates that all atoms are located at the same vertical distance relative to the Bragg plane, while a value of $f_c = 0$ indicates that the atoms are randomly distributed.

In addition, the coherent position p_c is a measure of the average vertical distance of the adsorbed atoms relative to the Bragg plane and can take values between 0 and 1. A value of $p_c = 0$ indicates that the atoms are located at the Bragg plane, while a value of $p_c = 0.5$ indicates that the atoms are located halfway between two Bragg planes. The coherent position p_c can be converted into the adsorption height z of the adsorbed atoms relative to the substrate surface by the following equation:

$$z = (n + p_c)d_{\text{hkl}}, \quad (3.5)$$

where n is an integer and equals 1 for the (200) Bragg reflection of the Ag(100) substrate used in this work.

However, Equation 3.4 does not take into account non-dipolar effects, which can lead to a deviation of the measured photoelectron yield. These non-dipolar effects can be accounted for by the following equation:

$$\frac{I}{I_0} = 1 + S_R R(E_\gamma) + 2|S_I| \sqrt{R(E_\gamma)} f_c \cos(\nu(E_\gamma) - 2\pi p_c + \psi), \quad (3.6)$$

where S_R , $|S_I|$ and ψ are non-dipolar correction parameters, which can be calculated from the scattering phase shifts δp and δd .^[58,59]

4 Experimental

4.1 The Ag(100) crystal surface

As substrate for the adsorption of QA molecules, a silver crystal with a (100) surface orientation provided by the team of the I09 beamline of the Diamond Light Source is used. Silver crystallizes in a face-centered cubic (fcc) structure with a lattice constant of $a_{\text{Ag}} = 4.0856 \pm 0.0020 \text{ \AA}$ at a temperature of $25 \text{ }^{\circ}\text{C}$.^[60–62] The (100) surface is created by cutting the crystal along the plane defined by the [100] and [010] directions. This results in a square surface unit cell with a lattice constant of $a_{\text{Ag}(100)} = a_{\text{Ag}}/\sqrt{2} = 2.8890 \text{ \AA}$. The unit cell of the Ag(100) surface and the distinct adsorption sites on the Ag(100) surface are illustrated in Figure 4.1.

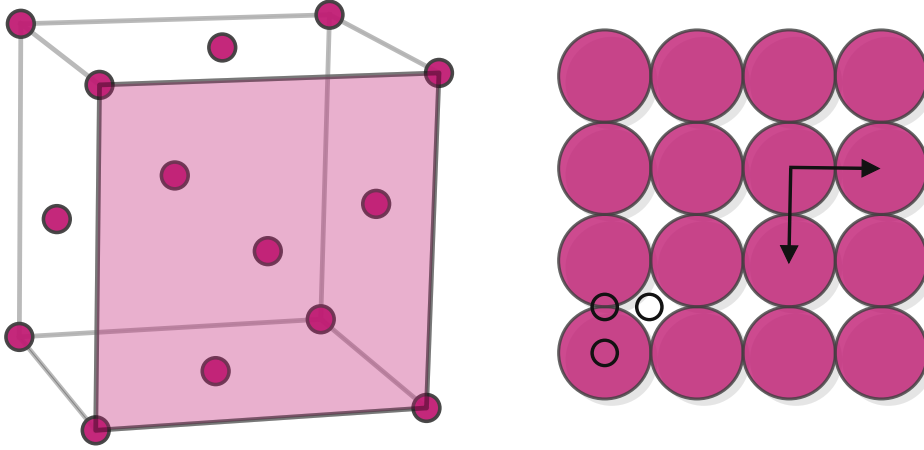


Figure 4.1: Unit cell of the silver crystal with marked (100) plane and the Ag(100) surface with the unit cell vectors (black arrows) and the distinct adsorption sites (black circles).

The Ag(100) crystal was cleaned in ultra-high vacuum (UHV) conditions by repeated cycles of Ar^+ ion sputtering and annealing. For the Ar sputtering, argon gas was dosed into the chamber to a pressure of $5 \cdot 10^{-5} \text{ mbar}$ and the Ar^+ ions were accelerated with an voltage of 1 keV. The annealing was done at approximately 825 K for 45 minutes to heal defects on the Ag(100) surface. The cleanliness of the Ag(100) surface is verified by LEED and XPS measurements before the deposition of QA molecules.

4.2 Experimental setup

The XPS and NIXSW measurements were performed at the end-station in EH2 of the I09 beamline at the Diamond Light Source in Didcot, UK.^[63] A schematic drawing of the experimental end-station is shown in Figure 4.2. At the end-station in EH2, an ultra-high vacuum (UHV) chamber with a base pressure of $\approx 10^{-10}$ mbar is available for surface science experiments, equipped with various preparation and analysis tools.

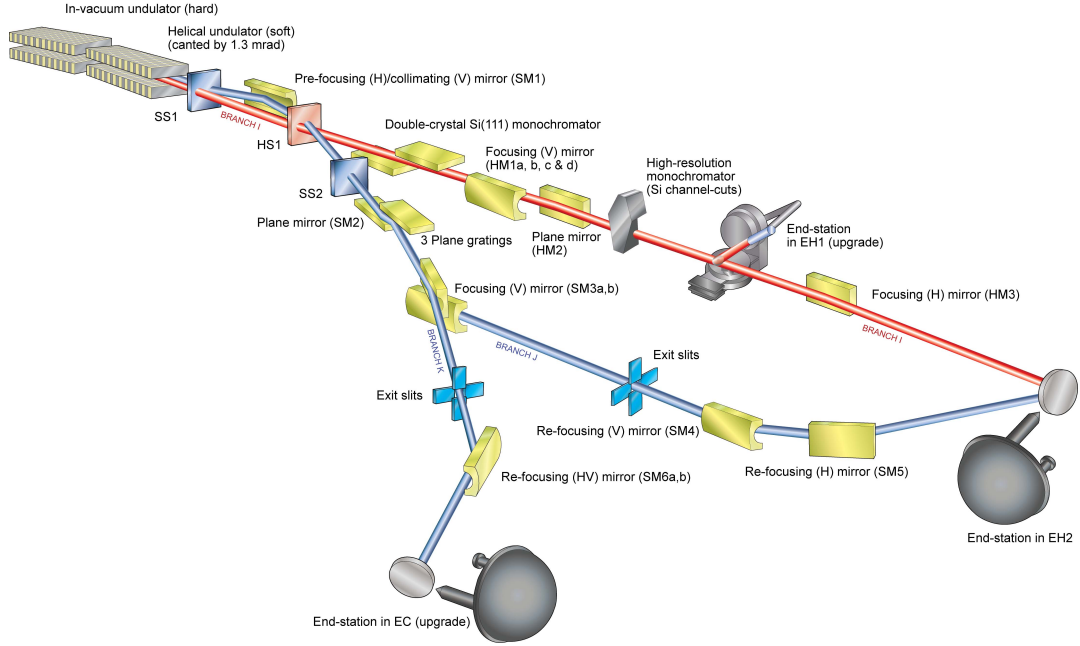


Figure 4.2: Schematic drawing of the I09 endstation of the Diamond Light Source in Didcot, UK. Picture taken from reference [63].

The end-station in EH2 (Figure 4.2) utilizes synchrotron radiation from an undulator inside the electron storage ring of the Diamond Light Source. At the end-station in EH2, hard (2.1 – 18 keV) and soft (230 – 2000 eV) X-rays are available for the experiments. For the hard X-rays, a double-crystal Si(111) monochromator is used, while for the soft X-rays a plane grating monochromator is employed. The X-rays are then directed into the experimental chamber using different focusing mirrors.^[63]

The experimental chamber is equipped with a 5 axes manipulator that allows precise sample positioning in x,y,z direction as well as polar and azimuth angles and contains a cryostat for cooling the sample down with liquid nitrogen or liquid helium. In addition, there are different preparation chambers attached to the experimental chamber for sample preparation, such as QA deposition.^[63]

For the XPS and NIXSW measurements, a hemispherical electron energy analyzer (VG Scienta EW4000 HAXPES) with a 70 frame per second CCD camera is used that allows the detection of photoelectrons with kinetic energies up to 10 kV. The analyzer has an acceptance cone of 56° and the analyzer-incident beam angle is fixed at 90° or 60° .^[63] The sample geometries for the XPS and NIXSW measurements are illustrated in Figure 4.3.

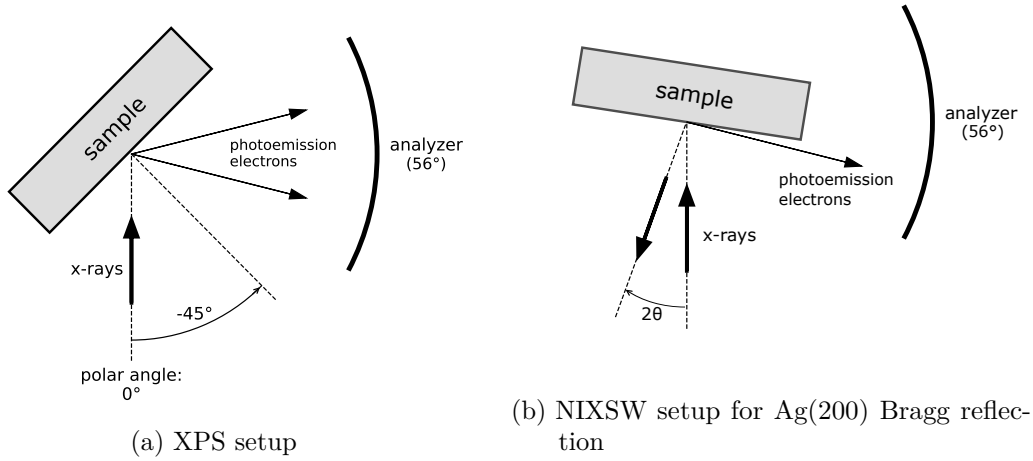


Figure 4.3: Top view of the sample geometries for the XPS and NIXSW measurements for the Ag(200) Bragg reflection at the I09 endstation of the Diamond Light Source in Didcot, UK. The angle θ denotes the deviation of the incident X-ray beam from the normal incidence with respect to the Ag(200) Bragg plane. Illustrated as in reference [64].

In addition to the aforementioned components for XPS and NIXSW measurements, the experimental chamber is also equipped with a LEED setup, an argon ion sputter gun, and an evaporator for the QA deposition.^[63]

4.3 XPS measurements

The XPS and NIXSW measurements of QA on Ag(100) were performed with the previously described experimental setup at the I09 endstation of the Diamond Light Source in Didcot, UK. The XPS measurements were conducted using photon energies of 500 eV for the C1s and N1s core levels and 630 eV for the O1s core level. The pass energy of the hemispherical electron energy analyzer was set to 20 eV. The step width for the spectra acquisition was set to 40 meV and 17 frames were recorded

for each core level and then averaged. The experimental parameters for the XPS measurements are summarized in Table 4.1.

Table 4.1: Experimental parameters for XPS measurements at the I09 endstation of the Diamond Light Source in Didcot, UK.

	photon energy E_γ	pass energy E_{pass}	step width	number of frames
C1s	500 eV	20 eV	40 meV	17
O1s	630 eV	20 eV	40 meV	17
N1s	500 eV	20 eV	40 meV	17

4.4 Data processing for the XPS spectra

The measured XPS spectra were processed and analyzed using the CasaXPS software.^[65] First, the XPS spectra with same experimental parameters and same experimental preparation were averaged to increase the signal-to-noise ratio.

Then, the BE of the averaged XPS spectra were calibrated against the Fermi edge, which was measured for each preparation after the XPS measurements of QA on the Ag(100) surface. This was done by fitting a XPS spectrum of the Fermi edge with the Fermi function

$$f(E) = \frac{a}{1 + e^{\frac{E-E_F}{k_B T}}} + b \quad (4.1)$$

where a is the amplitude, b the offset, E_F the Fermi energy, k_B the Boltzmann constant, and $T = 300$ K the temperature. The BE scale of the XPS spectra of QA on Ag(100) was then calibrated by aligning the fitted Fermi energy E_F to 0 eV accordingly.

After the calibration of the BE scale, the background of the XPS spectra was subtracted using a linear background or a Shirley background^[66] depending on the individual spectrum.

Finally, the different core level peaks of the XPS spectra were fitted using a numerical convolution of Lorentzian and a Gaussian to produce a Voigt profile. This function is called asymmetric Lorentzian lineshape $\text{LA}(\alpha, \beta, m)$, where α and β describe the asymmetry of the peak and m the width of the Gaussian.^[65] The asymmetric Lorentzian lineshape is defined as

$$LA(\alpha, \beta) = \begin{cases} [L(x)]^\alpha & x \leq E \\ [L(x)]^\beta & x > E \end{cases} \quad (4.2)$$

with

$$L(x) = \frac{1}{1 + 4 \left(\frac{(x-E)}{\text{FWHM}} \right)^2} . \quad (4.3)$$

This lineshape was used because it shows the best description of the XPS spectra. In addition to the lineshape, the area of the peak can be constrained. This was used to match the number of C atoms in the molecule to the number of C atoms in the corresponding peak. The exact parameters used for the peak fitting in CasaXPS are summarized in Table 4.2.

Table 4.2: Parameters used in CasaXPS^[65] for the different peaks of the XPS spectra of QA on Ag(100).

peak	line shape	area ratio
C _{arom}	LA(1,8,800)	10
C _{NH}	LA(1,8,800)	4
C _{CO}	LA(1,8,800)	4
C _O	LA(1,8,800)	2
O1	LA(1,8,400)	1 (for β -phase)
O1 _{damage}	LA(1,8,400)	-
O2	LA(1,8,400)	1 (for β -phase)
N1	LA(1,8,400)	1 (for β -phase)
N1 _{damage}	LA(1,8,400)	-
N2	LA(1,8,400)	1 (for β -phase)

As explained in the previous chapter, the FWHM of the peaks in the XPS spectra is influenced by various effects, such as instrumental broadening and the lifetime of the photohole. To account for these, the FWHM of the different peaks was not constrained to a fixed value, but was allowed to vary within a certain range. The exact values of the FWHM for the different peaks are given in the following chapter for each fit.

4.5 NIXSW measurements

The NIXSW measurements were performed with the Ag(200) Bragg reflection. The Bragg energy of the Ag(200) reflection is $E_{\text{Bragg}}^{200} = 3035.54$ eV. For the NIXSW

measurements, energies of the X-ray beam E_γ around the Bragg energy were used to excite photoelectrons from the C1s, O1s, and N1s core levels. The photon energy E_γ is between $E_{\text{Bragg}}^{200} - 4.5$ eV and $E_{\text{Bragg}}^{200} + 4.5$ eV.

The different NIXSW spectra were measured at a fresh sample position to avoid radiation damage effects. Furthermore, before an NIXSW measurement was performed, the reflectivity curve of the Ag(200) Bragg reflection was recorded to ensure that the position is usable and for the data evaluation. The maximum FWHM of the reflectivity curve was set to 0.90 eV.

The energy of the X-ray beam can be further classified as the *out-of-Bragg* regions, which is 2.5 eV below and above the Bragg energy. For these two regions, 4 data points (photoemission spectra) were measured, because there is no standing wavefield in the out-of-Bragg region and thus only the photoemission yield of the measured orbitals. The region with an energy of $E_{\text{Bragg}}^{200} \pm 2.5$ eV was measured with 26 data points. The other experimental parameters for the NIXSW measurements are summarized in Table 4.3.

Table 4.3: Experimental parameters for NIXSW measurements at the I09 endstation of the Diamond Light Source in Didcot, UK.

	pass energy E_{pass}	step width	size	number of frames
C1s	200 eV	40 meV	150 meV	6
O1s	200 eV	40 meV	150 meV	5
N1s	200 eV	40 meV	150 meV	3

4.6 Data processing for the NIXSW spectra

The NIXSW data was processed and analyzed with CasaXPS^[65] and Torricelli 4.0.1..^[58]

First, all single XPS spectra of the NIXSW measurement were processed in the same way as described for the XPS spectra in the section 4.4, but without the calibration of the BE to the Fermi edge and the area constraint. The relative positions and the FWHM of the peaks were constrained to the values from the fit model for the XPS spectra for all measurements and only the peak areas were allowed to vary. The errors of the integrated peak areas were computed by means of the Monte Carlo-based procedure implemented in CasaXPS^[65] and were used for the error calculation of the NIXSW data in Torricelli 4.0.1..^[58]

Afterwards, the peak areas of the different core levels for each photon energy E_γ were then further processed in Torricelli 4.0.1..^[58] Torricelli computes the ideal reflectivity curve and the phase of the standing wavefield for the given Bragg plane (here: Ag(200)) for the temperature of the experiment (here: $T = 300$ K) and the Bragg angle. Then,

the experimental reflectivity curve is fitted. Finally, the photoemission yield is fitted with taking the non-dipolar correction parameters from Table 4.4 into account to determine the coherent fraction and the coherent position. The normalized errors of the photoemission yield were calculated by a Monte Carlo analysis that is implemented in Torricelli 4.0.1.^[58] The final results of the NIXSW measurements are the coherent fraction f_c and the coherent position p_c with their respective errors.

Table 4.4: Non-dipolar correction parameters for the Ag(200) Bragg reflection.^[59]

core level	E_{kin}	δp	δd	$\Delta = \delta d - \delta p$
C1s	2748.74	0.76647	0.56969	-0.19678
O1s	2496.96	0.9676	0.69386	-0.27374
N1s	2633.74	0.86944	0.63570	-0.23374

5 XPS of QA on the Ag(100) surface

The XPS spectra of QA on the Ag(100) surface in the α -phase and in the β -phase are shown in this chapter, divided in the C1s, N1s and O1s core-level. In contrast to the XPS spectra with coverages of 1 ML, XPS spectra of a QA multilayer on the Ag(100) surface are presented. An overview of all XPS spectra is given in the Appendix (chapter 9).

Furthermore, a series of C1s XPS spectra while the sample is heated to 500 K to induce the phase transition from the α -phase to the β -phase is shown. Finally, C1s and N1s XPS spectra of QA preparation on the Ag(100) surface that was heated to 600 K is shown.

5.1 C1s XPS spectra

The C1s XPS spectra of QA on the Ag(100) surface in the α - and β -phase are shown in Figure 5.1. The preparation for the α -phase corresponds to a coverage of 1 ML, because this was the maximum coverage that could be achieved without the formation of a second layer. The coverages of the other preparations are calculated using the integrated peak areas of the C1s spectra with respect to the 1 ML preparation. This results in a coverage of approximately 0.54 ML for the β -phase preparation.

The C1s XPS spectra for both phases were fitted with 4 different peaks and have the same nomenclature. The main component is the C_{arom} peak, that originates from the 10 aromatic C atoms. The second peak at a lower BE is assigned to the 4 C atoms bonded to the N atoms from the amine group (C_{NH} peak). The third peak, namely C_{CO} , at higher BE corresponds to the 4 C atoms next to the carbonyl group (CO). The last peak at the highest BE is assigned to the 2 C atoms bonded to the O atom in the carbonyl group (C_{O} peak).

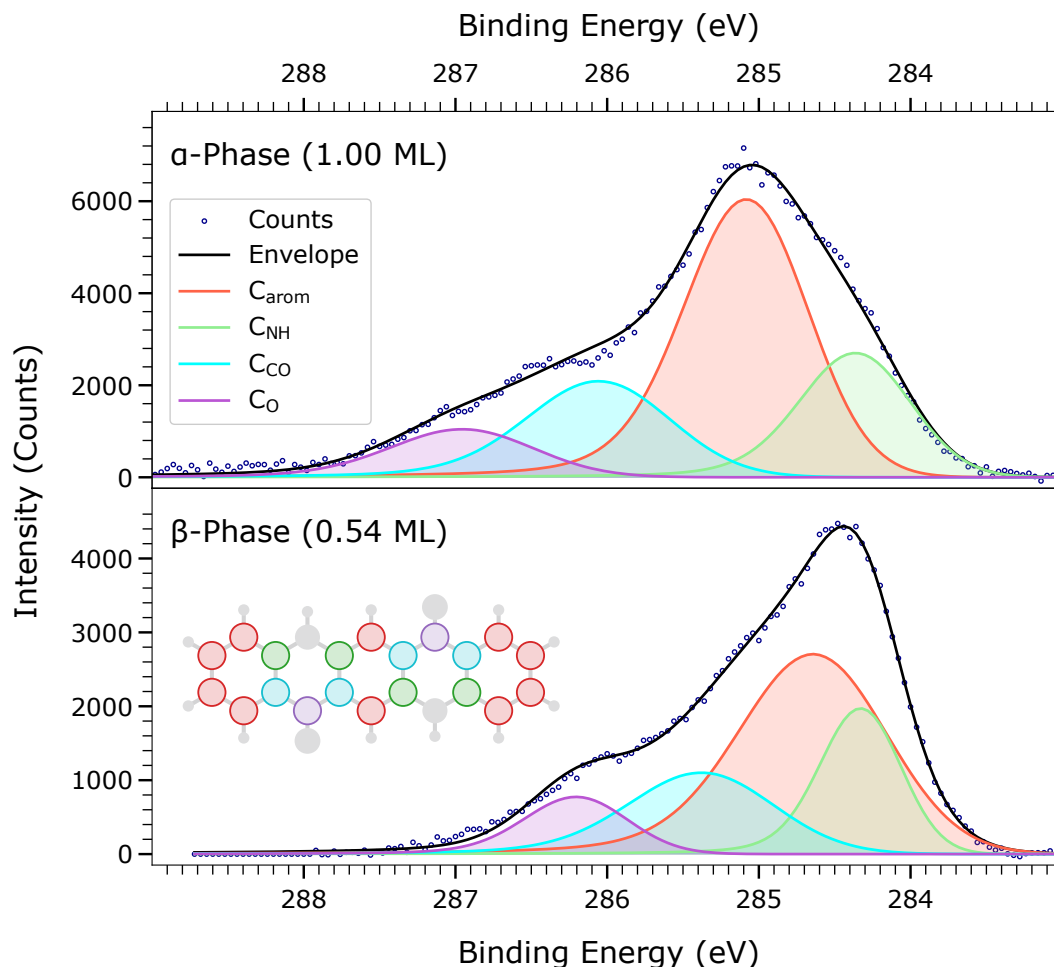


Figure 5.1: Comparison of the C1s XPS spectra of QA on the Ag(100) surface in the α -phase and in the β -phase with the counts (blue circles), C_{arom} (red), C_{NH} (green), C_{CO} (lightblue), C_{O} (purple) and the resulting envelope (black). The used fit parameters are summarized in Table 5.1.

Figure 5.1 shows a clear shift of all peaks to lower BE when going from the α -phase to the β -phase. The C_{arom} peak shifts from 295.00 eV in the α -phase to 284.54 eV in the β -phase, resulting in a shift of 0.46 eV. The C_{CO} peak shifts by 0.68 eV from 285.96 eV to 285.28 eV, while the C_{O} peak shows a shift of 0.73 eV from 286.86 eV to 286.13 eV. In total, these 3 peaks show similar shifts of approximately 0.6 eV.

The C_{NH} peak exhibits a different behavior, as it only shifts by 0.02 eV from 284.29 eV in the α -phase to 284.27 eV in the β -phase. The BE shift of the C_{NH} peak indicates a very unique change in the chemical environment during the phase transition compared to the other C atoms.

The FWHM of the peaks also changes from the α - to the β -phase. The C_{arom} peak broadens from 0.95 eV in the α -phase to 1.13 eV in the β -phase. The FWHM of the C_{CO} peak shows a almost no change, while the C_{O} peak narrows from a FWHM of 1.1 eV to 0.79 eV. The FWHM of the C_{NH} peak significantly decreases from 0.85 eV in the α -phase to 0.62 eV in the β -phase.

The fit parameters for the C1s XPS spectra of QA on the Ag(100) surface in the α - and β -phase are summarized in Table 5.1.

Table 5.1: BE, FWHM and area ratio for the different peaks of the C1s XPS spectra of QA on the Ag(100) surface in the α - and β -phase.

	peak	BE (eV)	FWHM (eV)	area ratio
α -phase	C_{arom}	285.00	0.95	10
	C_{NH}	284.29	0.85	4
	C_{CO}	285.96	1.1	4
	C_{O}	286.86	1.1	2
β -phase	C_{arom}	284.54	1.13	10
	C_{NH}	284.27	0.62	4
	C_{CO}	285.28	1.11	4
	C_{O}	286.13	0.79	2

The values for the BE and the FWHM can be compared to literature values of similar molecules on metal surfaces. For example, Peryleno[3,4-cd:9,11-c'd']dipyran-3,5,10,12-tetrone (PTCDA) on the Ag(100) surface shows different peaks for the following C atoms: $C_{\text{C-C}}$, $C_{\text{C-H}}$, and C_{funct} . The $C_{\text{C-C}}$ peak with a BE of 284.07 eV and the $C_{\text{C-H}}$ peak with a BE of 284.39 eV for PTCDA are comparable to the C_{arom} peak for QA. The BE for PTCDA is similar to the BE for QA, but slightly lower by approximately 0.2-0.5 eV.^[57]

The $C_{\text{C-CO}}$ peak for PTCDA with a BE of 284.99 eV exhibits a 0.6 eV higher BE than the $C_{\text{C-H}}$ peak and a 0.72 eV higher BE than the $C_{\text{C-C}}$ peak.^[57] This finding is similar to the BE difference between the C_{arom} peak and the C_{CO} peak for QA in the α -phase (0.96 eV) and in the β -phase (0.74 eV).

The C_{funct} peak for PTCDA with a BE of 286.96 eV shows a 2.9 eV higher BE than the $C_{\text{C-C}}$ peak and a 2.6 eV higher BE than the $C_{\text{C-H}}$ peak.^[57] This differences in the BE are slightly lower than the BE difference between the C_{arom} peak and the C_{O} peak for QA in the α -phase (1.86 eV) and in the β -phase (1.59 eV). However, the general trend of increasing BE for C atoms bonded to more electronegative atoms is the same for both molecules.

Another comparable molecule is indigo due to the presence of the aromatic ring, the N atoms, and the O atoms. The C1s XPS spectrum of indigo in UHV but on a

non-reactive MoS₂ substrate shows 2 different peaks and two shake-up features.^[67] The main peak contains the C atoms from the aromatic rings and has a BE of approximately 286 eV. The second peak at a higher BE of approximately 287.5 eV corresponds to the C atoms bonded to the N and O atoms.^[67]

The relative position of the peaks is similar to the C1s XPS spectra of QA on the Ag(100) surface, where the C_{arom} peak has a lower BE than the C_{CO} and C_O peaks. However, the absolute BE values for indigo are approximately 1-2 eV higher than for QA, which can be explained by the very different substrate effects and molecular structures.

5.1.1 Fit model for the NIXSW analysis in the β -phase

The C1s XPS spectra of QA on the Ag(100) surface in the β -phase were also fitted with a model containing only 3 peaks, where the C atoms from the C_{NH} peak are included in the C_{arom} peak. This fit model was used for the NIXSW analysis in the subsection 6.1.1, because the C_{NH} peak shows only a very small shift in its BE during the phase transition and therefore strongly overlaps with the C_{arom} peak. This leads to a large uncertainty in the fit parameters for the NIXSW spectra of the C_{NH} peak in the β -phase. The resulting fit is shown in Figure 5.2.

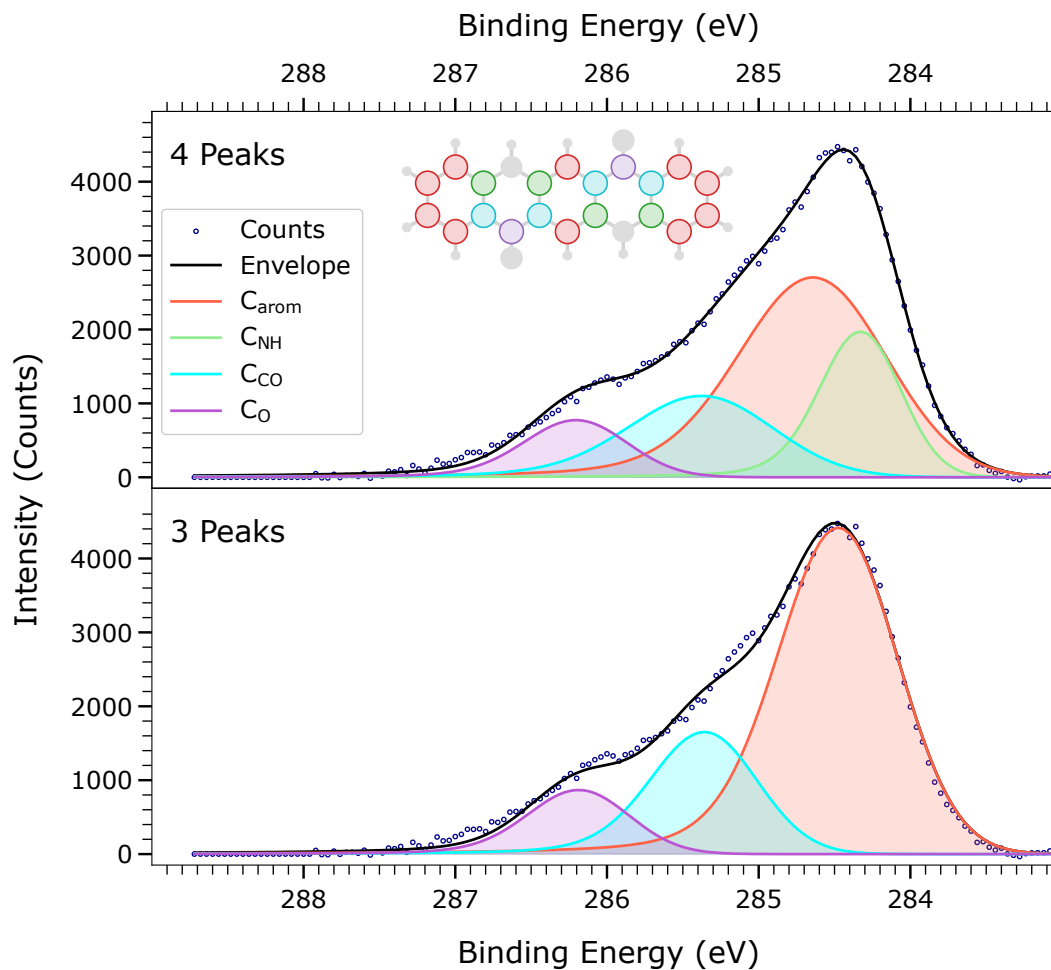


Figure 5.2: Comparison of the C1s XPS spectra of QA on the Ag(100) surface in the β -phase with 4 and 3 peaks fitted. The figure contains the counts (blue circles), C_{arom} (red), C_{NH} (green), C_{CO} (lightblue), C_{O} (purple) and the resulting envelope (black). The C_{NH} peak is omitted in the fit with 3 peaks. The used fit parameters are summarized in Table 5.2.

The fit parameters for the C1s XPS spectra of QA on the Ag(100) surface in the β -phase with both fit models are summarized in Table 5.2.

Table 5.2: BE, FWHM and area ratio for the different peaks of the C1s XPS spectra of QA on the Ag(100) surface in the β -phase with 4 and 3 peaks fitted.

	peak	BE (eV)	FWHM (eV)	area ratio
β -phase - 4 peaks	C _{arom}	284.54	1.13	10
	C _{NH}	284.27	0.62	4
	C _{CO}	285.28	1.11	4
	C _O	286.13	0.79	2
β -phase - 3 peaks	C _{arom} + C _{NH}	284.39	0.92	14
	C _{CO}	285.29	0.82	4
	C _O	286.12	0.78	2

As seen in Figure 5.2 and Table 5.2, both fit models describe the experimental data very well. The FWHM of the combined C_{arom} + C_{NH} peak is 0.92 eV, which is in between the FWHM of the individual peaks (1.13 eV and 0.62 eV). The BE of the combined peak is 284.39 eV, which is also in between the individual BE values of 284.54 eV and 284.27 eV. The area ratio of the combined peak is 14, which is the sum of the individual area ratios of 10 and 4.

The C_{CO} and C_O peaks show almost no change in their BE when comparing both fit models. Nevertheless, the FWHM of the C_{CO} peak decreases from 1.11 eV to 0.82 eV when using the fit model with 3 peaks, but the FWHM of the C_O peak remains unchanged. The change in the FWHM of the C_{CO} peak can be explained by the larger separation with respect to the combined C_{arom} + C_{NH} peak compared to the individual C_{arom} peak.

5.1.2 C1s XPS spectra for the Multilayer

In addition to the C1s XPS spectra of QA on the Ag(100) surface in the α - and β -phase, a C1s XPS spectrum of a QA multilayer on the Ag(100) surface is compared to the preparation with a coverage of 1 ML in Figure 5.3.

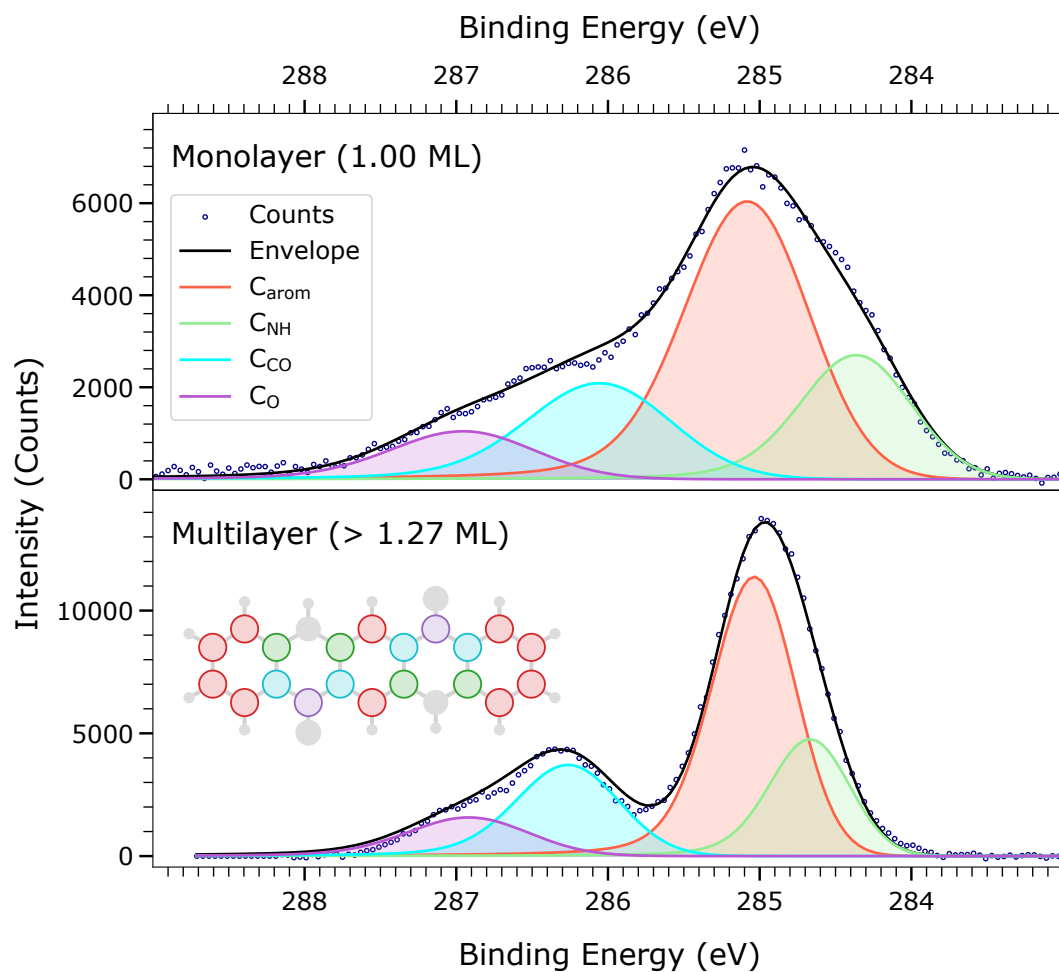


Figure 5.3: Comparison of the C1s XPS spectra of QA on the Ag(100) surface in the α -phase for 1.00 ML and 1.27 ML coverage with the counts (blue circles), C_{aryl} (red), C_{NH} (green), C_{CO} (lightblue), C_{O} (purple) and the resulting envelope (black). The used fit parameters are summarized in Table 5.3.

The C1s XPS spectrum of the QA multilayer was fitted with the same model as used for the α -phase and the fit parameters are summarized in Table 5.3.

Table 5.3: BE, FWHM and area ratio for the different peaks of the C1s XPS spectra of QA with a ML and Multilayer coverage on the Ag(100) surface in the α -phase.

	peak	BE (eV)	FWHM (eV)	area ratio
1.00 ML	C _{arom}	285.00	0.95	10
	C _{NH}	284.29	0.85	4
	C _{CO}	285.96	1.1	4
	C _O	286.86	1.1	2
1.27 ML	C _{arom}	284.98	0.64	10
	C _{NH}	284.61	0.61	4
	C _{CO}	286.19	0.78	4
	C _O	286.84	0.92	2

As seen in Figure 5.3 and Table 5.3, the C1s XPS spectrum of the QA multilayer shows a very different shape compared to the 1 ML preparation. The FWHM of all peaks decreases by 0.2-0.3 eV from the monolayer to the multilayer. Nevertheless, the BE of the C_{arom} and the C_O peak remain almost unchanged, while the other 2 peaks shift to higher BE by approximately 0.3 eV.

This again can be compared to PTCDA on the Ag(100) surface. The C1s XPS spectrum of a PTCDA multilayer on the Ag(100) surface shows also a shift of the C_{C-C}, C_{C-H} and C_{C-CO} peaks to higher BE by approximately 0.5-0.9 eV. The C_{funct} peak, however, shows an even larger shift to higher BE by approximately 2 eV.^[57]

This trend is similar to the C1s XPS spectra of QA on the Ag(100) surface in the monolayer and multilayer, where the peaks show no or only a small increase of 0.3 eV in the BE for C atoms. Nevertheless, the changes are not comparable in detail, because the molecular structures of QA and PTCDA are very different and the differences in the coverages of the compared spectra are not the same.

5.2 N1s XPS spectra

The N1s XPS spectra of QA on the Ag(100) surface in the α - and β -phase are shown in Figure 5.4. The preparation are the same as for the C1s spectra, with a coverage of 1 ML for the α -phase and 0.54 ML for the β -phase.

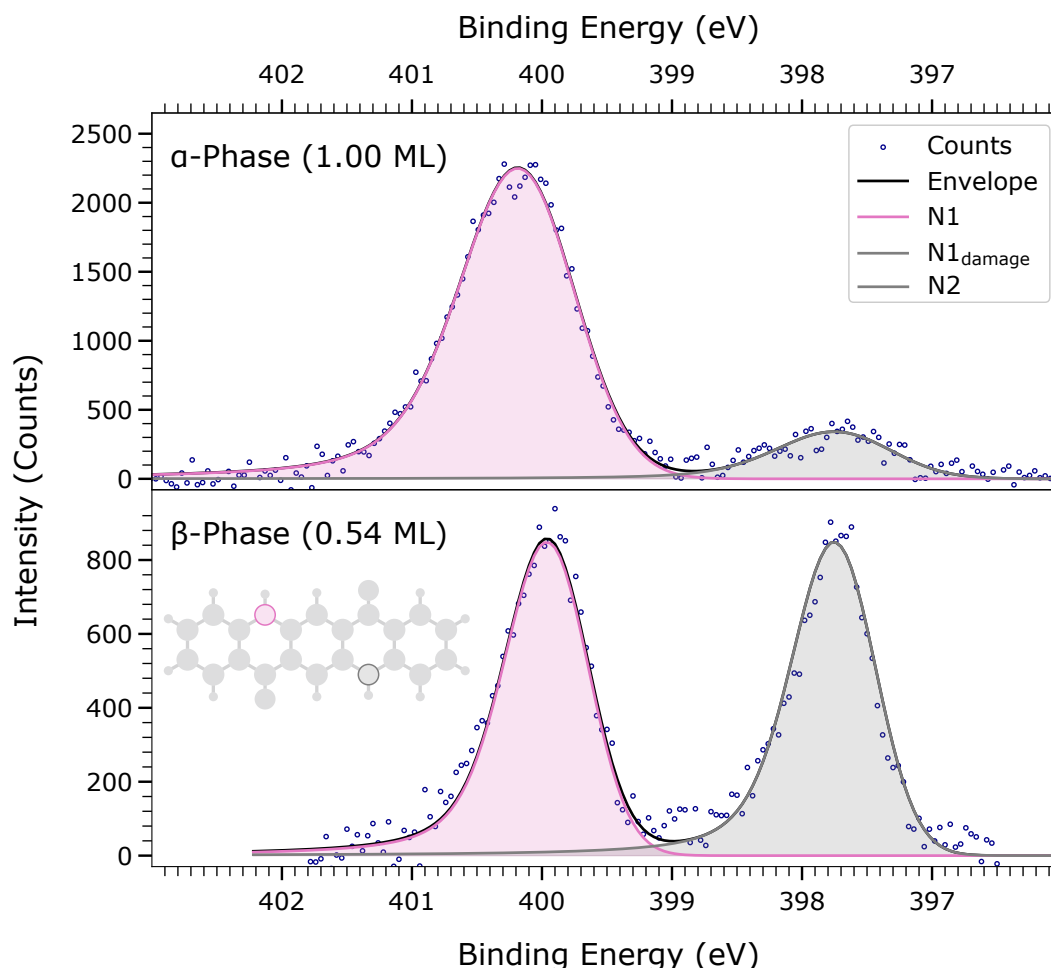


Figure 5.4: Comparison of the N1s XPS spectra of QA on the Ag(100) surface in the α -phase and in the β -phase with the counts (blue circles), N1 (purple), N1_{damage} (grey), N2 (grey) and the resulting envelope (black). The used fit parameters are summarized in Table 5.4.

The N1s XPS spectra for both phases were fitted with different models, both having 2 peaks. The main component for the α -phase is the N1 peak, that originates from the 2 N atoms in the amine group forming hydrogen bonds with the O atoms with the carbonyl groups of the neighboring molecules. In addition, a second peak at lower

BE was observed, which is assigned to N atoms from the QA molecules that have different chemical environments due to radiation damage (N1_{damage} peak). This peak is not related to the phase transition, but occurs due to prolonged exposure of the sample to the X-ray beam during the measurement. It was observed that the intensity of this peak changes for different measurements, which are shown in Figure ??.

In the β -phase, there are also 2 peaks observed in the N1s XPS spectrum. The main component is again the N1 peak from one N atom in the amine group. However, the N2 peak at lower BE is assigned to a different chemical environment of one N atom in the β -phase. The detailed fit parameters are summarized in Table 5.4.

Table 5.4: BE, FWHM and area ratio for the different peaks of the N1s XPS spectra of QA on the Ag(100) surface in the α - and β -phase.

	peak	BE (eV)	FWHM (eV)	area ratio
α -phase	N1	400.05	1.02	1
	N1 _{damage}	397.66	1.02	-
β -phase	N1	399.87	0.73	1
	N2	397.66	0.73	1

As seen in Figure 5.4 and Table 5.4, the N1s XPS spectrum of the β -phase shows only a minor change of 0.18 eV in the BE of the N1 peak. In contrast to the α -phase, the β -phase exhibits a second peak (N2) with the same area as the main peak, which indicates that both N atoms in the amine group have different chemical environments in the β -phase. The BE of the N2 peak is 397.66 eV, which is the same as the N1_{damage} peak in the α -phase. This indicates that the chemical environment of one N atom in the β -phase is similar to the damaged N atoms in the α -phase.

Furthermore, the FWHM of both peaks decreases from 1.02 eV in the α -phase to 0.73 eV in the β -phase, indicating a more uniform chemical environment for both N atoms in the amine group in the β -phase.

The BE of the N1s XPS spectra of QA on the Ag(100) surface can also be compared to literature values of similar molecules on metal surfaces. For example, the N1s XPS spectrum of pyridine shows a peak at a BE of approximately 400.18 eV.^[68] Another example is 6,6'-dibromoindigo, that also shows a N1s peak at a BE of 399.9 eV.^[69] Both literature values are very similar to the BE of the N1 peak of QA on the Ag(100) surface in both phases (400.05 eV and 399.87 eV).

The interpretation of the N2 peak in the N1s XPS spectrum of QA on the Ag(100) surface in the β -phase more complex due to the very large change in its BE. One possible explanation of this finding is the formation of an N-Ag bond. A similar behavior was observed for e.g. bis-(4-bromophenyl)amine on the Ag(111) surface.^[70] The formation of N-Ag bonds for bis-(4-bromophenyl)amine leads to a second N1s

peak at a lower BE of 397.1 eV, which is very similar to the BE of the N2 peak of QA in the β -phase (397.66 eV).

However, the N-Ag bond formation in bis-(4-bromophenyl)amine occurs with the loss of the H atoms from the amine group.^[70] Similar deprotonation processes were also observed for other amine-containing molecules, e.g. glyphosate on different substrates.^[71] There, the N1s XPS spectra show also a second peak at 2 eV lower BE after the deprotonation of the amine group. In contrast to the aforementioned example, no N-Ag bond is formed. Another result of this study was, that the deprotonation also occurs when the molecules are exposed to the X-ray beam.^[71] Further studies of polyaniline films and polypyrrole show that the deprotonation of amine groups leads to a change in the BE of the N1s peak of approximately 2 eV and that the peak of the deprotonated amine group is located at 397.7 eV.^[72–74]

There are even more examples like n-butylamine on Au and Pt, where the deprotonation of the amine group leads to a second N1s peak at a 2 eV lower BE than the main peak and a third peak at a 2 eV higher BE corresponds to a protonated amine.^[75]

All of these literature examples show that the deprotonation of amine groups leads to a second N1s peak at approximately 2 eV lower BE than the main peak. This is very similar to the N1s XPS spectrum of QA on the Ag(100) surface in the β -phase, where the N2 peak is also located at a 2.2 eV lower BE than the N1 peak. Therefore, it is very likely that the formation of deprotonated amine groups occurs in the QA monolayer in the β -phase. In addition, the deprotonation also occurs when the sample is exposed to the X-ray beam, which could explain why the N1_{damage} peak is located at the same BE as the N2 peak.

5.2.1 Multilayer

On top of the N1s XPS spectra of QA on the Ag(100) surface in the α - and β -phase, a N1s XPS spectrum of a QA multilayer on the Ag(100) surface is compared to the α -phase preparation with a coverage of 1 ML in Figure 5.5.

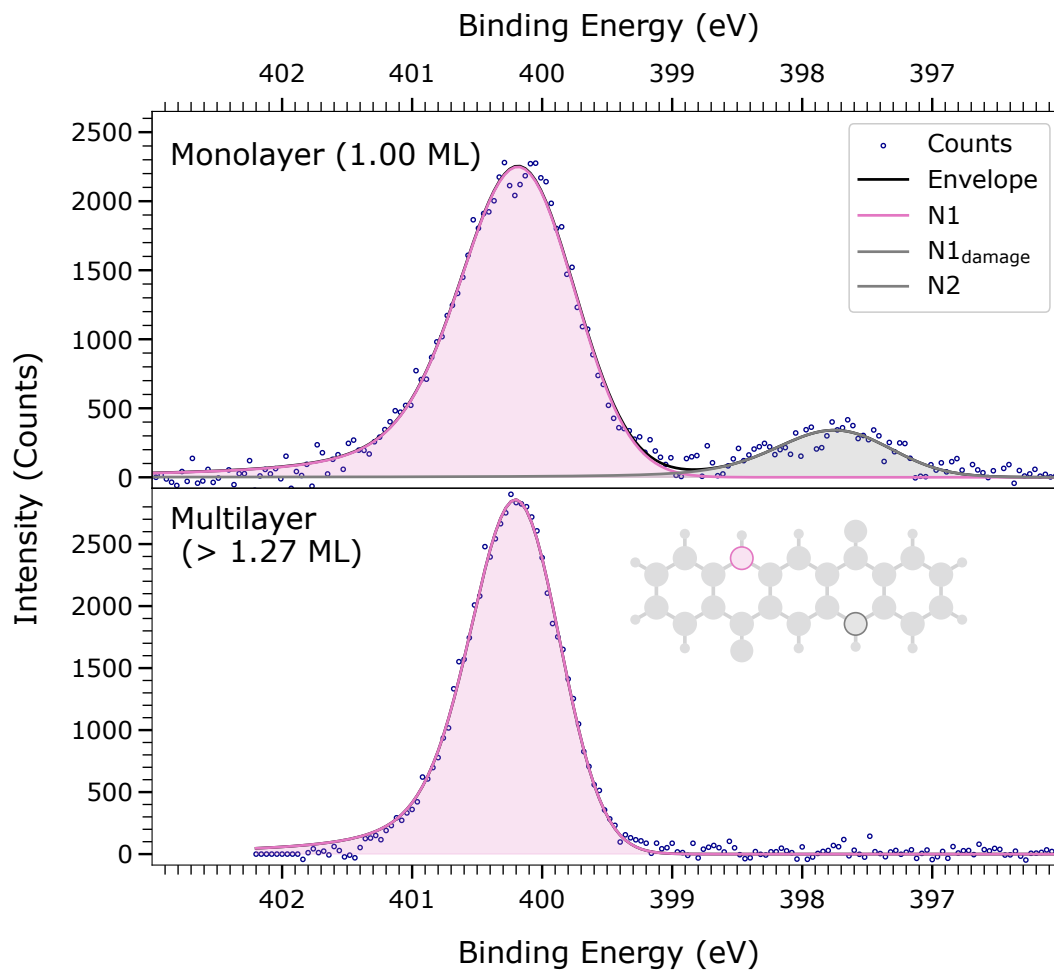


Figure 5.5: Comparison of the N1s XPS spectra of QA on the Ag(100) surface in the α -phase for 1.00 ML and 1.27 ML coverage with the counts (blue circles), N1 (purple), N1_{damage} (grey) and the resulting envelope (black). The used fit parameters are summarized in Table 5.5.

The corresponding fit parameters for the N1s XPS spectra of QA on the Ag(100) surface in the α -phase with both coverages are summarized in Table 5.5.

Table 5.5: BE, FWHM and area ratio for the different peaks of the N1s XPS spectra of QA with a ML and Multilayer coverage on the Ag(100) surface in the α -phase.

	peak	BE (eV)	FWHM (eV)	area ratio
1.00 ML	N1	400.05	1.02	1
	N1 _{damage}	397.66	1.02	-
1.27 ML	N1	400.10	0.79	1

The N1s XPS spectrum of the QA multilayer exhibits only one peak, which corresponds to the N1 peak from the amine group. The BE of this peak is 400.10 eV, which is very similar to the 1 ML preparation. However, the FWHM decreases from 1.02 eV to 0.79 eV.

5.3 O1s XPS spectra

The O1s XPS spectra of QA on the Ag(100) surface in the α - and β -phase are shown in Figure 5.6. The preparation are the same as for the C1s and N1s spectra, with a coverage of 1 ML for the α -phase and 0.54 ML for the β -phase.

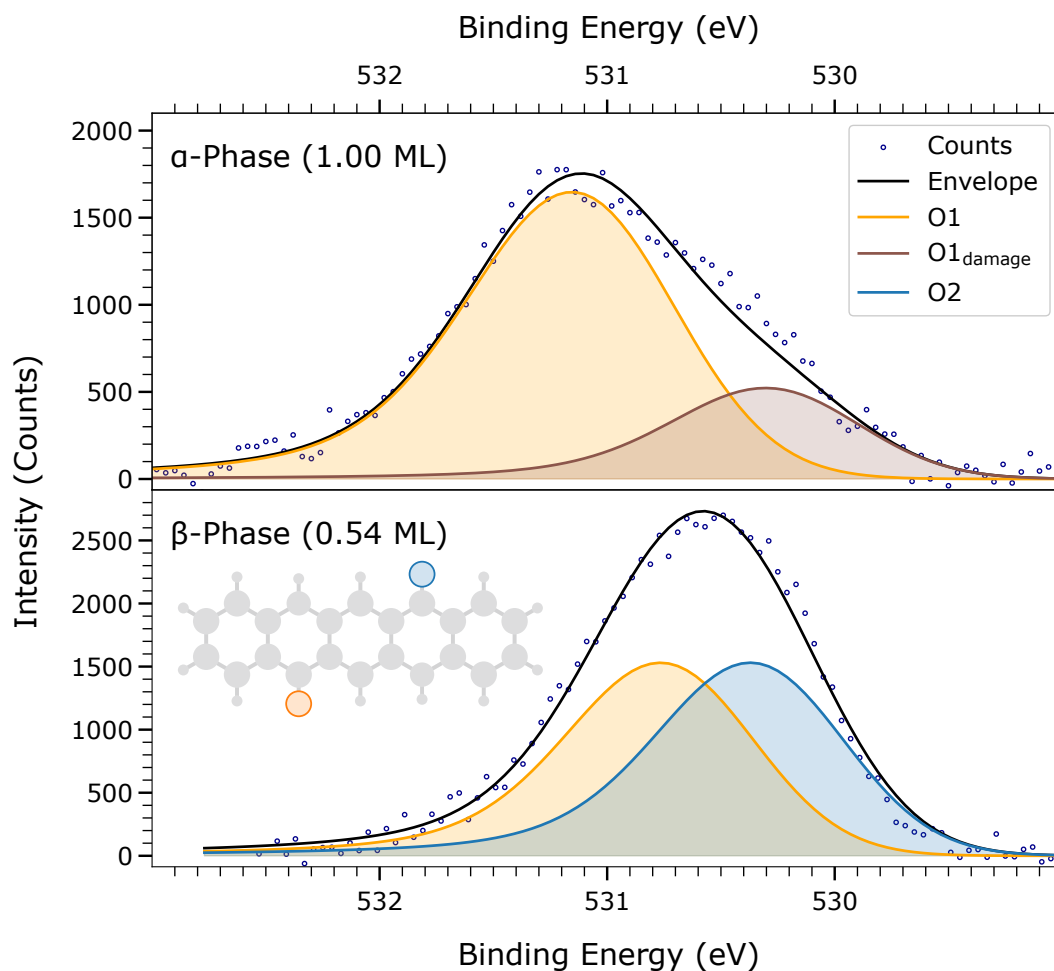


Figure 5.6: Comparison of the O1s XPS spectra of QA on the Ag(100) surface in the α -phase and in the β -phase with the counts (blue circles), O1 (yellow), O1_{damage} (brown), O2 (blue) and the resulting envelope (black). The used fit parameters are summarized in Table 5.6.

The O1s XPS spectra for both phases were fitted with different models, both having 2 peaks as the N1s XPS spectra. The 2 peaks of the O1s XPS spectrum and the 2 peaks of the N1s XPS spectrum are directly correlated, as both originate from functional groups that form hydrogen bonds with each other (amine and carbonyl group).

In the α -phase, the main component is the O1 peak, that originates from the 2 O atoms in the carbonyl groups forming hydrogen bonds with the N atoms in the amine groups of neighboring molecules. Similar to the N1s spectrum in the α -phase, a second peak at lower BE was observed, which is assigned to O atoms from the QA molecules that have different chemical environments due to radiation damage (O1_{damage} peak). This peak is not related to the phase transition and only occurs due to exposure of the sample to the X-ray beam during the measurement. It was observed that the intensity of this peak changes for different measurements, which are shown in Figure ??.

In the β -phase, there are also 2 peaks observed in the O1s XPS spectrum. The main component is again the O1 peak from one O atom in the carbonyl group. However, the O2 peak at lower BE is assigned to a different chemical environment of one O atom in the β -phase. Here, the O1 peak can be assigned to the O atom that is involved in hydrogen bonding with the N atom in the amine group (N1 peak), while the O2 peak occurs from the O atom that is not involved in hydrogen bonding and thus positioned opposite to the N atoms of a neighboring molecule that correspond to the N2 peak.

The detailed fit parameters are summarized in Table 5.6.

Table 5.6: BE, FWHM and area ratio for the different peaks of the O1s XPS spectra of QA on the Ag(100) surface in the α - and β -phase.

	peak	BE (eV)	FWHM (eV)	area ratio
α -phase	O1	531.02	1.05	1
	O1 _{damage}	530.18	0.95	-
β -phase	O1	530.64	0.95	1
	O2	530.25	0.95	1

The BE of the main O1 peak shifts by 0.38 eV from 531.02 eV in the α -phase to 530.64 eV in the β -phase. This shift is similar to the observed shifts for the N1s and C1s XPS spectra, where the N1 peak shifts by 0.18 eV and the C peaks shift by approximately 0.6 eV. The FWHM of the O1 peak decreases from 1.05 eV in the α -phase to 0.95 eV in the β -phase.

Again, the second peak in the β -phase (O2) has the same area ratio as the main peak, indicating that both O atoms in the carbonyl groups have different chemical environments in the β -phase. The BE of the O2 peak is 530.25 eV, which is slightly lower than the O1_{damage} peak in the α -phase (530.18 eV). This indicates that the chemical environment of one O atom in the β -phase is similar to the damaged O atoms in the α -phase. The behaviour of the O1s peaks is thus directly correlated to the behaviour of the N1s peaks in both phases.

The O1s XPS spectra of QA on the Ag(100) surface can be compared to literature values of e.g. PTCDA. The peak corresponding to the O atom from the keto group in PTCDA has a BE of 531.7 eV.^[57] Other O atoms in keto groups show also similar BE values of 530.7-532.7 eV.^[76-78] This is very similar to the BE of the O1 peak of QA on the Ag(100) surface in the α -phase and the β -phase (531.02 eV and 530.64 eV), confirming the plausibility of this assignment.

5.3.1 O1s XPS spectra of the Multilayer

Besides the O1s XPS spectra of QA on the Ag(100) surface in the α - and β -phase, a O1s XPS spectrum of a QA multilayer on the Ag(100) surface is compared to the α -phase preparation with a coverage of 1 ML in Figure 5.7.

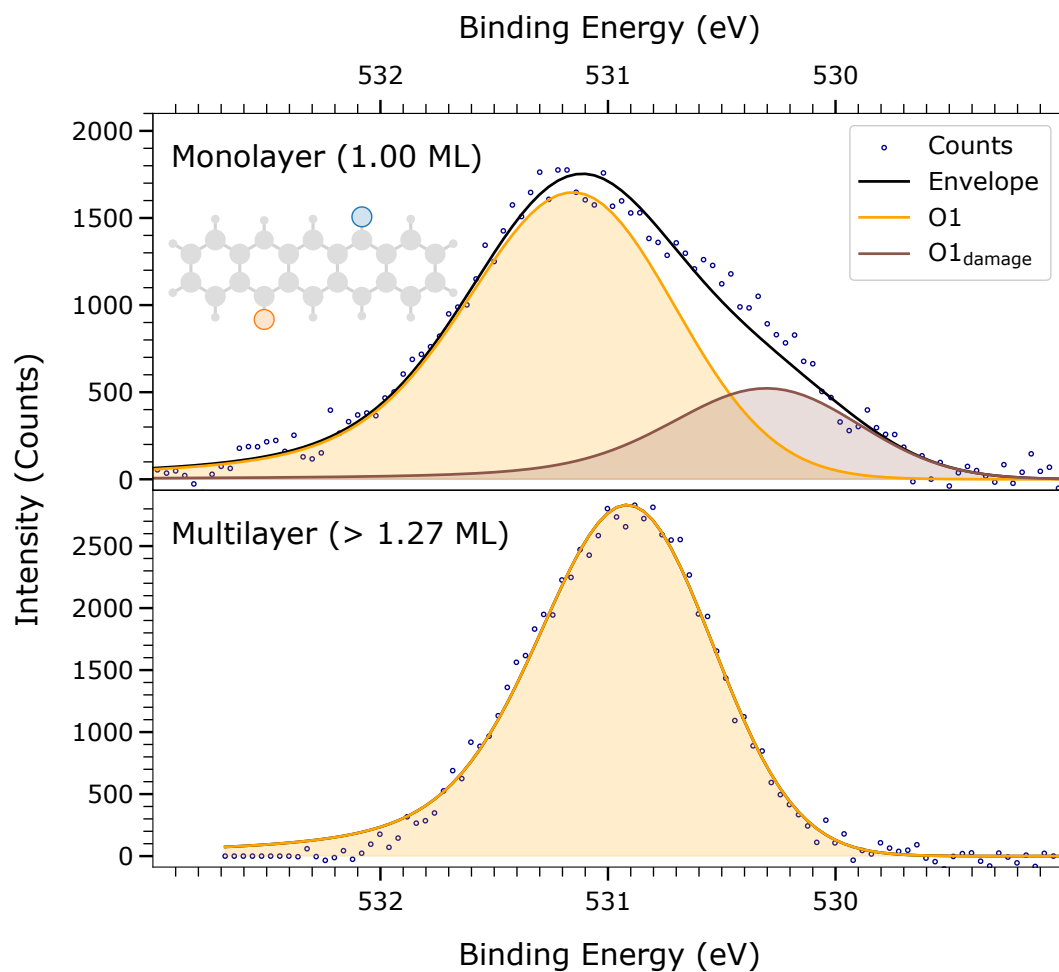


Figure 5.7: Comparison of the O1s XPS spectra of QA on the Ag(100) surface in the α -phase for 1.00 ML and 1.27 ML coverage with the counts (blue circles), O1 (yellow), O1_{damage} (brown) and the resulting envelope (black). The used fit parameters are summarized in Table 5.7.

The corresponding fit parameters for the O1s XPS spectra of QA on the Ag(100) surface in the α -phase with both coverages are summarized in Table 5.7.

Table 5.7: BE, FWHM and area ratio for the different peaks of the O1s XPS spectra of QA with a ML and Multilayer coverage on the Ag(100) surface in the α -phase.

	peak	BE (eV)	FWHM (eV)	area ratio
1.00 ML	O1	531.02	1.05	1
	O1 _{damage}	530.18	0.95	-
1.27 ML	O1	530.80	0.88	1

The O1s XPS spectrum of the QA multilayer exhibits only one peak, which corresponds to the O1 peak from the carbonyl groups. The BE of this peak is 530.80 eV, which is 0.22 eV lower than the 1 ML preparation. However, the FWHM decreases from 1.05 eV to 0.88 eV. This is the same trend as observed for the N1s XPS spectrum of the multilayer.

5.4 The phase transition

Figure 5.8 shows the C1s XPS spectra of QA on the Ag(100) surface while heating the sample from room temperature (~ 300 K) to 500 K to induce the phase transition from the α - to the β -phase. The fit parameters from Table 5.1 were used for the individual peaks and the area ratio for the peaks of one phase were fixed.

The C1s XPS spectra in Figure 5.8 indicate the gradual phase transition from the α - to the β -phase while heating the sample. The first spectrum at room temperature exhibits only the α -phase peaks. With increasing temperature, the intensity of the β -phase peaks increases, while the intensity of the α -phase peaks decreases. At the last XPS spectrum at a temperature of 500 K, only the β -phase peaks are visible. This indicates that the phase transition is complete.

However, the exact temperature of the sample during the measurement is not known, as the sample temperature was only measured at the sample holder far away from the sample. Nevertheless, the phase transition can be clearly observed in the C1s XPS spectra while heating the sample.

Furthermore, it gives information about the underlying process of the phase transition. As the β -phase forms in a gradual manner while heating the sample, it indicates that the transition proceeds via domain-wise switching of the adsorbate phase.

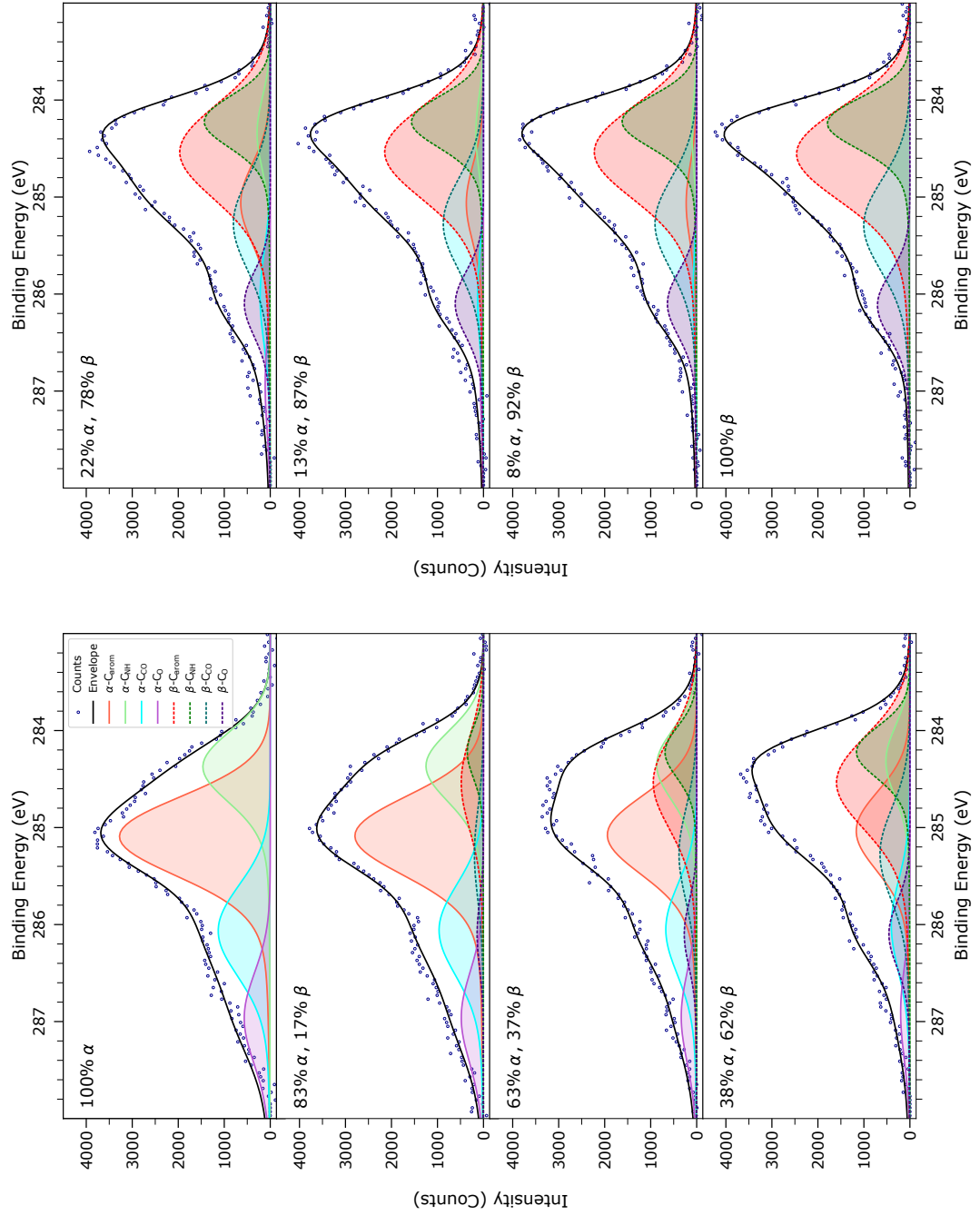


Figure 5.8: C1s XPS spectra of QA on the Ag(100) surface while heating the sample from room temperature to 500 K to induce the phase transition from the α - to the β -phase. The α -phase peaks have solid lines, while the β -phase peaks have dashed lines. The fit parameters from Table 5.1 were used for the individual peaks and the area ratio for the peaks of one phase were fixed.

5.5 XPS spectra of a damaged phase

In contrast to the XPS spectra of QA on the Ag(100) surface in the β -phase, where the sample was successfully heated to induce the phase transition, Figure 5.9 and Figure 5.10 show the C1s and N1s XPS spectra of a damaged QA layer on the Ag(100) surface. Here, the sample was overheated to temperatures of approximately 600 K, which resulted in a damaged QA layer instead of the formation of the β -phase.

The LEED pattern of this damaged phase is shown in Figure 9.5. This pattern shows no ordered structure, indicating that the QA layer is disordered due to the damage.

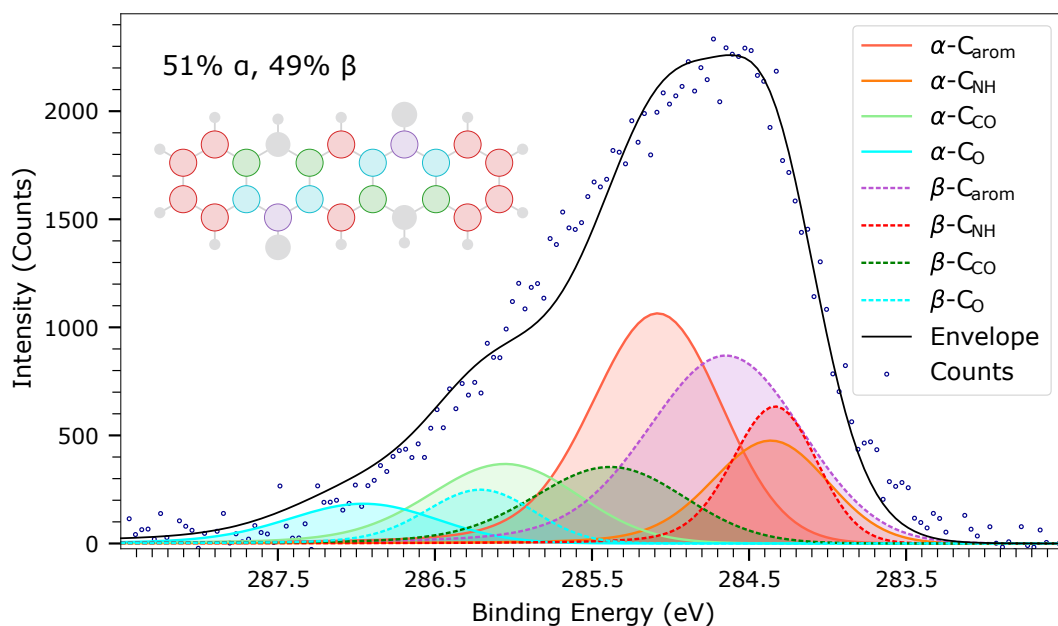


Figure 5.9: C1s XPS spectra of a damaged QA layer on the Ag(100) surface.

Even though the LEED pattern shows no ordered structure, the C1s XPS spectrum in Figure 5.9 is fitted with an 50:50 ratio of the α - and β -phase peaks from Table 5.1. The fit describes the experimental data very well, indicating that both phases are still present in the damaged layer.

This indicates that the damage of the QA molecules on the Ag(100) surface does not lead to a desorption of the molecules, but rather to a loss of the long-range order in the layer. Nevertheless, the chemical environment of the QA molecules remains analogous to a combination of the intact α - and β -phase.

Figure 5.10 shows the N1s XPS spectrum of the same damaged QA layer on the Ag(100) surface.

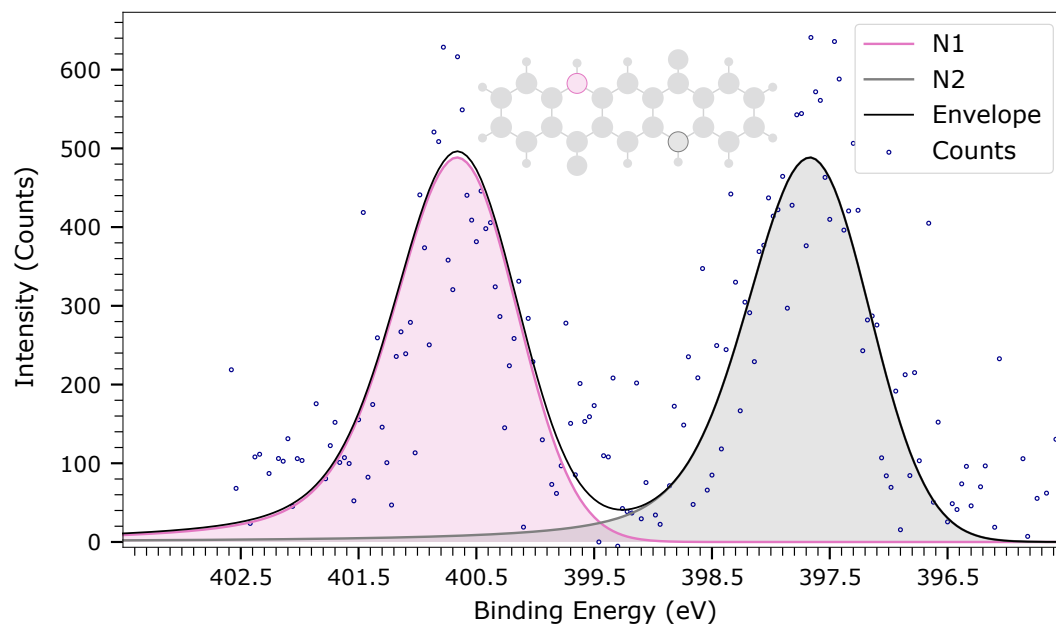


Figure 5.10: N1s XPS spectra of a damaged QA layer on the Ag(100) surface.

In contrast to the C1s XPS spectrum, the N1s XPS spectrum in Figure 5.10 is fitted only with the β -phase peaks, where the area ratio of the N1 and N2 peak is 1:1. Furthermore, the difference in BE between the N1 and N2 peak increases from 2.21 eV in the intact β -phase to 2.99 eV in the damaged layer. This indicates that the chemical environment of the N atoms in the damaged layer differs more significantly compared to the intact β -phase.

Even though the N1s XPS spectrum of the damaged layer exhibits two distinct peaks, the exact BE and area ratio of these peaks may be not precisely determined due to the low intensity of the N1s signal. This is clearly visible in the high noise level of the experimental data in Figure 5.10.

6 NIXSW of QA on the Ag(100) surface

The following sections show the NIXSW spectra and the resulting fit parameters for the C1s, O1s and N1s core levels of QA on the Ag(100) surface in the α - and β -phase. For each core level, one section is dedicated to the α -phase and one section to the β -phase, respectively.

To obtain the NIXSW spectra, a series of 31 XPS measurements at different photon energies around the (200) Bragg reflection of the Ag(100) surface were performed. Each XPS spectrum was fitted with the same model as used for the static XPS spectra in chapter 5, where the FWHM and BE of the individual peaks were fixed and only the peak area was varied. The resulting normalized photoelectron yield of the individual peaks were then plotted against the photon energy to obtain the NIXSW spectra.

6.1 The α -phase

In the following sections, the NIXSW spectra for the α -phase of QA on the Ag(100) surface are presented. They are divided into C1s, N1s, and O1s core levels.

6.1.1 C1s NIXSW spectra

The C1s NIXSW spectra of QA on the Ag(100) surface in the α -phase are shown in Figure 6.1.

At the bottom of Figure 6.1, the measured reflectivity curve of the Ag(100) surface around the (200) Bragg reflection is shown. The different data points and the fitted curve of the C1s NIXSW spectra are shown for the different XPS components from Table 5.1 in different colors. At the top of the figure, the sum of all C1s components is shown. For a better visibility, the different NIXSW spectra are shifted vertically by 1.

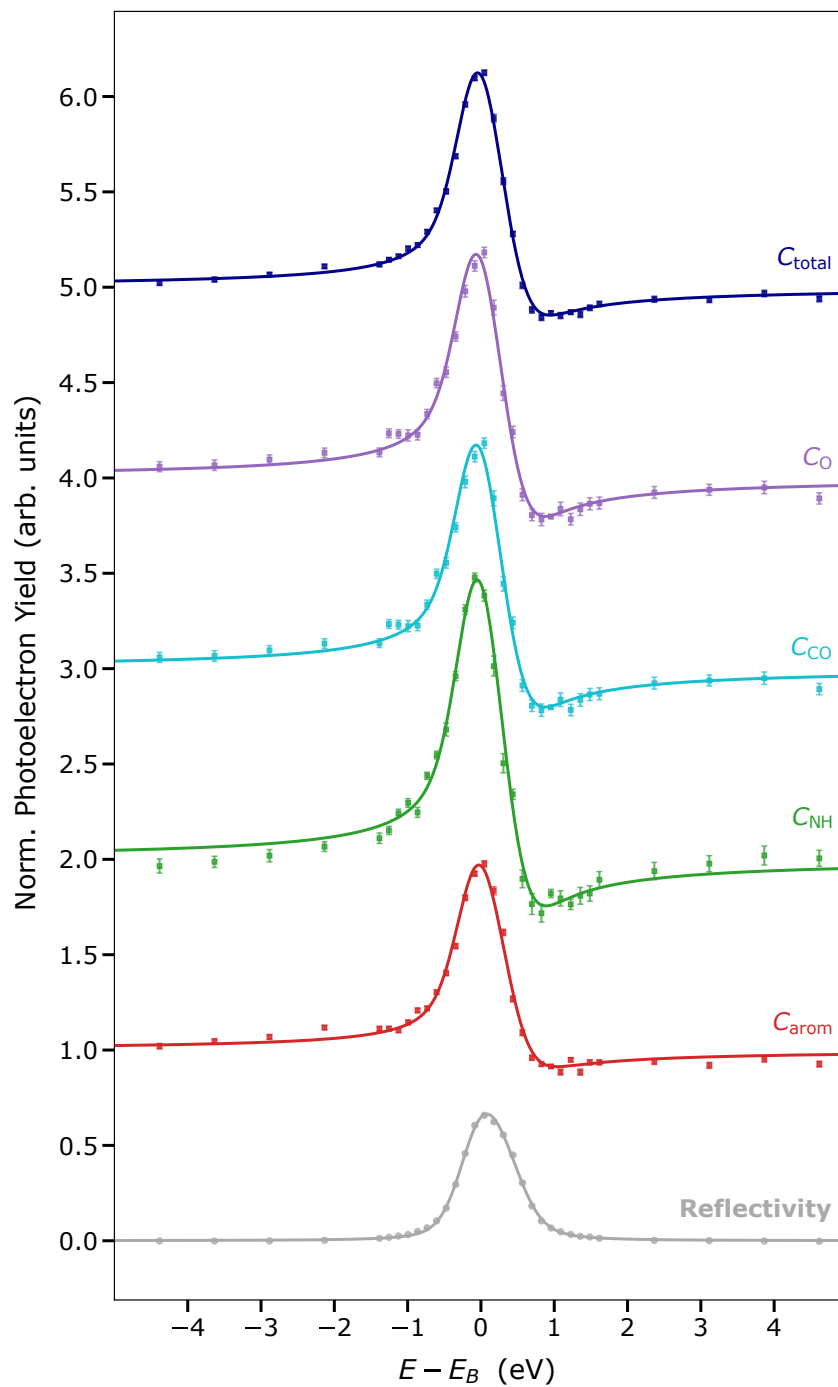


Figure 6.1: C1s NIXSW spectra of QA on the Ag(100) surface in the α -phase. The sum of all components is shown at the top, followed by the individual components C_{arom} , C_{NH} , C_{CO} and C_O and the reflectivity at the bottom. For a better visibility, the different NIXSW spectra are shifted vertically by 1. The fit parameter from Table 5.1 were used for the individual XPS spectra.

Figure 6.1 shows that the different C1s components have distinct NIXSW spectra. Consequently, the different C atoms in the QA molecule on the Ag(100) surface have distinct coherent fractions and positions. The resulting fit parameters for the different C1s components are summarized in Table 6.1.

Table 6.1: Resulting coherent positions p_c and coherent fractions f_c for the different C1s components of QA on the Ag(100) surface in the α -phase from Figure 6.1. The distance z_{200} is calculated from the coherent position p_c .

	p_c	f_c	z_{200} (Å)
C _{arom}	0.459 ± 0.007	0.369 ± 0.015	2.981 ± 0.014
C _{NH}	0.438 ± 0.006	0.811 ± 0.031	2.938 ± 0.012
C _{CO}	0.469 ± 0.007	0.615 ± 0.020	3.001 ± 0.014
C _O	0.458 ± 0.011	0.436 ± 0.029	2.979 ± 0.022
C _{total}	0.455 ± 0.004	0.523 ± 0.012	2.973 ± 0.008

Table 6.1 shows that the coherent fractions f_c vary between 0.369 and 0.811, with a coherent fraction f_c of 0.523 for the total C1s signal. The C_{NH} peak has the highest coherent fraction f_c of 0.811, followed by the C_{CO} peak with a coherent fraction f_c of 0.615. The C_{arom} and C_O peaks have lower coherent fractions f_c of 0.369 and 0.436, respectively. The two peak with the highest coherent fractions f_c correspond to the C atoms that are not bond to hydrogen atoms and are part of two 6 membered rings in the QA molecule.

Furthermore, the different C1s components have coherent positions p_c between 0.438 and 0.469, which results in a coherent position p_c of 0.455 for the total C1s signal. These values correspond to distances z_{200} between 2.938 Å and 3.001 Å, with a distance of 2.973 Å for the total C1s signal.

In total, there is only a deviation of 0.063 Å in the distance z_{200} . Taking the errors of the distance z_{200} into account, the deviations of the adsorption height may be even smaller. This indicates that all C atoms in the QA molecule on the Ag(100) surface in the α -phase are located at similar heights above the surface, with only small differences between the different C atoms.

However, the averaged distance of the C atoms to the Ag(100) surface of around 2.973 Å is lower than the sum of the van der Waals radii for C/Ag, which is 3.42 Å.^[79] In comparison, the binding distance of the C atoms in the α -phase are 13 % closer to the surface than the van der Waals radii. The binding distance of the C atoms in the α -phase is also comparable to other π -conjugated molecules on the Ag(100) surface, such as PTCDA with a binding distance of 2.84 Å.^[57] Additionally, the binding distance of pentacene on Ag(111), which is between 2.99 Å and 3.13 Å,^[80] is a little higher than the binding distance of the C atoms in the α -phase. The differences between QA and pentacene are the functional groups in QA, namely the amine and

carbonyl groups, which can interact with the Ag(100) surface and thus lead to a stronger interaction between the C atoms and the Ag(100) surface in QA compared to pentacene, which does not have any functional groups.

6.1.2 N1s NIXSW spectra

In the following, the N1s NIXSW spectra of QA on the Ag(100) surface in the α -phase are presented in Figure 6.2. At the bottom of the figure, the measured reflectivity curve of the (200) Bragg reflection is shown. The data points and the fitted curve of the N1s NIXSW spectrum for the N1 component are shown above the reflectivity. For a better visibility, the different curves are shifted vertically by 1.

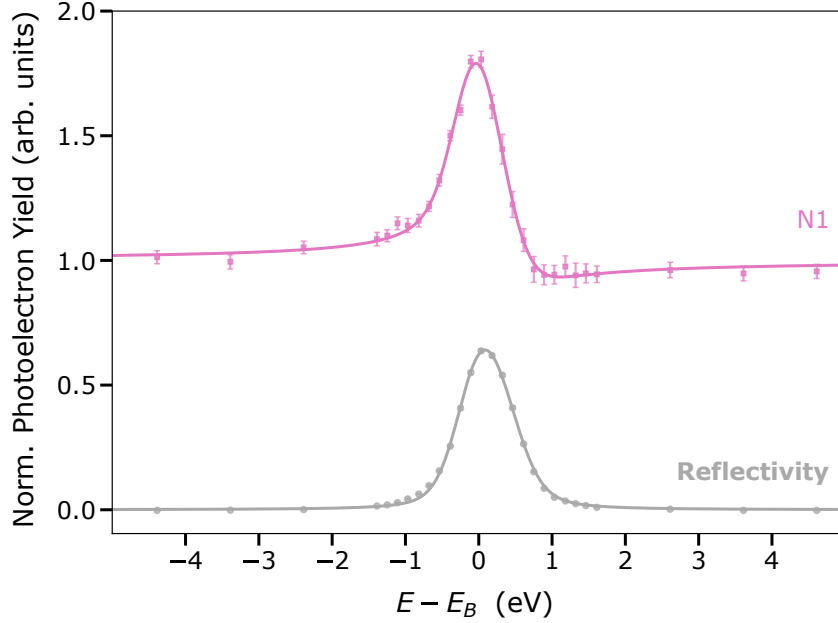


Figure 6.2: N1s NIXSW spectra of QA on the Ag(100) surface in the α -phase. The component N1 is shown at the top and the reflectivity at the bottom. For a better visibility, the different NIXSW spectra are shifted vertically by 1. The fit parameter from Table 5.4 were used for the individual XPS spectra.

Figure 6.2 shows that the N1s NIXSW spectrum of the N1 component exhibits a similar peak as the C1s NIXSW spectra. The resulting fit parameters and the distance z_{200} for the N1s component are summarized in Table 6.2.

Table 6.2: Resulting coherent positions p_c and coherent fractions f_c for the different N1s components of QA on the Ag(100) surface in the α -phase from Figure 6.2. The distance z_{200} is calculated from the coherent position p_c .

	p_c	f_c	z_{200} (Å)
N1	0.499 ± 0.009	0.288 ± 0.013	3.062 ± 0.018

Table 6.2 shows that the N1s component has a coherent position p_c of 0.499, which corresponds to a distance z_{200} of 3.062 Å. This distance is higher than the distance of the C atoms to the Ag(100) surface of 2.973 Å, showing that the N atoms are located farther away from the Ag(100) surface than the C atoms in the QA molecule on the Ag(100) surface in the α -phase. The sum of the van der Waals radii for N/Ag, which is 3.30 Å,^[79] is larger than the binding distance of the N atoms, however, the binding distance of the N atoms in the α -phase is only 7 % closer to the surface than the van der Waals radii.

The coherent fraction of the N1 component f_c is 0.288. This coherent fraction is significantly lower than the coherent fractions of the C1s components, which are between 0.369 and 0.811. Compared to the merocyanine HB238 on the Ag(100) surface, which has a coherent fraction of 0.234 for the N1s component,^[64] the N atoms in the QA molecule on the Ag(100) surface in the α -phase have a higher degree of order. However, there are also organic molecules like azobenzen on the Ag(111) surface, that exhibit coherent fractions of almost 0.5,^[81] which is significantly higher than the coherent fraction of the N atoms in the QA molecule on the Ag(100) surface in the α -phase.

6.1.3 O1s NIXSW spectra

In the following, the O1s NIXSW spectra of QA on the Ag(100) surface in the α -phase are presented in Figure 6.3. Again, the reflectivity curve of the (200) Bragg reflection is shown at the bottom and the data points and fitted curve of the O1s NIXSW spectrum for the O1 component at the top. For a better visibility, the different curves are shifted vertically by 1.

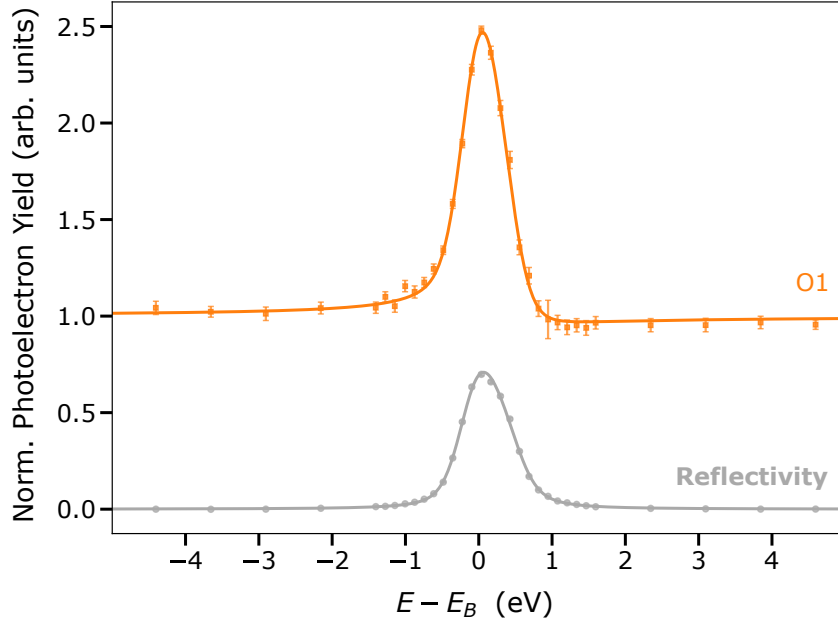


Figure 6.3: O1s NIXSW spectra of QA on the Ag(100) surface in the α -phase. The component O1 is shown at the top and the reflectivity at the bottom. For a better visibility, the different NIXSW spectra are shifted vertically by 1. The fit parameter from Table 5.6 were used for the individual XPS spectra.

The O1s NIXSW spectrum of the O1 component in Figure 6.3 also shows a similar behaviour as the C1s and N1s NIXSW spectra. The resulting fit parameters for the O1s component are summarized in Table 6.3.

Table 6.3: Resulting coherent positions p_c and coherent fractions f_c for the different O1s components of QA on the Ag(100) surface in the α -phase from Figure 6.3. The distance z_{200} is calculated from the coherent position p_c .

	p_c	f_c	z_{200} (Å)
O1	0.332 ± 0.004	0.546 ± 0.016	2.721 ± 0.008

Table 6.3 shows that the O1 component has a coherent position p_c of 0.332, which corresponds to a distance z_{200} of 2.721 Å. In comparison to the C and N atoms, the O atoms are located closer to the Ag(100) surface in the α -phase. The sum of the van der Waals radii for O/Ag, which is 3.24 Å,^[79] is 16 % larger than the binding distance of the O atoms. This is a significant difference compared to the C and N atoms, which are only 13 % and 7 % closer to the surface than the van der Waals radii, respectively.

In comparison to PTCDA with a coherent position p_c of 0.24, resulting in a adsorption height of 2.53 Å^[57] the O atoms in the QA molecule on the Ag(100) surface in the α -phase are located farther away from the surface. This is the same results as for the C atoms in QA and PTCDA, which also show a higher binding distance for QA compared to PTCDA.

The coherent fraction of the O1 component f_c is 0.546, which is significantly higher than the coherent fraction of the N1 component of 0.288. Furthermore, the coherent fraction of the O1 component is also higher than the coherent fractions of the C_{arom} and C_O components, which are 0.369 and 0.436, respectively. However, the coherent fraction of the O1 component is lower than the coherent fractions of the C_{NH} and C_{CO} components, which are 0.811 and 0.615. This shows that the degree of order for the O1 component is located in between the different C1s components and the N1s component, resulting in a medium degree of order for the O atoms in the QA molecule on the Ag(100) surface in the α -phase.

6.1.4 Discussion of the α -phase

In the following, the coherent positions p_c and coherent fractions f_c for the different core levels of QA on the Ag(100) surface in the α -phase will be discussed in detail and compared to each other.

As an overview of the resulting coherent positions and fractions for the different core levels of QA on the Ag(100) surface in the α -phase, an Argand representation is shown in Figure 6.4. Here, the coherent position p_c is represented as the angle and the coherent fraction f_c as the distance from the origin.

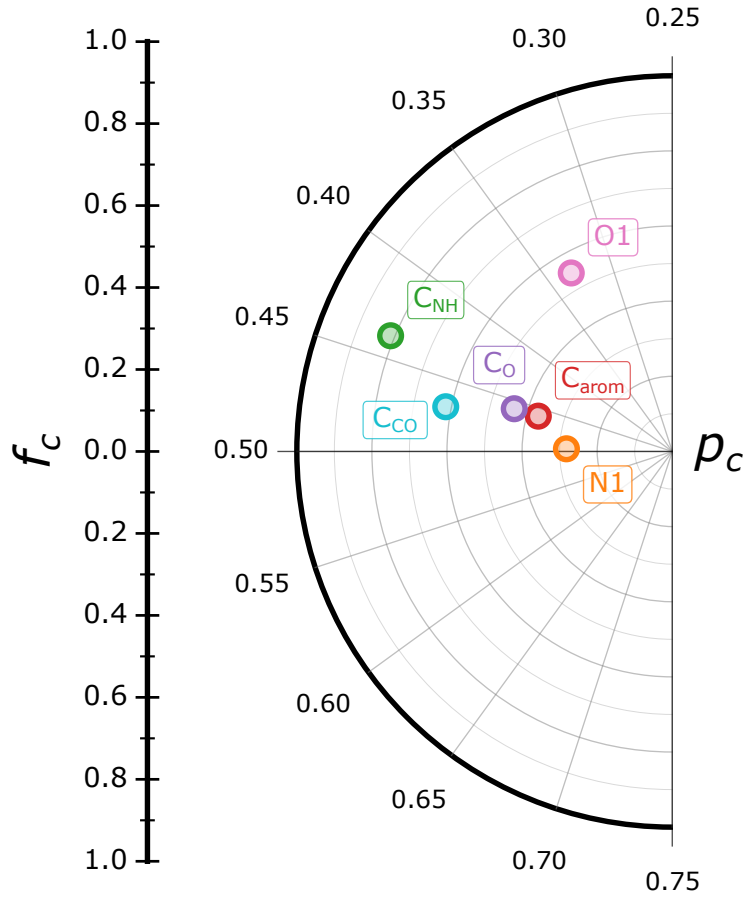


Figure 6.4: Argand representation for QA on the Ag(100) surface in the α -phase for the C1s, N1s, and O1s core levels.

As the coherent positions p_c are directly proportional to the adsorption heights, the following discussion of the coherent positions p_c is equivalent to a discussion of the adsorption heights of the different atoms in the QA molecule on the Ag(100) surface in the α -phase.

Figure 6.4 shows that the different core levels of QA on the Ag(100) surface in the α -phase have similar coherent positions p_c . However, the O1 component has the by far lowest coherent position p_c of 0.332 ($z_{200} = 2.721$ Å) and does not align with the other components, which have coherent positions p_c between 0.438 and 0.499 ($z_{200} = 2.938 - 3.062$ Å). This shows in general, that the C and N atoms in the QA molecule on the Ag(100) surface in the α -phase are located at similar heights above the surface, while the O atoms are located significantly closer to the surface.

As the QA molecule is adsorbed with the aromatic ring parallel to the surface, all C atoms are expected to have similar distances z_{200} to the Ag(100) surface. The small differences in the adsorption heights of the different C1s components could be explained by functional groups, which can alter the adsorption behavior of the neighbored atoms. However, these differences are of the order of the error bars and thus not significant.

The different adsorption height of the O1 component shows that the O atoms have a special role in the adsorption of QA on the Ag(100) surface in the α -phase, as they are located closer to the surface and thus have a stronger interaction with the surface compared to the other atoms in the QA molecule.

This may be explained by the altered interactions of the different atoms in the QA molecule with the Ag(100) surface. The O atoms are part of the carbonyl group and have a high electron density due to the double bond to the C atom and the 2 electron lone pairs on the O atom. This high electron density can lead to a stronger interaction between the O atoms and the Ag(100) surface due to a donation of electron density from the O atom to the Ag(100) surface, which could explain the closer distance of the O atoms to the surface compared to the C and N atoms.

In addition, the unique position of the O atoms in the QA molecule, as they are located outside of the ring structure, needs to be pointed out in contrast to the C and N atoms, which are located in the ring structure. This also explains why the O atoms are located closer to the surface, as they can approach the Ag(100) surface and thus interact more strongly with the surface due to their position.

The N atoms have the highest distance to the Ag(100) surface. This shows that the interaction between the N atoms and the Ag(100) surface is less relevant for the structural arrangement of the QA molecule on the Ag(100) surface in the α -phase compared to the interaction between the O atoms and the Ag(100) surface.

As the hydrogen bonds between the amine groups and the carbonyl groups of the neighboring molecules are important for the structural arrangement of the QA molecule on the Ag(100) surface in the α -phase, the heights of these two types of atoms need to be discussed together. The O atoms are located at a distance $z_{200} = 2.721$ Å to the Ag(100) surface, while the N atoms are located at a distance $z_{200} = 3.062$ Å. This difference in the adsorption heights of the O and N atoms of around 0.34 Å can be

explained by the hydrogen bond between the O atom and the H atom bond to the N atom of the neighboring molecule.

The O atom mainly interacts with the Ag(100) surface, leading to a lower adsorption height of the O atom. In contrast, the adsorption height of the N atom is mainly influenced by the hydrogen bond to the O atom of the neighboring molecule, which leads to a higher adsorption height of the N atom. The difference in the adsorption heights of the O and N atoms can thus be explained by the different types of interactions of these atoms with the Ag(100) surface and with each other.

Now, the coherent fraction f_c for the different core levels of QA on the Ag(100) surface in the α -phase also needs to be taken into account to understand the structural arrangement of the QA molecule on the Ag(100) surface in the α -phase. The coherent fraction f_c is a measure for the degree of order of the different atoms in the QA molecule on the Ag(100) surface in the α -phase, with a higher coherent fraction f_c indicating a higher degree of order and thus a more well-defined adsorption height for the corresponding atoms.

The high coherent fractions f_c of the C_{NH} and C_{CO} components could be explained by the position of these atoms in the QA molecule, as they are part of two 6 membered rings in the QA molecule and bond to 3 hetero atoms each, which leads to lower flexibility of these C atoms in comparison to the other C atoms and thus a higher degree of order. In contrast, the lower coherent fractions f_c of the C_{arom} and C_O components could be explained by the fact that these atoms are located at the edges and corners of the QA molecule, which leads to a higher flexibility of these atoms and thus a lower degree of order.

The coherent fraction f_c of the N1 component is 0.288, indicating a very low degree of order for the N atoms compared to the other atoms in the QA molecule on the Ag(100) surface in the α -phase. This is in agreement with the fact that the N atoms are located at the edges of the QA molecule, which leads to a higher flexibility of these atoms and thus a lower degree of order compared to the C atoms. In addition, the amine groups are involved in intermolecular hydrogen bonds, which could also lead to a higher flexibility, as the N atom can be located at different positions depending on the hydrogen bond and the exact position of the O atom in the carbonyl group of the neighboring molecule.

Furthermore, as already explained before, the interaction between the N atoms and the Ag(100) surface is less relevant for the structural arrangement of the QA molecule on the Ag(100) surface in the α -phase, which means that the N atoms are not strongly influenced by the Ag(100) surface and thus can be located at different positions above the surface.

Last, the coherent fraction f_c of the O1 component is 0.546. This is a medium degree of order for the O atoms in the QA molecule on the Ag(100) surface in the α -phase,

which is in between the low ordered N atoms and the higher degree of order of some C atoms. Again, this can be explained by the position of the O atoms in the molecule and the interactions of the O atoms with the Ag(100) surface and with the neighboring molecules.

On the one hand, the location outside of the ring allows a higher flexibility of the O atoms, which leads to a lower degree of order. On the other hand, the stronger interaction between the O atoms and the Ag(100) surface and also the hydrogen bonds to the neighboring molecule lead to a higher degree of order for the O atoms. In total, these two effects lead to a medium degree of order for the O atoms in the QA molecule on the Ag(100) surface in the α -phase.

6.2 The β -phase

Next, the NIXSW spectra for the β -phase of QA on the Ag(100) surface are presented. They are divided into C1s, N1s, and O1s core levels.

6.2.1 C1s NIXSW spectra

First, the C1s NIXSW spectra of QA on the Ag(100) surface in the β -phase are shown in Figure 6.5. At the bottom of the figure, the measured reflectivity curve of the (200) Bragg reflection is shown. The data points and the fitted curve of the C1s NIXSW spectra are shown for the three XPS components from Table 5.2 in different colors. At the top of the figure, the sum of all C1s components is shown. For a better visibility, the NIXSW spectra are shifted vertically by 1.

The NIXSW spectra in Figure 6.5 show that the different C1s components exhibit only small changes in the NIXSW spectra. This indicates that the different C atoms in the QA molecule on the Ag(100) surface in the β -phase have similar coherent fractions and positions. The resulting fit parameters for the different C1s components are summarized in Table 6.4.

Table 6.4: Resulting coherent positions p_c and coherent fractions f_c for the different C1s components of QA on the Ag(100) surface in the β -phase from Figure 6.5. The distance z_{200} is calculated from the coherent position p_c .

	p_c	f_c	z_{200} (Å)
$C_{\text{arom}} + C_{\text{NH}}$	0.387 ± 0.003	0.758 ± 0.017	2.834 ± 0.006
C_{CO}	0.350 ± 0.002	0.660 ± 0.025	2.758 ± 0.004
C_{O}	0.387 ± 0.007	0.564 ± 0.026	2.834 ± 0.014
C_{total}	0.378 ± 0.002	0.712 ± 0.012	2.815 ± 0.004

Table 6.4 shows that the different C1s components have coherent positions p_c between 0.350 and 0.387, which results in a coherent position p_c for the total C1s signal of 0.378. These values correspond to distances z_{200} between 2.758 Å and 2.834 Å, with a distance of 2.815 Å for the total C1s signal. It is important to note that the $C_{\text{arom}} + C_{\text{NH}}$ and C_{O} components have the same coherent position p_c of 0.387, but different coherent fractions f_c of 0.758 and 0.564, respectively.

In comparison to the α -phase, the C atoms in the β -phase are located closer to the Ag(100) surface, with a difference in the distance z_{200} of around 0.16 Å (5 %) for the total C1s signal. This means that the distance of the C atoms to the Ag(100) surface in the β -phase is lower than the sum of the van der Waals radii for C/Ag, which is 3.42 Å,^[79] by 18 %. Compared to the α -phase, where the binding distance of the C atoms is only 13 % closer to the surface than the van der Waals radii, the binding

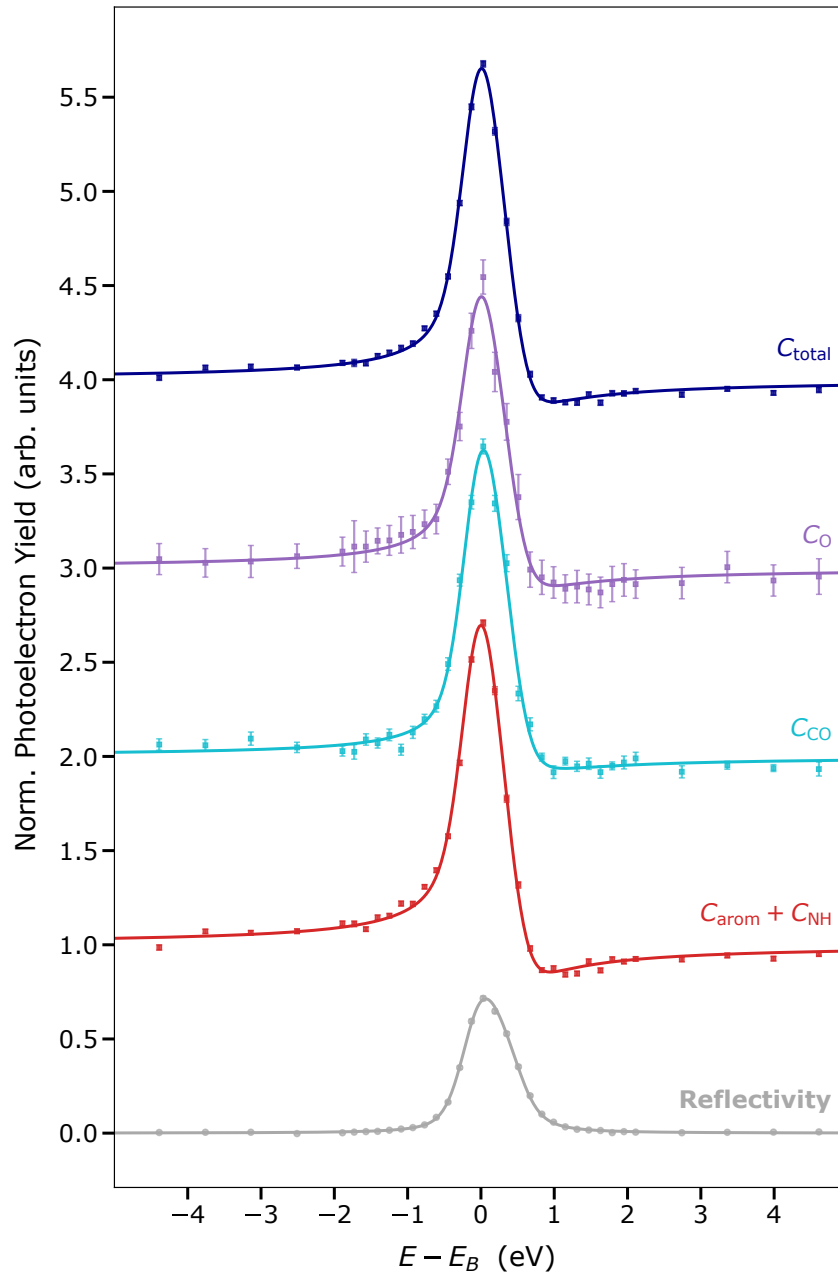


Figure 6.5: C1s NIXSW spectra of QA on the Ag(100) surface in the β -phase. The sum of all components is shown at the top, followed by the individual components $C_{\text{arom}} + C_{\text{NH}}$, C_{CO} and C_{O} and the reflectivity at the bottom. For a better visibility, the different NIXSW spectra are shifted vertically by 1. The fit parameter from Table 5.2 were used for the individual XPS spectra.

distance of the C atoms in the β -phase is significantly closer to the surface than in the α -phase.

The binding distance of the C atoms in the β -phase is also significantly closer to the surface than the binding distance of the C atoms in PTCDA on the Ag(100) surface, which is 2.84 Å.^[57] As the merocyanine HB238 on the Ag(100) surface has a binding distance of 2.958 Å,^[64] the binding distance of the C atoms of QA to the Ag(100) surface in the β -phase is even more significantly smaller.

The coherent fractions f_c of the different C1s components in the β -phase are between 0.564 and 0.758, with a coherent fraction f_c of 0.712 for the total C1s signal. These values are significantly higher than the coherent fractions of the C1s components in the α -phase, which are between 0.369 and 0.811, with a coherent fraction f_c of 0.523 for the total C1s signal.

In comparison to PTCDA on the Ag(100) surface, which has a coherent fraction of 0.51 for the total C1s component,^[57] the C atoms in the QA molecule on the Ag(100) surface in the β -phase have a significantly higher degree of order. Compared to the merocyanine HB238 on the Ag(100) surface, which has a coherent fraction of 0.242 for the C1s component,^[64] the C atoms in the QA molecule on the Ag(100) surface in the β -phase have a much higher degree of order.

6.2.2 N1s NIXSW spectra

The N1s NIXSW spectra of QA on the Ag(100) surface in the β -phase are shown in Figure 6.6. At the bottom of the figure, the measured reflectivity curve of the (200) Bragg reflection is shown. The data points and the fitted curve of the N1s NIXSW spectra are shown for the two XPS components from Table 5.4 in different colors. At the top of the figure, the sum of all N1s components is shown. For a better visibility, the NIXSW spectra are shifted vertically by 1.

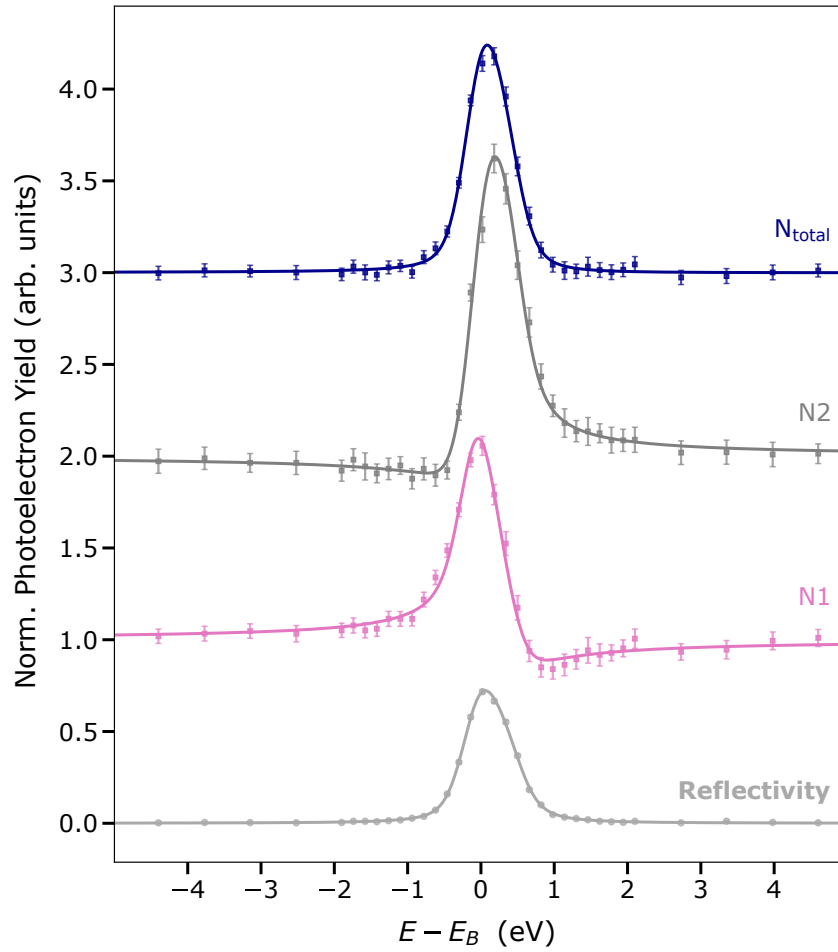


Figure 6.6: N1s NIXSW spectra of QA on the Ag(100) surface in the β -phase. The sum of all components is shown at the top, followed by the individual components N1 and N2 and the reflectivity at the bottom. For a better visibility, the different NIXSW spectra are shifted vertically by 1. The fit parameter from Table 5.4 were used for the individual XPS spectra.

Figure 6.6 shows very different NIXSW spectra for the two N1s components. The protonated N atom (N1) has a coherent position p_c close to 0.5, which corresponds to a maximum followed by a minimum in the NIXSW spectrum. In contrast, the deprotonated N atom (N2) has a coherent position p_c close to 0.2, which corresponds to a minimum followed by a maximum in the NIXSW spectrum. This opposite behavior indicates that the two N atoms in the QA molecule on the Ag(100) surface in the β -phase are located at significantly different heights above the surface and explains the shape of the N_{total} peak with only one maximum.

The resulting fit parameters for the different N1s components are summarized in Table 6.5.

Table 6.5: Resulting coherent positions p_c and coherent fractions f_c for the different N1s components of QA on the Ag(100) surface in the β -phase from Figure 6.6. The distance z_{200} is calculated from the coherent position p_c .

	p_c	f_c	z_{200} (Å)
N1	0.486 ± 0.008	0.374 ± 0.021	3.036 ± 0.016
N2	0.184 ± 0.004	0.645 ± 0.034	2.419 ± 0.008
N_{total}	0.289 ± 0.005	0.645 ± 0.034	2.633 ± 0.010

The values in Table 6.5 show that the N1s components have coherent positions p_c of 0.486 and 0.184, which correspond to distances z_{200} of 3.036 Å and 2.419 Å, respectively. This indicates a significant difference in height of 0.6 Å between the two N atoms in the QA molecule on the Ag(100) surface in the β -phase. This also means that the total N1s signal has a coherent position p_c of 0.289, which corresponds to a distance z_{200} of 2.633 Å, which is in between the distances of the two N atoms. However, as the two N atoms have very different coherent positions p_c , the results of the total N1s signal are not very meaningful.

In comparison to the α -phase, where the N atoms have a coherent position p_c of 0.499, which corresponds to a distance z_{200} of 3.062 Å, the N1 component in the β -phase has a similar coherent position p_c and thus a similar distance to the surface as the N atoms in the α -phase. As already stated above, the deprotonated N2 component has a significantly distance z_{200} of 2.419 Å. This means that the deprotonated N atom in the β -phase is located significantly closer to the Ag(100) surface than the not deprotonated N atoms in the α and β -phase, with a difference in the distance z_{200} of around 0.64 Å (21 %).

Again, this the distance of the deprotonated N atom to the Ag(100) surface in the β -phase is significantly smaller than the sum of the van der Waals radii for N/Ag, which is 3.30 Å,^[79] by 27 %. This is a significant difference compared to the N atoms in the α -phase, which are only 7 % closer to the surface than the van der Waals radii.

The coherent fractions f_c are 0.374 for the N1 component and 0.645 for the deprotonated N2 component, indicating very different degrees of order for the two N atoms. Compared to the α -phase, where the N atoms have a coherent fraction f_c of 0.288, the N1 component in the β -phase has a higher degree of order, while the deprotonated N2 component has a significantly higher degree of order than the N atoms in the α -phase.

6.2.3 O1s NIXSW spectra

The O1s NIXSW spectra of QA on the Ag(100) surface in the β -phase are shown in Figure 6.7. At the bottom of the figure, the measured reflectivity curve of the (200) Bragg reflection is shown. The data points and the fitted curve of the O1s NIXSW spectra are shown for the two XPS components from Table 5.6 in different colors. At the top of the figure, the sum of all O1s components is shown. For a better visibility, the NIXSW spectra are shifted vertically by 1.

As seen in Figure 6.7, the different O1s components exhibit almost identical NIXSW spectra. This indicates that the different O atoms in the QA molecule on the Ag(100) surface in the β -phase have very similar coherent fractions and positions.

The resulting coherent positions p_c and coherent fractions f_c for the different O1s components of QA on the Ag(100) surface in the β -phase from Figure 6.7 are summarized in Table 6.6. The distance z_{200} is calculated from the coherent position p_c .

Table 6.6: Resulting coherent positions p_c and coherent fractions f_c for the different O1s components of QA on the Ag(100) surface in the β -phase from Figure 6.7. The distance z_{200} is calculated from the coherent position p_c .

	p_c	f_c	z_{200} (Å)
O1	0.272 ± 0.007	0.658 ± 0.056	2.599 ± 0.014
O2	0.291 ± 0.011	0.559 ± 0.072	2.638 ± 0.022
O _{total}	0.282 ± 0.006	0.598 ± 0.042	2.619 ± 0.012

Table 6.6 shows that the O1s components have coherent positions p_c of 0.272 and 0.291, which correspond to distances z_{200} of 2.599 Å and 2.638 Å, respectively. Taking the errors of these two values into account, the two O atoms in the QA molecule on the Ag(100) surface in the β -phase are located at similar heights above the surface. The total O1s signal has a coherent position p_c of 0.282, which corresponds to a distance z_{200} of 2.619 Å, which is in between the distances of the two O atoms.

In comparison to the α -phase, where the O atoms have a distance z_{200} of 2.721 Å, the O1 component in the β -phase has a lower distance to the Ag(100) surface than the O atoms in the α -phase, with a difference in the distance z_{200} of around 0.12 Å (4 %). This difference in the distance z_{200} is similar to the difference of the C atoms between the α -phase and the β -phase, which is around 0.16 Å (5 %). This shows that both the C atoms and the O atoms are located closer to the Ag(100) surface in the β -phase compared to the α -phase.

The distance of the O atoms to the Ag(100) surface in the β -phase is also significantly smaller than the sum of the van der Waals radii for O/Ag, which is 3.24 Å,^[79] by around 20 %.

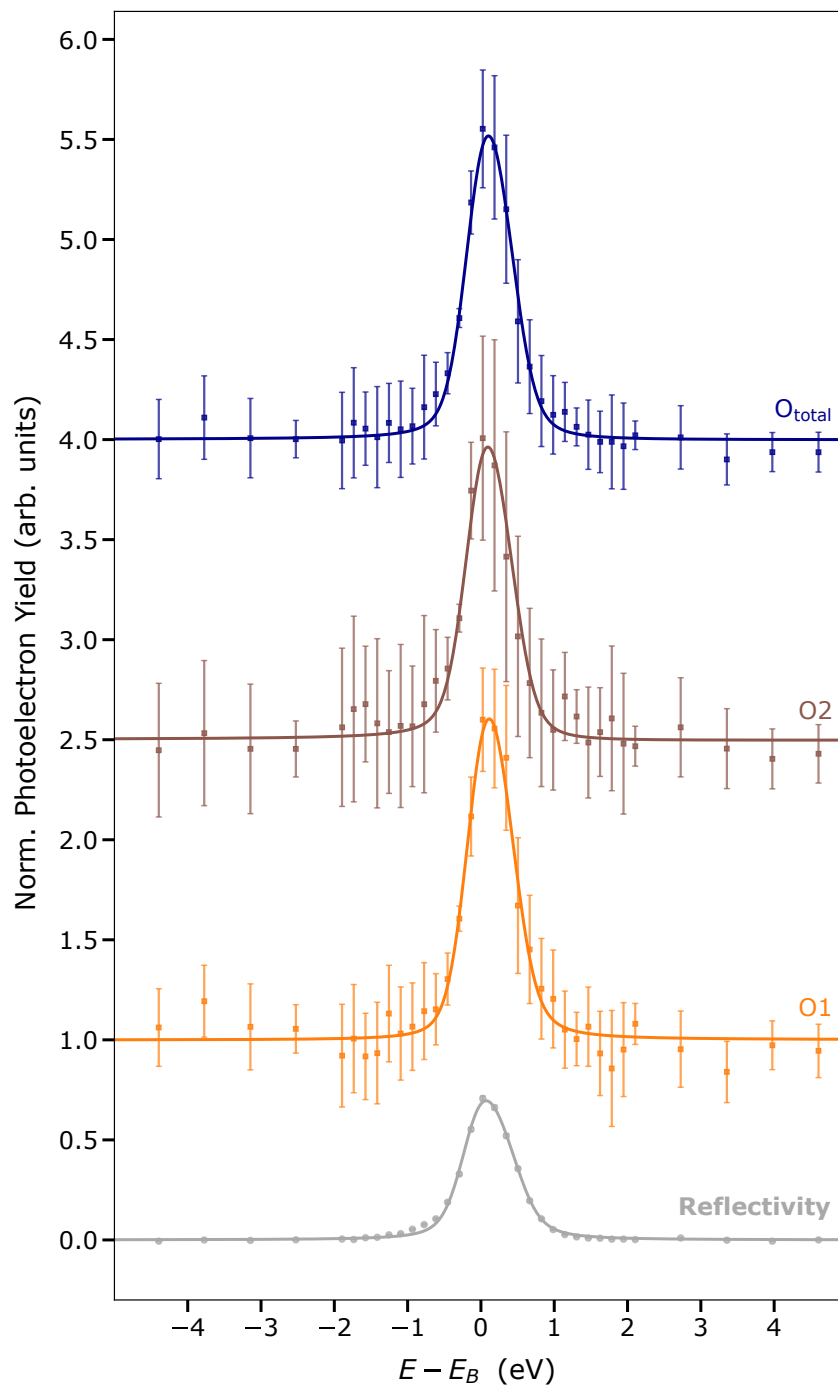


Figure 6.7: O1s NIXSW spectra of QA on the Ag(100) surface in the β -phase. The sum of all components is shown at the top, followed by the individual components O1 and O2 and the reflectivity at the bottom. For a better visibility, the different NIXSW spectra are shifted vertically by 1. The fit parameter from Table 5.6 were used for the individual XPS spectra.

For the N atoms, two different heights of the two N atoms above the surface were observed, whereas the O atoms are located at similar heights above the surface. This is a significant difference between the N and O atoms, which form hydrogen bonds between the neighboring molecules with the N1/O1 component and do not form hydrogen bonds between the N2/O2 component due to the deprotonation of the N2 component.

In comparison to PTCDA on the Ag(100) surface, which has a distance of 2.53 Å for the O atoms,^[57] the O atoms in the QA molecule on the Ag(100) surface in the β -phase are located farther away from the surface by 0.09 Å.

The coherent fractions f_c of the O1 and O2 components are 0.658 and 0.559, respectively. The total O1s signal has a coherent fraction f_c of 0.598. This shows that the O atoms have similar degrees of order, which is in agreement with the similar heights of the O atoms.

Compared to the α -phase, where the O atoms have a coherent fraction f_c of 0.546, the O1 and O2 components in the β -phase have a higher degree of order. Furthermore, the O atoms in the QA molecule on the Ag(100) surface in the β -phase have also a higher degree of order than the O atoms in PTCDA on the Ag(100) surface, which have a coherent fraction of 0.51.^[57]

6.2.4 Argand representation for the β -phase

To summarize the resulting coherent positions and fractions for the different core levels of QA on the Ag(100) surface in the β -phase, an Argand representation is shown in Figure 6.8. Here, the coherent position p_c is represented as the angle and the coherent fraction f_c as the distance from the origin.

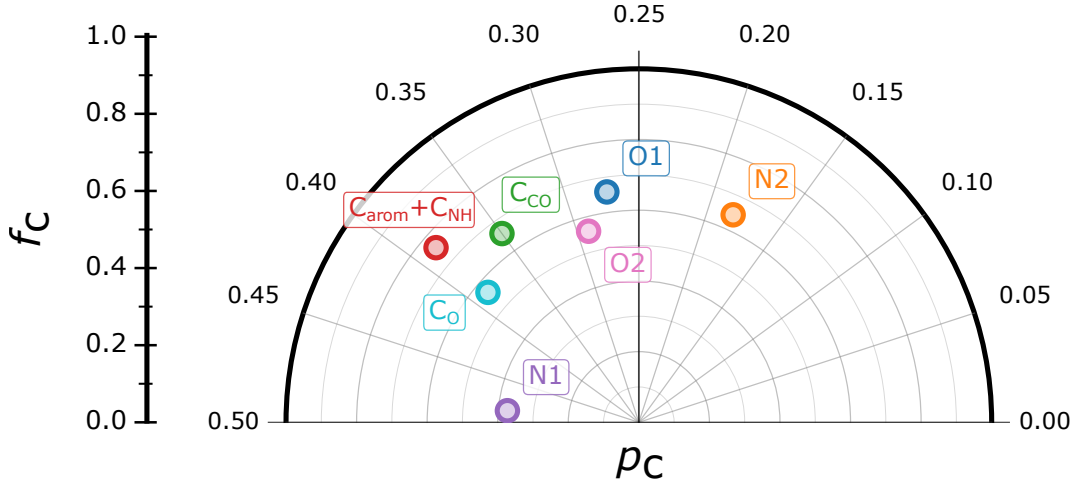


Figure 6.8: Argand representation for QA on the Ag(100) surface in the β -phase for the C1s, N1s, and O1s core levels.

Figure 6.8 shows that the different core levels of QA on the Ag(100) surface in the β -phase have more varying coherent fractions f_c and coherent positions p_c compared to the α -phase, indicating that the different atoms in the QA molecule are located at more diverse heights above the Ag(100) surface.

All C1s components have similar coherent positions p_c and thus similar heights above the surface. However, there are slight differences between the $C_{\text{arom}} + C_{\text{NH}}$, C_{CO} and C_{O} components. The C_{CO} component has the lowest coherent position p_c of 0.350, but the $C_{\text{arom}} + C_{\text{NH}}$ and C_{O} components have the same coherent position p_c of 0.387, which is higher than the coherent position p_c of the C_{CO} component. This shows that the C atoms in the QA molecule are not in a perfect plane but slightly distorted.

The C1s component also have different coherent fractions f_c , with the $C_{\text{arom}} + C_{\text{NH}}$ component having the highest coherent fraction f_c of 0.758, followed by the C_{CO} component with a coherent fraction f_c of 0.660 and the C_{O} component with a coherent fraction f_c of 0.564. This shows that the $C_{\text{arom}} + C_{\text{NH}}$ component has the highest degree of order, while the C_{O} component has the lowest degree of order. However, the differences in the coherent fractions f_c are also affected by the different signal-to-noise ratios of the different components, which can lead to higher errors for the components

with lower signal-to-noise ratios and thus to a less reliable determination of the coherent fractions f_c .

The N1 and N2 components have very different coherent positions p_c , with the N1 component having a coherent position p_c of 0.486 and the N2 component having a coherent position p_c of 0.184. It is important to note that the N2 component is the deprotonated N atom, which does not form a hydrogen bond to the neighboring molecule, while the N1 component is the protonated N atom, which forms a hydrogen bond to the neighboring molecule. This shows that the deprotonation of the N2 component leads to a significantly different adsorption behavior of this N atom compared to the protonated N1 component.

The deprotonated N atom is located significantly closer to the Ag(100) surface than the protonated N atom. This could be explained by the fact that the deprotonated N atom has a high electron density due to the electron lone pair, which can lead to a stronger interaction between the deprotonated N atom and the Ag(100) surface due to a donation of electron density from the deprotonated N atom to the Ag(100) surface. In addition, this shows that the adsorption height of the deprotonated N atom is mostly influenced by the interaction with the Ag(100) surface.

In contrast, the protonated N atom has a similar distance to the surface as the N atoms in the α -phase. This shows that the interaction between the protonated N atom and the Ag(100) surface is not significantly altered by the phase transition from the α -phase to the β -phase. Furthermore, it also shows that the adsorption height of the protonated N atom is mostly influenced by the hydrogen bond to the neighboring molecule.

The coherent fractions f_c of the N1 and N2 components are also very different, with the N1 component having a coherent fraction f_c of 0.374 and the N2 component having a coherent fraction f_c of 0.645. This shows that the deprotonated N2 component has a significantly higher degree of order than the protonated N1 component, which could be explained by its stronger interaction with the Ag(100) surface, which leads to a more well-defined adsorption height for the deprotonated N2 component compared to the protonated N1 component.

The protonated N1 component has a higher coherent fraction f_c by 0.9 as the N atoms in the α -phase, which shows that the degree of order of the protonated N1 component is altered by the phase transition from the α -phase to the β -phase, but not as much as for the deprotonated N2 component. This indicates that the adsorption height of the protonated N1 component is mostly influenced by the hydrogen bond to the neighboring molecule.

Last, the O1 and O2 components have similar coherent positions p_c and thus similar heights above the surface, which is closer to the surface than in the α -phase. This shows that the O atom with and the O atom without a hydrogen bond to the

neighboring molecule have similar adsorption heights. Consequently, the interaction between the O atoms and the Ag(100) surface is more important than the interaction by intermolecular hydrogen bonds for the adsorption height.

This behaviour is the opposite to the N atoms, where the adsorption height strongly depends on the presence of a hydrogen bond to the neighboring molecule. This can be explained by the fact that the electron density of the O atoms with or without a hydrogen bond does not change as significant as the electron density of the N atoms between the protonated and deprotonated state.

The coherent fractions f_c of the O1 and O2 components are different by 0.099, with the O1 component having a coherent fraction f_c of 0.658 and the O2 component having a coherent fraction f_c of 0.559. This shows that the O1 component, which forms a hydrogen bond to the neighboring molecule, has a higher degree of order than the O2 component, which does not form a hydrogen bond to the neighboring molecule. This indicates that the hydrogen bond to the neighboring molecule has a small influence on the degree of order of the O atoms in the QA molecule, even though the adsorption height of the O atoms is mostly influenced by the interaction with the Ag(100) surface.

7 Structure Models of QA on the Ag(100) surface

The results of the XPS and NIXSW measurements, that were discussed beforehand, have led to new insights regarding the adsorption height of QA on the Ag(100) surface and especially the role of the hydrogen bonds in the β -phase. Based on these findings, revised structure models with the heights from the NIXSW measurements for both the α - and the β -phase are proposed in the following sections.

However, the lateral positions of the atoms in the structure models are still based on the structure models from the literature, which were discussed in chapter 2 and based on LEED and STM measurements.^[28,30] The covalent radii of the atoms are taken from reference [82], while the van der Waals radii of the atoms are taken from reference [83]. The structure models were created with ASE.^[84] As the adsorption heights of the H atoms could not be determined from the NIXSW measurements, they were estimated based on the adsorption heights of the other atoms. Nevertheless, the H atoms are shown in the structure models for completeness, and their height is set to the same value as the height of the C_{arom} atoms.

7.1 Structure model for the α -phase

First, the structure model for the α -phase of QA on the Ag(100) surface is explained in more detail. The top view of the revised structure model is shown in Figure 7.1, while the long and short side views of the QA molecule are shown in Figure 7.2 and Figure 7.3, respectively.

Figure 7.1 shows the top view of the current structure model for the α -phase of QA on the Ag(100) surface. The unit cell and its structure is based on reference [28], which was discussed in chapter 2. However, it is important to note that the α -phase consists of parallel homochiral chains of QA molecules, which are stabilized by hydrogen bonds between the amine and carbonyl groups of neighboring molecules.^[28] Each QA molecule forms 4 hydrogen bonds with its two neighbors, which are indicated by the white ellipses in Figure 7.1.

The α -phase of QA on the Ag(100) surface forms a pol structure, which means that the molecules are not adsorbed at specific adsorption sites on the surface, but rather in a way that optimizes the intermolecular interactions, such as the hydrogen bonds. This explains the more or less similar adsorption heights of the different atoms, as

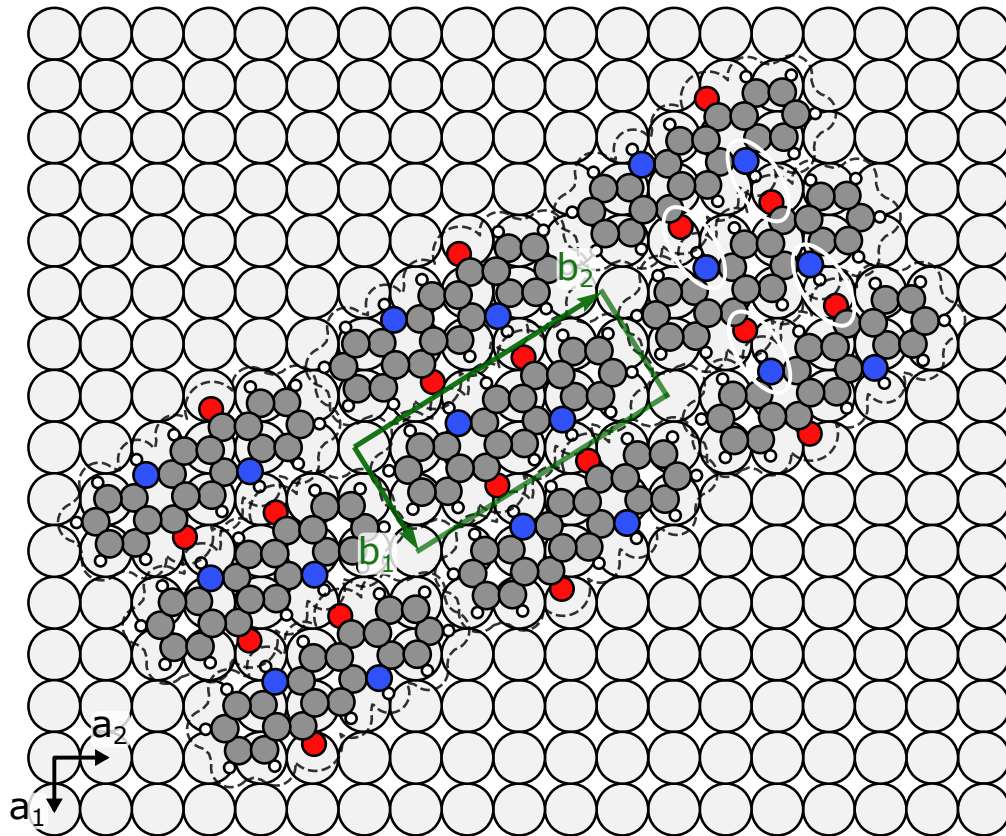


Figure 7.1: Top view of the structure model for the α -phase of QA on the Ag(100) surface. The van der Waals radii of the atoms are shown as dashed grey lines, and the 4 hydrogen bonds of one QA molecule to its two neighbors are indicated by white ellipses. The unit cell is indicated by the green rectangle with the corresponding unit cell vectors $\mathbf{b}_1$ and $\mathbf{b}_2$. The unit cell and its structure is based on reference [28].

there is no specific adsorption site for e.g. the O or N atoms. The adsorption height of the QA molecule in the α -phase is thus mainly determined by the intermolecular interactions, such as the hydrogen bonds, rather than by the interaction with the surface.

The distance of N and O atoms that form the hydrogen bonds in the α -phase is approximately 2.7 Å, which indicates a strong hydrogen bond, as the typical distance for hydrogen bonds is approximately 2.9 Å.^[83,85] This strong hydrogen bonding is supposed to be the driving force for the formation of the α -phase and the stabilization of the structure.

The structure of the α -phase is not significantly different from the structure model from the literature,^[28] which is shown in Figure 2.2. The main difference is that the adsorption height of the QA molecule is now based on the results of the NIXSW measurements, which are discussed in section 6.1.

This leads to the new images of the long and short side views of the QA molecule in the α -phase, which are shown in Figure 7.2 and Figure 7.3, respectively.

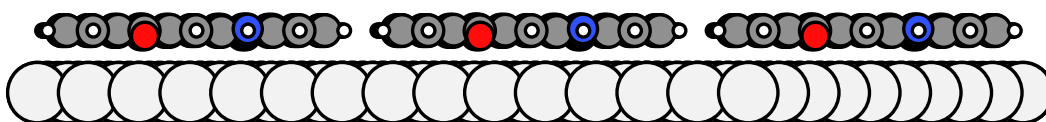


Figure 7.2: Long side view of the revised structure model for the α -phase of QA on the Ag(100) surface.

The long side view of the structure model for the α -phase of QA on the Ag(100) surface in Figure 7.2 shows the adsorption height of the QA molecule, which is based on the results of the NIXSW measurements. It shows that the QA molecule is adsorbed face-on on the Ag(100) surface, with the O atoms of the carbonyl groups pointing towards the surface. The other atoms of the QA molecule are more or less at the same height, and only exhibit minor differences in the adsorption height.

The different heights of the O and N atoms, which are the atoms that are involved in the hydrogen bonds, is clearly visible in the long side view. The H atom, that is part of the amine group and thus forms a hydrogen bond with the keto group of a neighboring molecule, is expected to be located at a height between the heights of the N and O atoms. However, the adsorption height of the H atoms are not known.

The long side view also shows the distance between two 1D chains of QA molecules, which is described by the unit cell vector $\mathbf{b}_2$. As there are no hydrogen bonds between the 1D chains, the distance between the chains is larger than the distance between the molecules within the chains, which is described by the unit cell vector $\mathbf{b}_1$. The

distance between the molecules within the chains is visible in the short side view of the structure model for the α -phase of QA on the Ag(100) surface in Figure 7.3.

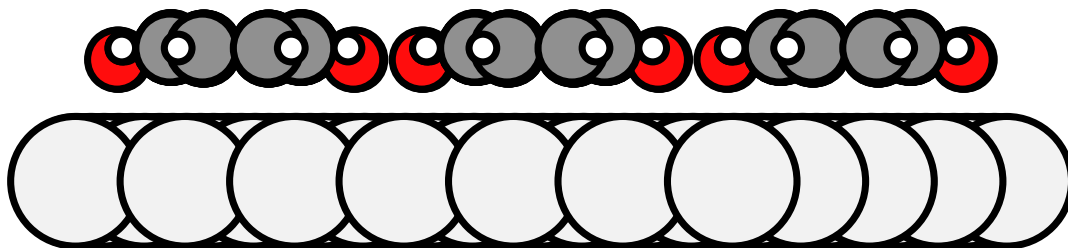


Figure 7.3: Short side view of the revised structure model for the α -phase of QA on the Ag(100) surface.

Furthermore, the short side view of the structure model for the α -phase of QA on the Ag(100) surface in Figure 7.3 shows the approach of the O atoms of the carbonyl groups towards the surface. The O atoms are closer to the surface than the other atoms of the QA molecule. This view also shows that the O atoms are located outside of the ring, allowing them to be more flexible in their adsorption height and thus to interact more strongly with the surface. The N atoms of the amine groups, on the other hand, are located within the ring and thus are more constrained in their adsorption height.

7.2 Structure model for the β -phase

In contrast to the structure model for the α -phase of QA on the Ag(100) surface, the structure model for the β -phase significantly differs from the literature, which is shown in Figure 2.3. However, the unit cell and the positions of the molecules within the unit cell are still based on the LEED and STM measurements from literature.^[28]

The main difference of the structure model for the β -phase of QA on the Ag(100) surface compared to the literature is the deprotonation of 50 % of the N atoms, which leads to only 1 hydrogen bond per molecule instead of the previously proposed 1.5 hydrogen bonds per molecule.

Furthermore, it is important to note that different models of the β -phase of QA on the Ag(100) surface are possible with the current knowledge. In principal, three different models are possible, which will all be discussed in the following. The difference between these models is the chirality of the molecules and thus different possibilities for the formation of the hydrogen bonds.

However, all three models have in common that there are only 1 hydrogen bond per molecule, have the same unit cell and the same positions of the molecules within the unit cell. This is needed to match the experimental results of the LEED and STM measurements, as well as the XPS and NIXSW measurements.

7.2.1 Homochiral β -phase

The homochiral structure model for the β -phase of QA on the Ag(100) surface is shown in Figure 7.4. In this structure model, all QA molecules have the same handedness, which means that there are no dimers with alternating handedness as in the structure model from the literature. The unit cell and its structure is based on reference [28].

The homochiral structure model for the β -phase of QA on the Ag(100) surface in Figure 7.4 shows that all QA molecules have the same handedness. The hydrogen bonds in this structure model are formed between one molecule with one of its neighbors, while there are no hydrogen bonds to the other neighbor. This leads to a total of one hydrogen bond per molecule.

However, these hydrogen bonds are very long with a distance of 3.5 Å between the N and the O atoms. This distance is larger than the most often observed distance for hydrogen bonds from literature, which is approximately 2.9 Å.^[83,85] Nevertheless, the distance of 3.4 Å is still in the range of possible hydrogen bonds, which goes up to a distance of 4.0 Å.

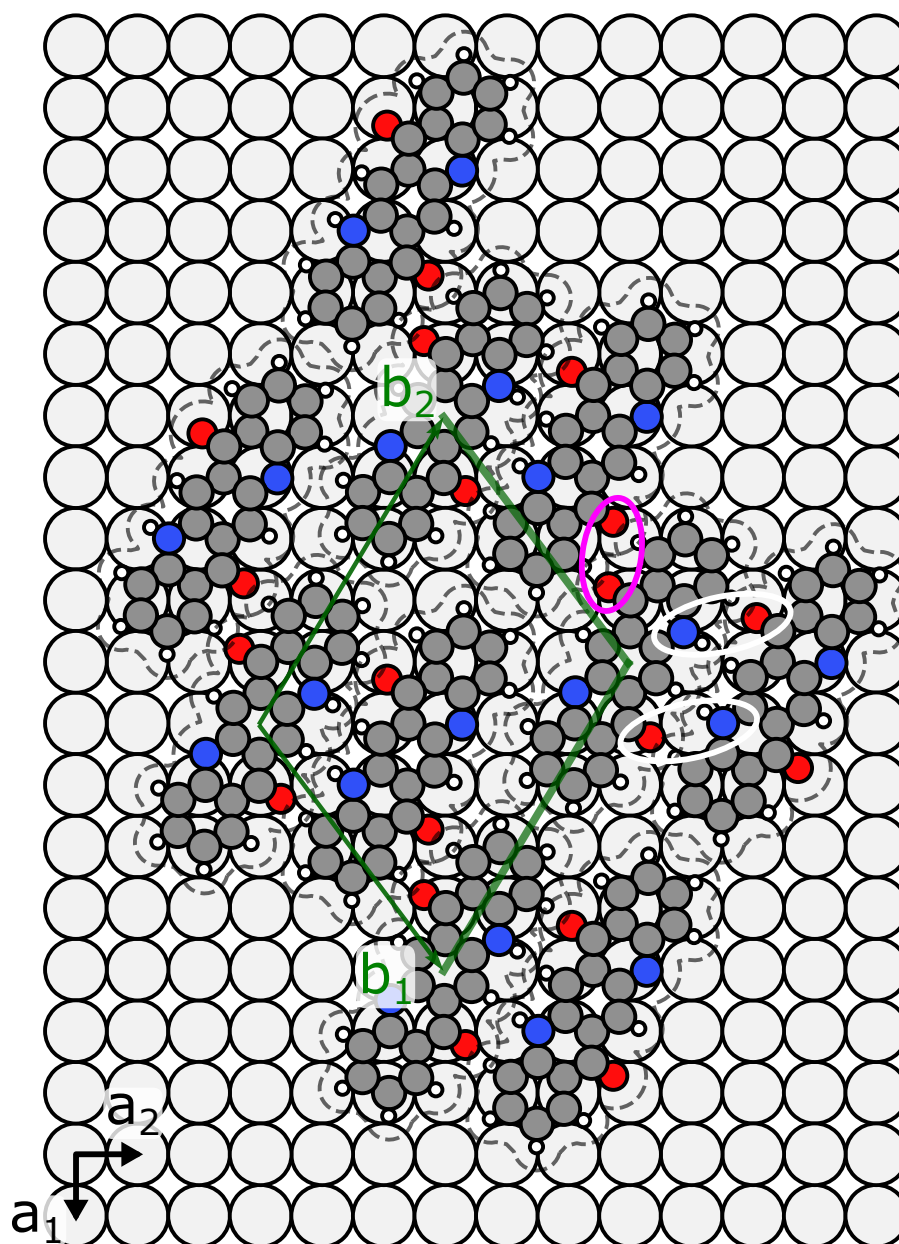


Figure 7.4: Top view of the structure model for the β -phase of QA on the Ag(100) surface. The van der Waals radii of the atoms are shown as dashed grey lines, and the 2 hydrogen bonds of one QA molecule to one of its two neighbors are indicated by white ellipses. The small distance between the O atoms in neighboring molecules are marked by a pink ellipse. The unit cell is indicated by the green rectangle with the corresponding unit cell vectors $\mathbf{b}_1$ and $\mathbf{b}_2$. The unit cell and its structure is based on reference [28].

As the O atoms are located outside of the ring, they are more flexible in their lateral position and thus can be located closer to the N atoms. This only leads to a slightly decreased distance between the O and N atoms, as the O and N atoms are almost located in the best way possible for the given molecular arrangement.

Furthermore, as the hydrogen bonds only exist between one molecule and one of its neighbors, there is not a strong driving force for the molecules to arrange in the long 1D molecular chains that are observed in the STM images.

In addition, there is one more problem with the homochiral structure model. The two neighboring molecules that do not form a hydrogen bond with each other, especially their O atoms, are located very close to each other with a distance of approximately 3.1 Å, which is close to the sum of the van der Waals radii of the O atoms, which is approximately 3.0 Å.^[83] This leads to a strong repulsion between the O atoms and make the positions of the O atoms very unlikely. Consequently, the O atoms are expected to be located further apart from each other. This is also possible, because the O atoms are located outside of the ring and thus are more flexible in their lateral position.

However, these two O atoms are located opposite to H atoms from the aromatic ring. In general, weak hydrogen bonds between the H atoms of the aromatic ring and the O atoms of the carbonyl groups are possible, which could lead to a stabilization of the structure.^[86]

Beside these problems, the homochiral structure model is consistent with the molecular arrangement, the hydrogen bonds and the unit cell from the LEED pattern even with respect to the handedness of the molecules, which was not possible for the previous structure model from literature.

7.2.2 Heterochiral β -phase with dimers

The heterochiral structure model for the β -phase of QA on the Ag(100) surface with dimers is shown in Figure 7.5. This structure model is the most similar to the structure model from the literature,^[28] as it also consists of dimers with alternating handedness, but the number of hydrogen bonds is reduced to one hydrogen bond per molecule, which is in agreement with the results of the XPS measurements.

Figure 7.5 shows the heterochiral structure model for the β -phase of QA on the Ag(100) surface with dimers. In this structure model, two molecules with the same handedness form a dimer with two hydrogen bonds between the O and N atoms. The dimers with the same handedness are shown in the same underlying color. The handedness of the molecules in the dimers is alternating, which means that there are dimers with opposite handedness next to each other. This is the same as in the

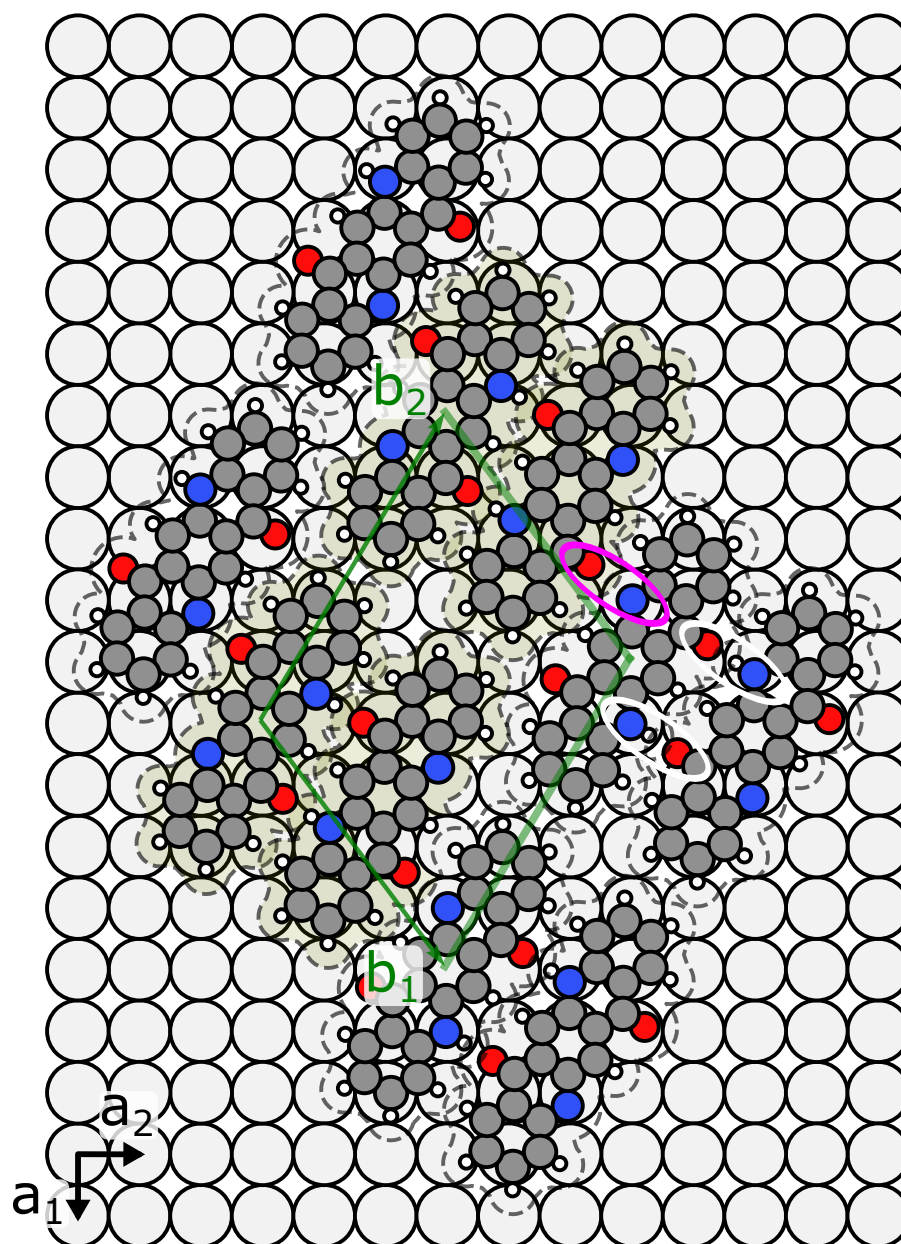


Figure 7.5: Top view of the structure model for the β -phase of QA on the Ag(100) surface. The van der Waals radii of the atoms are shown as dashed grey lines, and the 2 hydrogen bonds of one QA molecule to one of its two neighbors are indicated by white ellipses, while the missing hydrogen bond due to deprotonation is indicated by a pink ellipse. The unit cell is indicated by the green rectangle with the corresponding unit cell vectors $\mathbf{b}_1$ and $\mathbf{b}_2$. The unit cell and its structure is based on reference [28].

structure model from the literature,^[28] which also consists of dimers with alternating handedness.

However, in contrast to the structure model from the literature, there are no hydrogen bonds between the neighboring molecules in different dimers. This is due to the deprotonation of 50 % of the N atoms, which leads to only one hydrogen bond per molecule. The missing hydrogen bond between the neighboring molecules in different dimers is indicated by the pink ellipse in Figure 7.5.

It may be possible that the N atom that is deprotonated in the β -phase is not the N atom that form hydrogen bonds between different dimers, but the N atom that form the hydrogen bonds within the dimers. However, this is less likely, as the deprotonation is related to an energy loss and the formation of hydrogen bonds is related to an energy gain and there should be no reason to deprotonate the N atoms within the dimer, but still form a dimer.

Nevertheless, this structure model has the same problem as the homochiral structure model, as the formation of long 1D molecular chains in the STM images is not well explained by this structure model, as there are only hydrogen bonds within the dimers, but no hydrogen bonds between the dimers. Thus, there is no strong driving force for the formation of the long 1D molecular chains.

Furthermore, the rearrangement from the α - to the β -phase for which a change in the handedness of the different dimers is needed, may be difficult to understand, as the molecules in the α -phase are homochiral and thus all have the same handedness. There is also no mechanism for the change in the handedness of the molecules in the literature.^[28,30]

However, this structure model also has the advantage that the distance of the N and O atoms that form hydrogen bonds is 2.6 Å, which is 0.3 Å shorter than the typical distance for hydrogen bonds (2.9 Å).^[83,85] This shows that the hydrogen bonds in this structure model are very strong, which could be a driving force for the formation of the β -phase and the energy compensation for the deprotonation of the N atoms.

The N-O distance of 2.6 Å is also the distance between the O atoms and the deprotonated N atoms of a neighboring molecule, which are located opposite to each other. This is smaller than the sum of the van-der Waals radii of the O and N atoms, which is approximately 3.2 Å.^[83]

Consequently, the O atoms with a high electron density and the deprotonated N atoms with also a high electron density are located close to each other and therefore should repel each other. This may cause a lateral displacement of the O atoms, which are located outside of the aromatic ring and thus more flexible, in the structure model to avoid the repulsion. In advance, this may lead to a weak hydrogen bond to a H-C

atom of the aromatic ring, which could stabilize the structure. However, this is only a speculation without any evidence.

7.2.3 Heterochiral β -phase with alternating handedness

The heterochiral structure model for the β -phase of QA on the Ag(100) surface with alternating handedness is shown in Figure 7.6. In this structure model, neighboring molecules always have different handedness, which means that there are no dimers with alternating handedness as in the structure model from the literature. The unit cell and its structure is based on reference [28].

Figure 7.6 shows the heterochiral structure model for the β -phase. In this structure model, neighboring molecules always have different handedness (e.g., R L R L). The molecules in the structure model with the yellow color have the same handedness, while the other molecules have the opposite handedness as the yellow ones.

The hydrogen bonds in this structure model are formed between neighboring molecules with different handedness, which leads to a total of one hydrogen bond per molecule. In contrast to the homochiral structure model, the hydrogen bonds in the heterochiral structure model with alternating handedness are always with both neighboring molecules. Thus, the intermolecular hydrogen bonds are the driving force for the formation of the long 1D molecular chains that are observed in the STM images, which is a big advantage of this structure model compared to the homochiral and the dimer structure model.

However, the distance of the N and O atoms that form the hydrogen bonds in this structure model is even larger with a distance of 3.8 Å. This is even larger than the distance of 3.5 Å in the homochiral structure model, which is already quite large for a hydrogen bond. Thus, the hydrogen bonds in this structure model are very weak, but still possible in comparison to literature values for hydrogen bonds.^[83,85]

As already mentioned for the homochiral structure model, the O atoms are located outside of the ring and thus are more flexible in their lateral position. This allows them to be located closer to the N atoms, which leads to a smaller distance between the O and N atoms. This may decrease the N-O distance from 3.8 Å to a more reasonable distance for hydrogen bonds. However, there is no evidence that this occurs.

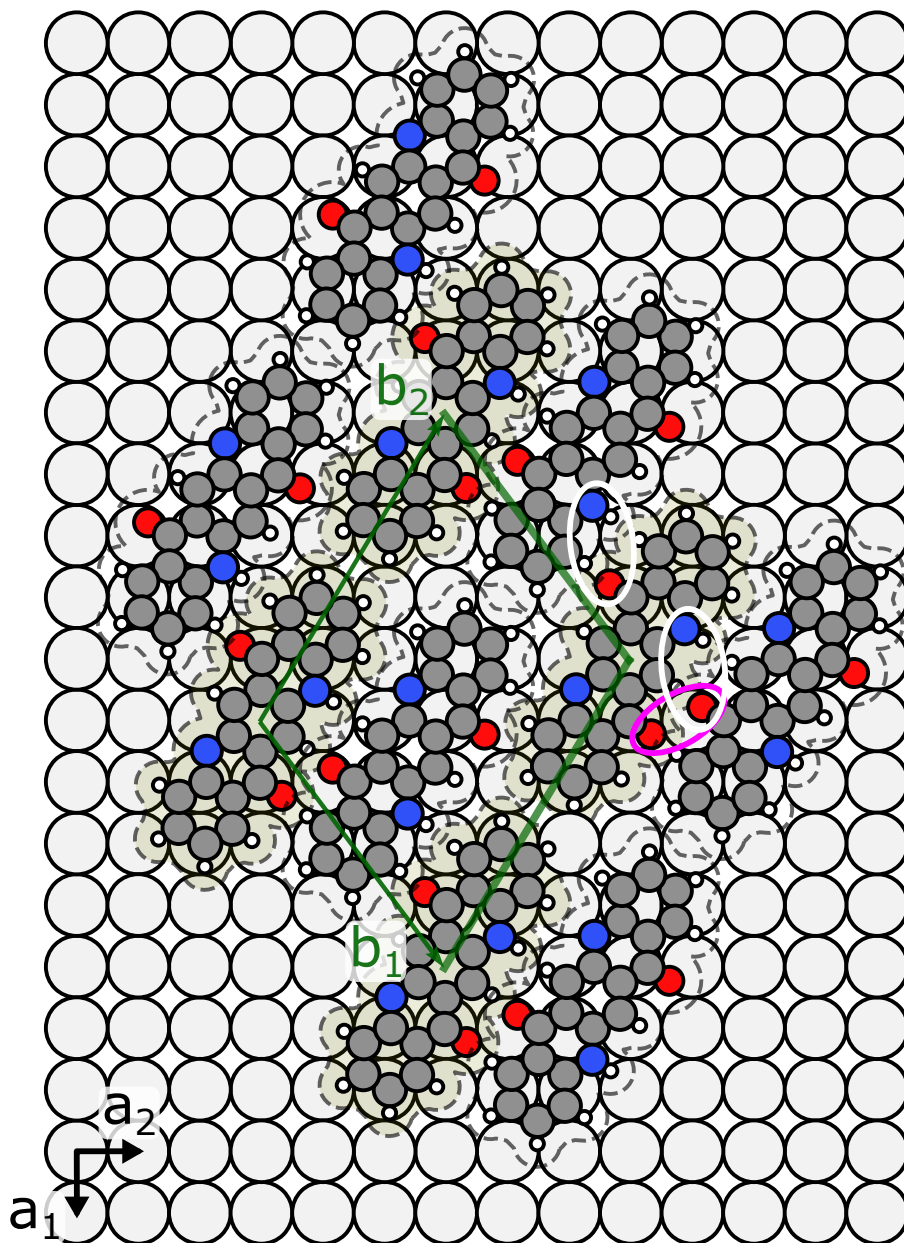


Figure 7.6: Top view of the structure model for the β -phase of QA on the Ag(100) surface. The van der Waals radii of the atoms are shown as dashed grey lines, and the 2 hydrogen bonds of one QA molecule to its two neighbors are indicated by white ellipses. The small distance between the O atoms in neighboring molecules are marked by a pink ellipse. The unit cell is indicated by the green rectangle with the corresponding unit cell vectors $\mathbf{b}_1$ and $\mathbf{b}_2$. The unit cell and its structure is based on reference [28].

7.3 Structure model for the β -phase

In contrast to the structure model for the α -phase of QA on the Ag(100) surface, the structure model for the β -phase significantly differs from the literature, which is shown in Figure 2.3.

As already observed in the literature,^[28] the β -phase of QA on the Ag(100) surface is not homochiral, but consists of dimers with opposite chirality. In between the QA molecules in the dimers, there are 2 hydrogen bonds. The hydrogen bonds are formed between the O1 and N1 components of the neighboring molecules. These two hydrogen bonds are shown in Figure 7.7 as white ellipses.

However, between the different dimers, there are no hydrogen bonds at all. This is in contrast to the previous structure model from the literature, which is shown in Figure 2.3, where it was assumed that there is one hydrogen bonds between the neighboring molecules in different dimers. The missing hydrogen bond between the neighboring molecules in different dimers is indicated by the orange ellipse in Figure 7.7.

As the XPS measurements have shown, the N2 component is deprotonated in the β -phase and is thus not able to form a hydrogen bond with the O2 component of a neighboring molecule. As the number of the N atoms in the N1 and N2 component are the same, there are only two hydrogen bonds per molecule to its two neighbors in the β -phase, which are formed between the O1 and N1 components of neighboring molecules in the dimers.

Furthermore, the β -phase exhibits a commensurate structure, which is described by the unit cell vectors $\mathbf{b}_1$ and $\mathbf{b}_2$ in Figure 7.7. The unit cell and its structure is based on reference [28]. This shows that the QA molecules in the β -phase are adsorbed at specific adsorption sites on the Ag(100) surface, which is in contrast to the α -phase, where the molecules are adsorbed in a pol structure.

The atom that is most likely adsorbed at a specific adsorption site is the deprotonated N2 component, which is not involved in any hydrogen bonds and located very close to the surface. The N2 component is thus expected to be adsorbed at a specific adsorption site on the Ag(100) surface, which is most likely a on-top site, as almost all N atoms of the N2 component are located above the Ag atoms in the structure model in Figure 7.7.

Nevertheless, some of the N atoms of the N atoms are not located above the Ag atoms. This could be due to the fact that the unit cell of the β -phase is very large and contains two QA molecules and 27 Ag atoms per unit cell. Thus, it is possible that the positions of the N atoms are not perfectly accurate in the structure model in Figure 7.7, which is based on LEED and STM measurements from the literature.^[28] In

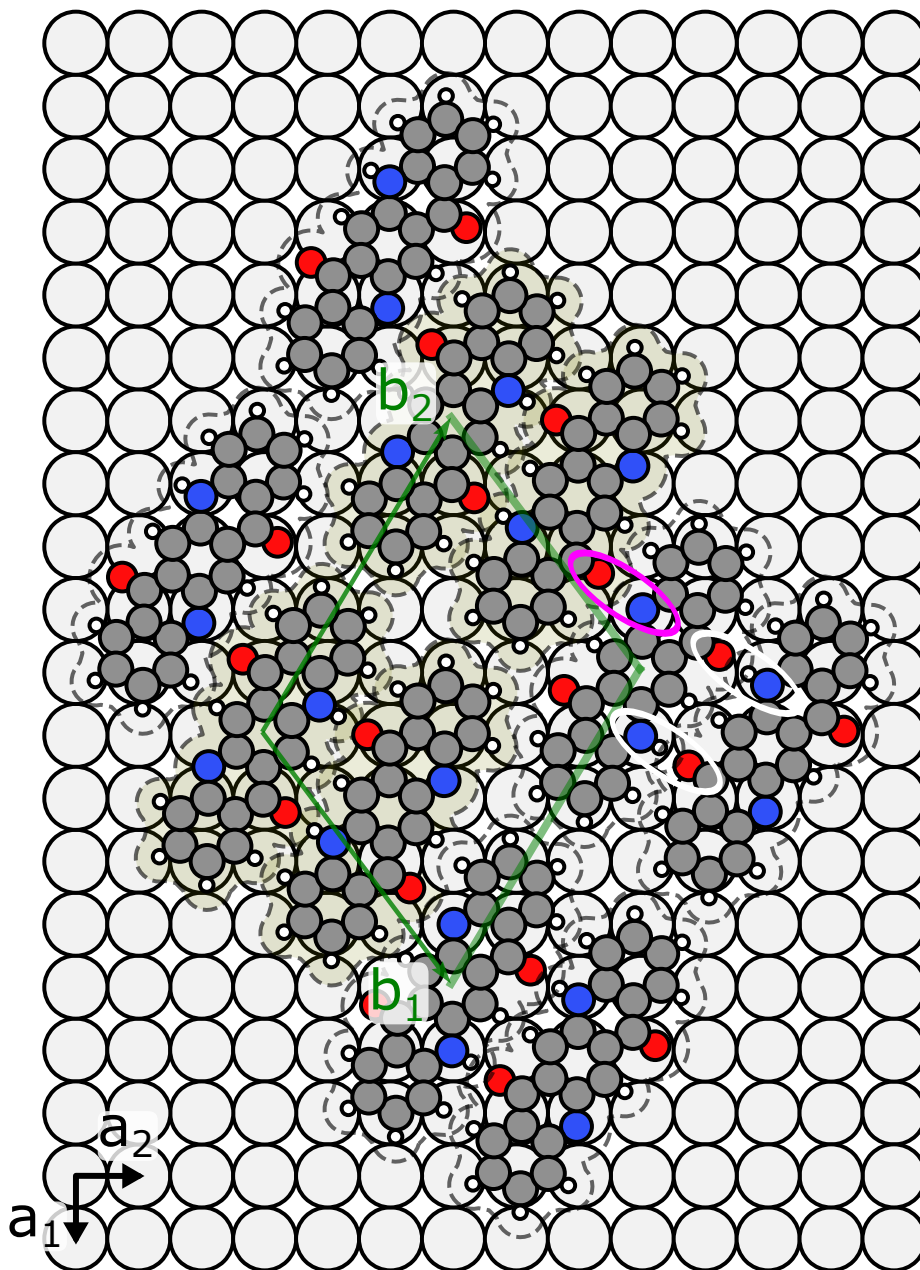


Figure 7.7: Top view of the structure model for the β -phase of QA on the Ag(100) surface. The hydrogen bonds between the O1 and N1 components are marked with white ellipses, while the missing hydrogen bond due to deprotonation between the O2 and N2 component is marked with an orange ellipse. The van der Waals radii of the atoms are shown as dashed grey lines. The unit cell is indicated by the green rectangle with the corresponding unit cell vectors $\mathbf{b}_1$ and $\mathbf{b}_2$ and is based on reference [28].

addition, the bonds in the QA molecule are not perfectly rigid, and thus the positions of the atoms could be slightly different from the positions in the structure model.

Another interesting aspect of the structure model for the β -phase of QA on the Ag(100) surface is the shift between the different dimers along the unit cell vector $\mathbf{b}_2$. This shift can be seen in the long side view of the structure model in Figure 7.8, where the QA molecule at the back is shifted to the right compared to the dimer at the front.

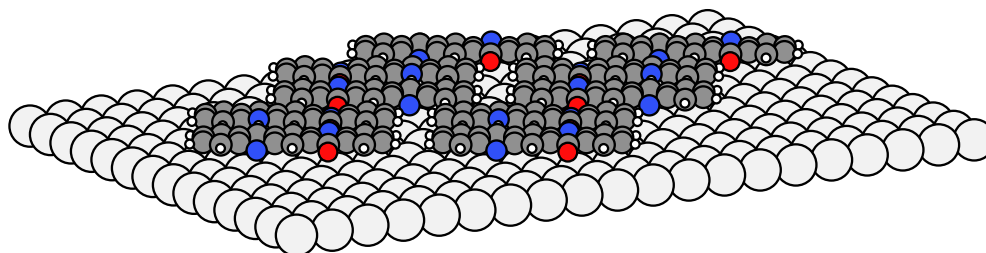


Figure 7.8: Long side view of the revised structure model for the β -phase of QA on the Ag(100) surface. The underlying Ag atoms at the corners of Figure 7.7 are deleted for better visibility.

As already observed for the α -phase, the QA molecules in the β -phase form molecular chains in one direction, which is described by the unit cell vector $\mathbf{b}_1$. The distance between the different molecular chains are described by the unit cell vector $\mathbf{b}_2$ and are clearly visible in Figure 7.8.

Furthermore, this view already indicates the different adsorption heights of the different atoms in the QA molecule in the β -phase, which are based on the results of the NIXSW measurements. The N2 component is the closest atom to the surface. For a more detailed view of the different adsorption heights, the long side view of only one QA molecule is shown in Figure 7.9 from both sides of the molecule.

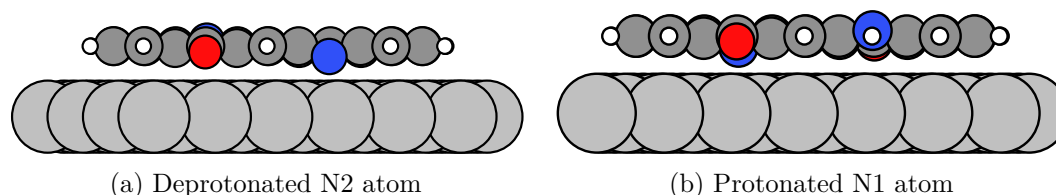


Figure 7.9: Long side views of the revised structure model for the β -phase of QA on the Ag(100) surface. The left image shows the side of the deprotonated N2 component, while the right image shows the side of the N1 component.

Figure 7.9 shows the different adsorption heights of the different atoms in the QA molecule in the β -phase. The deprotonated N2 component is the closest atom to the

surface, while the protonated N1 component is the highest atom, which is even higher than the C atoms. The O atoms are located closer to the surface than the C atoms, but are not as close to the surface as the N2 component. Furthermore, the different C atoms have different adsorption heights as well, which can be seen at the C_{CO} and C_O components, that are closer to the surface than the other C atoms and thus have an adsorption height between the other C atoms and the O atoms. The different adsorption heights can be explained by the different interactions of the atoms with the surface, which was done in detail in section 6.2.

This behavior of the different adsorption heights of the different atoms in the QA molecule can also be seen in the short side view of the structure model for the β -phase on the Ag(100) surface in Figure 7.10.

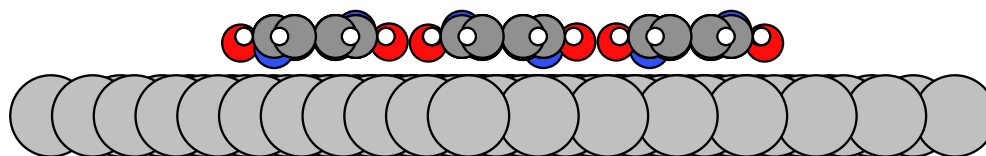


Figure 7.10: Short side view of the revised structure model for the β -phase of QA on the Ag(100) surface. The two QA molecules on the left are one dimer and form two hydrogen bonds, while the QA molecule on the right is part of the neighboring dimer without hydrogen bonds to the shown QA molecule.

Again, it is visible that the O atoms are closer to the surface than the C atoms, while the N2 component is the closest atom to the surface and the N2 component is the highest atom. This view enable a more detailed view at atoms involved in the hydrogen bonds, as the hydrogen bonds are perpendicular to the view. The hydrogen bonds are formed between the left and the middle QA molecule, which are part of the same dimer.

The N1 component is the located higher than the O1 component. However, the difference is only about 0.1 Å. The small difference in the height can be compensated by the H atom of the amine group, which is expected to be located at a height between the N1 and O1 component.

8 Summary and Outlook

The XPS measurements of QA on the Ag(100) surface revealed significant differences between the two phases. First, the C1s spectra revealed distinct components for chemically different C atom, which match to the stoichiometry of the QA molecule. The same holds true for the N1s and O1s spectra regarding existence of the hydrogen bonds. This allows to assign the different components of the spectra to specific atoms in the molecule for both phases.

Second, the BE of all components, except N1, in the β -phase are significantly lower than in the α -phase, which indicates a stronger molecule-substrate interaction in the β -phase. This is in line with the fact that the β -phase has a commensurate structure, which is generally related to a stronger molecule-substrate interaction, while the α -phase only has a pol structure.

Third, the N1s spectra of the α -phase show only one component, which indicates that all N atoms have the same chemical environment and thus form hydrogen bonds. In contrast, the N1s spectra of the β -phase show two different components with an area ratio of 1:1. The N1 component in the β -phase is at the same BE as the N1 component in the α -phase, while the N2 component in the β -phase is shifted by 2.2 eV to lower BE. The N1 component in the β -phase can thus be assigned to the N atoms that form hydrogen bonds, while the N2 component can be assigned to the N atoms that do not form hydrogen bonds due to the deprotonation of the N atoms.

The same holds true for the O1s spectra, where the O1 component can be assigned to the O atoms that form hydrogen bonds, while the O2 component can be assigned to the O atoms that do not form hydrogen bonds due to the deprotonation of the N atoms. Consequently, the XPS results indicate that there are two hydrogen bonds per molecule in the α -phase, but only one hydrogen bond per molecule in the β -phase, which contradicts the previously proposed 1.5 hydrogen bonds per molecule in the β -phase.^[28,30]

Last, the second component of the N1s and O1s spectra are also present in the XPS spectra if the sample is exposed to the X-rays for a long time. This indicates that the X-rays cause a deprotonation of the N atoms, which is a known phenomenon in XPS measurements.

Furthermore, the NIXSW measurements revealed the adsorption heights of the different atoms in the QA molecule for both phases. In the α -phase, the adsorption heights of the C and N atoms are similar and approximately 3.0 Å, while the O atoms are significantly lower at approximately 2.7 Å. This indicates a stronger molecule-substrate

interaction for the O atoms, which may be the reason for the pol structure of the α -phase. Furthermore, it indicates that the hydrogen bonds are not this relevant for the adsorption height of the O atoms.

In the β -phase, the adsorption heights of all atoms (except N1 & N2) are closer to the surface than in the α -phase. The C atoms are approximately at 2.8 Å, while the O atoms are again located closer to the surface at approximately 2.6 Å. This is in line with the stronger molecule-substrate interaction in the β -phase, which is also indicated by the lower BE of all components in the XPS spectra and the commensurate structure.

The N atoms show the most significant change in adsorption height between the two phases. The N1 atoms, which form hydrogen bonds, are located at approximately 3.0 Å in the α -phase, and do not change their adsorption height in the β -phase. This indicates that the N atoms that form hydrogen bonds are not this relevant for the molecule-substrate interaction and their adsorption height is mainly influenced by the hydrogen bonds.

In contrast, the N2 atoms, which do not form hydrogen bonds, are located at approximately 2.4 Å in the β -phase. This indicates that the N atoms that do not form hydrogen bonds are the most relevant for the molecule-substrate interaction. This is supposed to be the reason for the commensurate structure of the β -phase, which is generally related to a strong molecule-substrate interaction. Furthermore, it indicates that the deprotonation of the N atoms is the main reason for the stronger molecule-substrate interaction in the β -phase, which compensates the energy loss due to the missing hydrogen bonds.

Finally, the results of the XPS and NIXSW measurements indicate that the previously proposed structure models for QA on the Ag(100) surface are not correct, especially for the β -phase. Thus, new structure models for both phases are proposed based on the results of this thesis, which are discussed in detail in chapter 7. The underlying unit cell and molecular arrangement of the two phases are still the same, but the deprotonated N atoms are now taken into account, which leads to a lower number of hydrogen bonds in the β -phase. In addition, the adsorption heights of the different atoms are now included in the structure models.

In conclusion, the results of this thesis provide new insights into the phase transition of QA on the Ag(100) surface, especially regarding the role of the hydrogen bonds and the molecule-substrate interaction. However, there are still open questions regarding the exact mechanism of the phase transition and the exact positions of the deprotonated N atoms in the β -phase.

The first question could be investigated by performing temperature-dependent STM measurements, which may show the rearrangement of the QA molecules while the phase transition occurs. The second question could be investigated by performing

NIXSW measurements for different Bragg planes, which allows the triangulation of the e.g. N atoms and thus the determination of their exact adsorption site.

9 Appendix

9.1 Overview of all XPS spectra

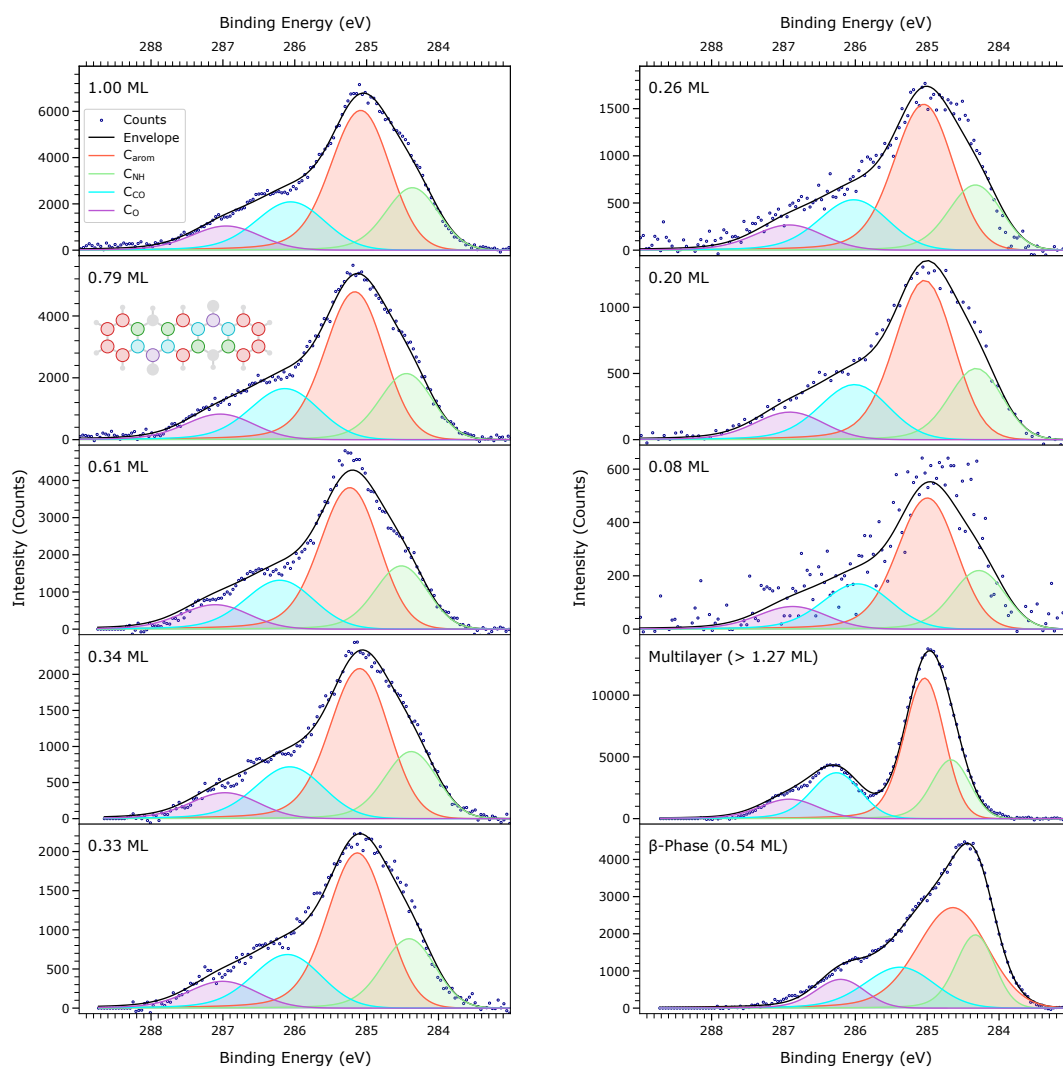


Figure 9.1: Overview of all C1s XPS spectra of QA on the Ag(100) surface.

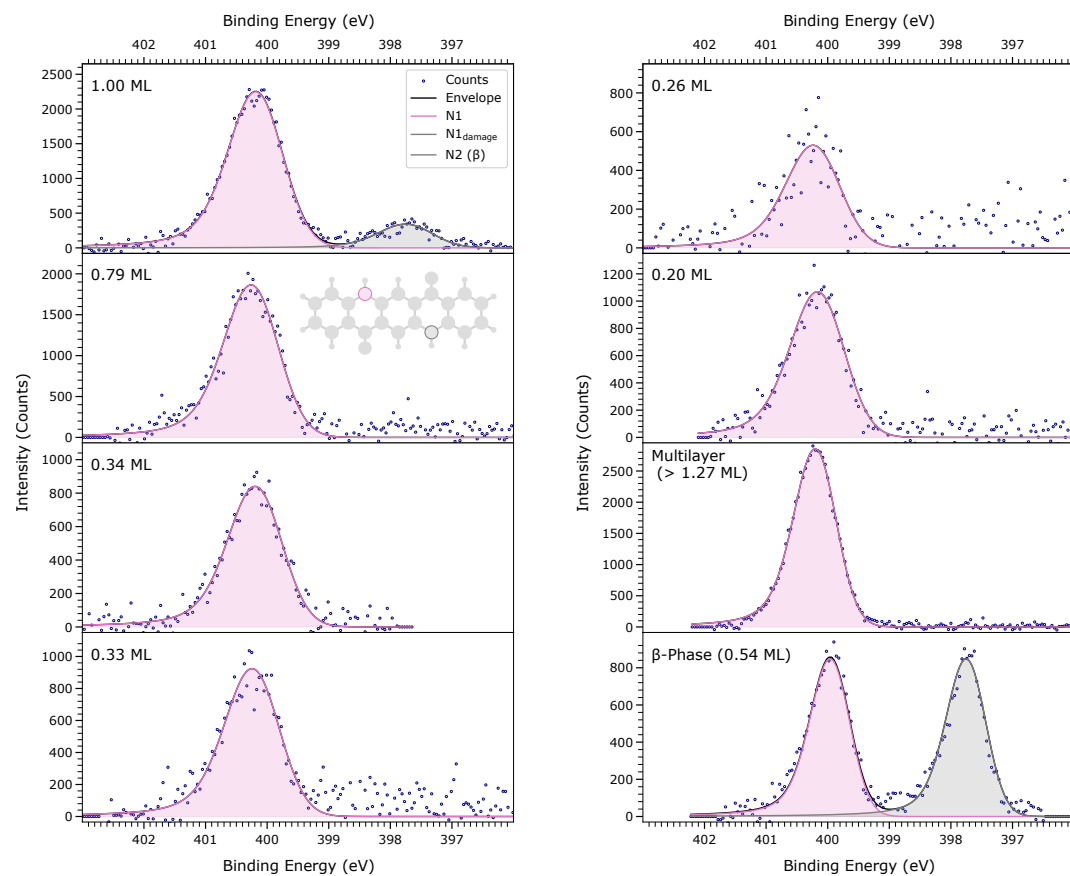


Figure 9.2: Overview of all N1s XPS spectra of QA on the Ag(100) surface.

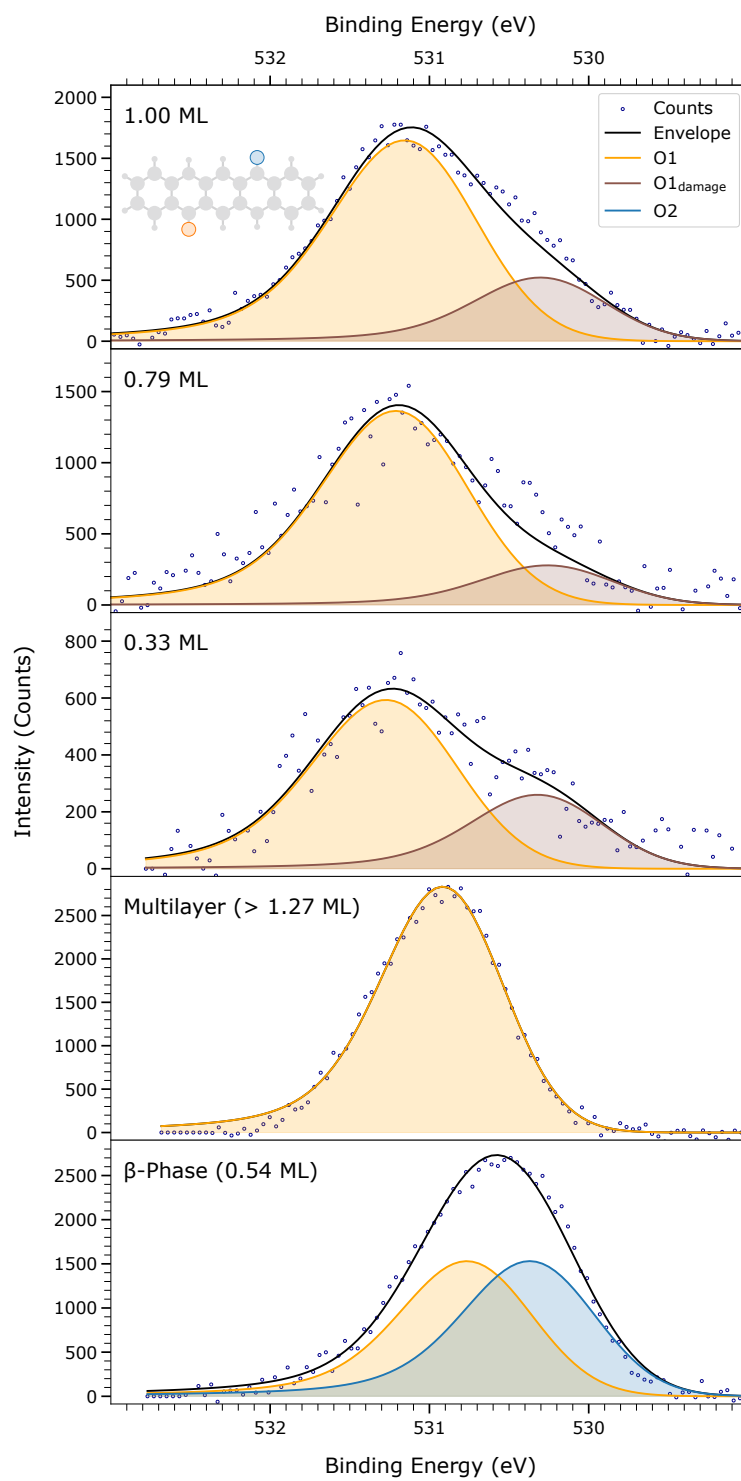
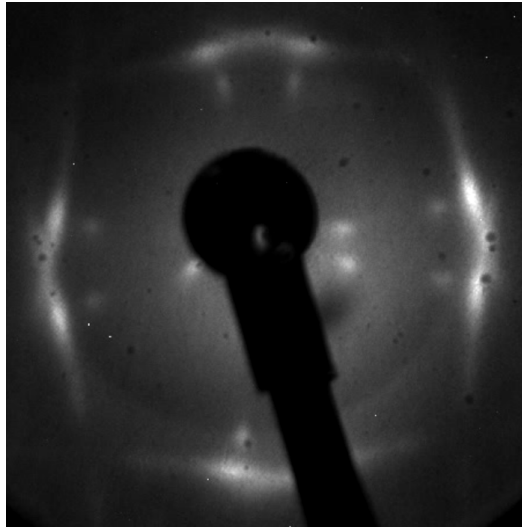
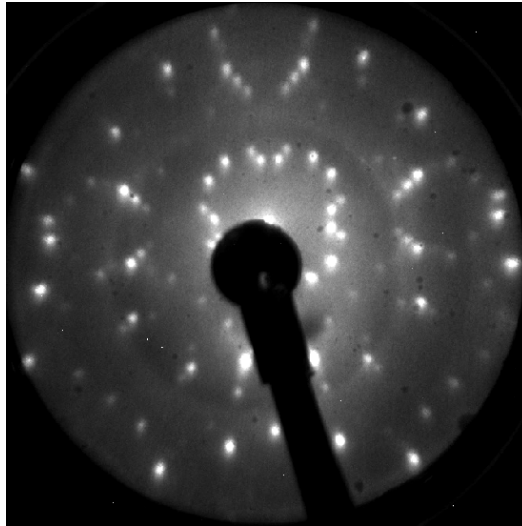


Figure 9.3: Overview of all O1s XPS spectra of QA on the Ag(100) surface.

9.2 LEED patterns of QA on the Ag(100) surface



(a) Point-on-line (POL) α -phase



(b) Commensurate β -phase

Figure 9.4: LEED patterns for the α - and the β -phase at 25.5 eV taken at the I09 beamline of the Diamond Light Source.

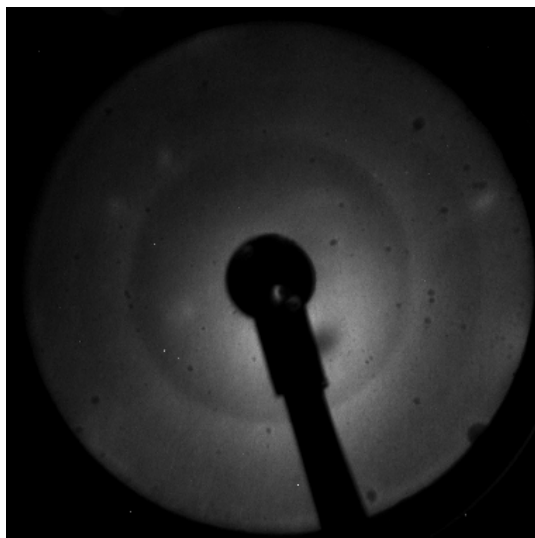


Figure 9.5: LEED patterns for burnt phase.

List of abbreviations

BE binding energy

FWHM full width at half maximum

pol point-on-line

LEED low energy electron diffraction

OLED organic light-emitting diode

OPV organic photovoltaic

OFET organic field-effect transistor

ML monolayer

NIXSW normal incidence x-ray standing wavefield

XSW x-ray standing wavefield

fcc face-centered cubic

POL point-on-line

QA quinacridone

PTCDA Peryleno[3,4-cd:9,11-c'd']dipyran-3,5,10,12-tetrone

UHV ultra-high vacuum

SPA-LEED spot profile analysis low energy electron diffraction

STM scanning tunneling microscopy

XPS x-ray photoelectron spectroscopy

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